E350 U3400

E350 Three Phase WC Meter

User Manual

Revision history

Version	Date	Comments
V4.1	02.07.2014	Initial Version including FW R4.1
V4.2	12.11.2014	Added comment for Pulsed Outputs
V4.5	30.03.2016	Includes all FW R4.5 enhancements
V4.6	14.06.2016	Includes R4.6 changes for element totalling
V4.6.1	28.07.2016	Correction to the meter dimension
V4.6.2	07.11.2016	Minor corrections
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V5.0	22.08.2018	Includes all FW R5.0 enhancements
V6.0	18.06.2019	Includes all FW R6.0 enhancements

Important Information
This document provides information relating to the E350 Three Phase WC Meter.

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1 About this document

Range of validity

The present user manual applies to the E350 Three Phase WC meter specified on the title page.

Purpose

The user manual contains all the information required for application of the meters for the intended purpose. This includes:

- Provision of knowledge concerning characteristics, construction and function of the meters
- Information about possible dangers, their consequences and measures to prevent any danger
- Details concerning the performance of all work throughout the service life of the meters (parameterization, installation, commissioning, operation, maintenance, shutting down and disposal)

Target group

The contents of this user manual are intended for technically qualified personnel of utilities (energy supply companies) responsible for the system planning, installation and commissioning, operation, maintenance, decommissioning and disposal of the meters.

Reference documents

In addition, the following documents provide more information related to the subject of this document:

- U3400 Factsheet
- EMPWin+ User Manual

Metering Conventions

To comply with AS 4140-1995 (Metering and utility information exchange), this manual uses the terms *Import metering* and *Export metering* based on the customer perception. These replace the previously used terms ‘delivered’ and ‘received’ based on the supply authority perception:

Energy imported by customer = Energy delivered by supply authority

Energy exported by customer = Energy received by supply authority

Terms and abbreviations

A list of terms and abbreviations used in this document is available in section 14 “Terms and abbreviations”.

2 Safety

2.1 Safety information

Attention is drawn as follows in the individual chapters of this user manual with classified word symbols and pictographs to the relevant danger level, i.e. the severity and probability of any danger:

Definition of Danger
For a possibly dangerous situation, which could result in severe physical injury or fatality.

Definition of Warning
For a possibly dangerous situation, which could result in minor physical injury or material damage.

Definition of Note
For general details and other useful information to simplify the work.

In addition to the danger level, all safety information also describes the type and source of the danger, its possible consequences and measures to counteract the danger.

2.2 Responsibilities

The owner of the meters – usually the utility company – is responsible that all persons engaged on work with meters:

- Have read and understood the relevant sections of the user manual.
- Are sufficiently qualified for the work to be performed.
- Strictly observe the safety regulations (according to section 2.3) and the operating information in the individual chapters.

In particular, the owner of the meters bears responsibility for the protection of persons, prevention of material damage and the training of personnel.

Landis+Gyr provides training courses for this purpose on specific equipment; please contact the relevant agent if interested.

2.3 Safety regulations

The following safety regulations must be observed at all times:

- The meter connections must not be live during installation or when opening. Contact with live parts is dangerous to life. The relevant main fuses should therefore be removed and kept in a safe place until the work is completed, so that other persons cannot replace them unnoticed.
- Local safety regulations must be observed. Only technically qualified and suitably trained personnel is authorised to install the meters.

- The meters must be held securely during installation. They can cause injuries if dropped.
- Meters which have fallen, must not be installed, even if no damage is apparent, but must be returned for testing to the service and repair department responsible (or the manufacturer). Internal damage can result in functional disorders or short-circuits.
- The meters must on no account be cleaned with running water or with compressed air devices. Water penetrating can cause short-circuits.

3 Description of Unit

3.1 Field of Application

The U3400 three phase meter is designed for 3x240 Volt whole current 50 Hertz distribution systems. It is a direct-connect meter with a maximum load of 100 amperes and a base current (I_b) of 10 amperes.

The meter is designed to support full AMI functionalities or operate as a standalone non-AMI meter. It supports ripple control functionality to control the meter internal load control relay.

The U3400 meter has been designed to have a plug in, field upgradeable communication module for WAN, LAN and HAN communications.

The meter is compliant to the relevant Australian Standards (see Appendix C – Standards”).

3.2 Major Features

- Class 1 accuracy
- Supports type 4, 5 and 6 installation
- NMI M6 pattern approval
- 7 digit Liquid Crystal Display
- Non-volatile memory
- Real Time Clock function
- Mains Clock synchronisation
- Four quadrant measurement
- load profile recording
    - Up to 594 Days of load profiling recording (4 channels 30 minutes interval) with standard memory.
    - Up to 1,320 Days of load profiling recording (4 channels 30 minutes interval) with “Extended 1” memory.
    - Up to 13,178 Days of load profiling recording (4 channels 30 minutes interval) with “Extended 2” memory.
    - Up to 4 channels per Load Profile group with Standard or Extended 1 memory.
    - Up to 24 channels per Load Profile group with Standard or Extended 2 memory. (96 LP channels in total)
    - Up to 4 Load Profile group with flexible interval of 5,10,15,30,60 minutes per group
- Time-of-Use tariff
- Up to 24 TOU registers.
    - 35 days of TOU and Total register snapshots with “Extended 1” memory.
    - 200 days of TOU and Total register snapshots with “Extended 2” memory.
- Event management
- Enhanced quality of supply monitoring

- The E350 U3400 HW is capable to support THD and Individual voltage harmonics as part of future FW upgrades.
- Disconnect relay (100A)
- Supply capacity control
- Integrated load control relays (1x31.5A)
- Fraud prevention and detection
- PredicTemp™
- Programmable meter configuration
- Plug - N - Play communication system (where supported by the communications vendor)
- Optical communication for local communication
- ANSI C12.18/21 protocol for communication module
- ANSI C12.19 data structures
- Self-registration of meters
- Ripple
- Transparent Modem support for remote reads via PSD or CSD

3.3 Catalogue Number

Please refer to the “E350 U3400 Catalogue Number” document.

Your local Landis+Gyr sales represent will provide you this document.

4 Basic meter function

The meter configuration is programmable, allowing it to be customised to meet the user's requirements.

4.1 Characteristics

4.1.1 Calendar Clock

The meter has a dedicated internal crystal to maintain the real time clock (RTC).

It also supports mains clock synchronisation as a programmable option.

Mains clock synchronisation is based on Phase A only

The clock is used as time stamp for event logging, Load Profile interval clocking, time-of-use tariffs, load control and leap year occurrence. During an outage the meter always uses this RTC chip for time keeping. If the meter is connected to an AMI communication network the network will update the meter time via the communication module.

The calendar makes the necessary changes for Leap Year. The meter time and calendar clock complies with AS 62054.21 (2006). Crystal time keeping deviation is less than 2ppm/day at 23°C.

A Load Profile group will stop profiling data when a time change backwards has occurred that causes the time to pass back through an already completed Load Profile interval.

MeterComm doesn't allow a change unless the difference is greater than 3 seconds. All other systems requesting a time change will log the change even if there is no change to the time.

Time and Date

The time and date of the calendar clock can be set remotely via the communication module or locally via EMPWin+ software. Time and date is stored as Universal Standard Time (UTC), and displayed as local standard time.

Only 24-h format is supported for time. Midnight is shown as 00:00. The date is in DD, MM, YY format.

Time Zone

The time zone selection gives the ability of selecting the respective geographical standard time location for where the meter will be installed. This enables the AMI Head-End System to correctly apply standard local time zone offsets if it is collecting data across different time zones.

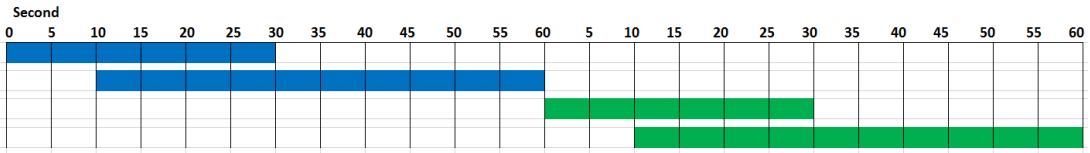
Timestamp The metrology data and energy information are logged with the time of occurrence. Time stamps are used when storing information in profiles, and event logs.

Battery While the meter is off, the power supply for the real-time clock is backed up by an on board battery. The battery typically provides 5 years reserve (meter off supply) and 10 years life (meter on supply). Also the battery life time is dependent on operating conditions such as temperature.

Smart Time Synchronisation

E350 U3400 meter with R5.0 F/W supports the Smart Time Synchronisation (**STS**) functionality. In all previous F/W versions a time change back across a completed LP interval boundary could occur and LP recording would be suspended.

- The **STS** functionality makes small time changes of up to 20 seconds backwards every 60 seconds. Therefore, the meter will take up to 80 seconds for every minute recorded by the meter until it has completed its time update.



- The **STS** method is not used for forward time changes.
- The **STS** method is not used if the requested time change backwards and is greater than one hour
- The **STS** method is not used if the backwards time change can be performed with in the interval boundary.

- If the **STS** functionality has started to make the small incremental changes and then a power outage occurs the process will be terminated and no “Time Change To” event will be created. The outage will leave the meters time partially adjusted so another time change request may need to be performed.
- If **STS** functionality is started, the required change in mm:ss are logged in the event log.

Events

Time	Description
11/05/2018 13:01:28	Time Changed (new time used as time stamp)
11/05/2018 12:55:19	Time Changed (old time used as time stamp)(Slow time adjustment mode backwards by 02:02)

Note: The STS time adjustment will typically take several minutes to complete, and the event times will look like a forward time change occurred. Please refer to the event log description for indication of STS event and required time change in mm:ss.

4.1.2 Low burden and low starting current

The use of low-power electronics means that the meter has a low power burden (less than 2 watts and 10 volt-amperes) and the starting current is well below the equivalent electromechanical meter at less than 40 milli-Amperes.

4.1.3 Meter Memory

The entire metrology data is recorded in a durable, non-volatile memory (FLASH). All the energy registers are stored in this memory.

4.1.4 Power Outage

A total power failure is detected directly from the power supply of the meter and works independently of the voltage measurement.

After a power failure, the internal power supply voltage starts decreasing. As soon as the power-down threshold is reached, all functions of the device are disabled and the power down procedure is initiated. The device operates until the power down threshold is reached.

The threshold is a parameter that is determined by the hardware and is related to the operating voltage requirement and therefore not programmable.

When a meter reaches a power down threshold limit, it triggers the power down procedure and notifies the upstream system of the supply failure upon detection.

Power ups and power downs trigger a power up/down event pair which gets logged together with the timestamp in the event log. In addition, the meter counts the number of power failures.

When a power outage event is recorded, the communication interface interrupt is signalled so that a communication module may take action to support a last gasp message to the Network Management System.

4.1.5 Programmable Display

The Liquid Crystal Display (LCD) of the meter is user-programmable to configure sequencing of the displayable items, number of energy digits required, scrolling options etc. See also section 12.2.11 “Normal Display”.

Various annunciators included on the LCD display, provide additional information on the displayed metering quantities. Others annunciators include element indicator, load control, disconnect relay status, WAN/LAN/HAN connectivity, direction of active and reactive energy etc. See also section 11.2 “Reading the Display”.

4.1.6 Four Quadrant metering

Energy imported by customer = Energy delivered by supply authority

Energy exported by customer = Energy received by supply authority

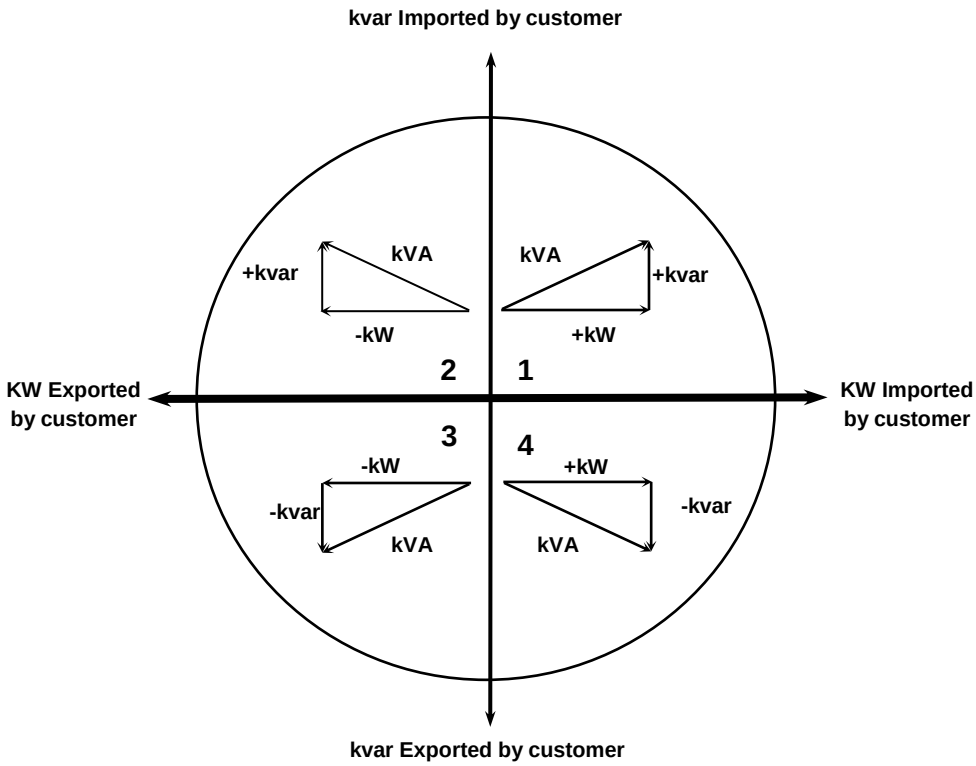


Figure 4.1 Four-quadrant geometric representation of active power (kW) apparent power (kVA) and reactive power (kvar)

The meter measure and record energy as follows

- Import Wh Quadrant 1 + 4
- Export Wh Quadrant 2 + 3
- Import varh Quadrant 1 + 2
- Export varh Quadrant 3 + 4

4.2 Measured Quantities

Registers

The active, reactive and apparent energy registers have a resolution of 1 Wh, 1 and 1 VAh respectively. Following energy registers are available to record different energy quantities. See also section 12.2.3 “Energy Registers”.

Register list
kWh Imported Phase A
kWh Imported Phase B
kWh Imported Phase C
kWh Imported Total
kWh Exported Phase A
kWh Exported Phase B
kWh Exported Phase C
kWh Exported Total
kVARh Imported Phase A (Q1,Q2)
kVARh Imported Phase B (Q1,Q2)
kVARh Imported Phase C (Q1,Q2)
kVARh Imported Total (Q1,Q2)
kVARh Exported Phase A (Q3,Q4)
kVARh Exported Phase B (Q3,Q4)
kVARh Exported Phase C (Q3,Q4)
kVARh Exported Total (Q3,Q4)
kVAh Phase A (Q1+Q4)
kVAh Phase B (Q1+Q4)
kVAh Phase C (Q1+Q4)
kVAh Total (Q1+Q4)
kVAh Phase A (Q2+Q3)
kVAh Phase B (Q2+Q3)
kVAh Phase C (Q2+Q3)
kVAh Total (Q2+Q3)

Measurement:

Each measurement element measures import and export energy. Values are continuously incremented every metering data update period. Depending on the energy flow direction, the meter increments the instantaneous values either in the import or export register.

Total registers

The Total Import/Export register can be a net or gross calculation depending on the configured metering type. The Total registers can be configured to total across any combination of elements, A+B+C being the default. Values are continuously incremented every metering data update period. The Calculation of values is shown below:

Example Import Energy Net Calculation (A+B+C)

The below table shows a load profile read with 30 minute interval.

	Phase A	**Phase B**	**Phase C**	**Total**
Import kWh	4	2	4	3
Export kWh	-2	-2	-3	0

Phase A reads 4kWh import and 2kWh export -> balance of 2kWh Import

Phase B reads 2kWh import and 2kWh export -> balance of 0kWh

Phase C reads 4kWh import and 3kWh export -> balance of 1kWh Import

The sum of above delivers a **3kWh Import** and **0kWh Export** which is shown under the column total

Example Import Energy Net Calculation (A+B)

	Phase A	Phase B	Phase C	Total
Import kWh	4	2	4	2
Export kWh	-2	-2	-3	0

Phase A reads 4kWh import and 2kWh export -> balance of 2kWh Import

Phase B reads 2kWh import and 2kWh export -> balance of 0kWh

Phase C reads 4kWh import and 3kWh export -> balance of 1kWh Import

The sum of above delivers a **2kWh Import** and **0kWh Export** due to the fact that the total is only across Phase A and Phase B, the total across these registers is shown under the column total.

Example Export Energy Net Calculation (A+B+C)

	Phase A	Phase B	Phase C	Total
Import kWh	2	2	3	0
Export kWh	-4	-2	-4	-3

Phase A reads 2kWh import and 4kWh export -> balance of 2kWh Export

Phase B reads 2kWh import and 2kWh export -> balance of 0kWh

Phase C reads 3kWh import and 4kWh export -> balance of 1kWh Export

The sum of above delivers a **0kWh Import** and **3kWh Export** which is shown under the column total

Example Export Energy Net Calculation (A+B)

	Phase A	Phase B	Phase C	Total
Import kWh	2	2	3	0
Export kWh	-4	-2	-4	-2

Phase A reads 2kWh import and 4kWh export -> balance of 2kWh Export

Phase B reads 2kWh import and 2kWh export -> balance of 0kWh

Phase C reads 3kWh import and 4kWh export -> balance of 1kWh Export

The sum of above delivers a **0kWh Import** and **2kWh Export** due to the fact that the total is only across Phase A and Phase B, the total across these registers is shown under the column total.

Example Export and Import Energy Gross Calculation (A+B+C)

The below table shows a load profile read with 30 minute interval.

	Phase A	Phase B	Phase C	Total
Import kWh	4	2	4	10
Export kWh	-2	-2	-3	-7

Phase A reads 4kWh import and 2kWh export

Phase B reads 2kWh import and 2kWh export

Phase C reads 4kWh import and 3kWh export

Based on above load, the kWh Imported Total Gross register will be 10kWh and the kWh Exported Total Gross register will be 7kWh

Example Export and Import Energy Gross Calculation (A+B)

The below table shows a load profile read with 30 minute interval.

	Phase A	Phase B	Phase C	Total
Import kWh	4	2	4	6
Export kWh	-2	-3	-3	-5

Phase A reads 4kWh import and 2kWh export

Phase B reads 2kWh import and 3kWh export

Phase C reads 4kWh import and 3kWh export

Based on above load, the kWh Imported Total Gross register will be 6kWh and the kWh Exported Total Gross register will be 5kWh, as the totalling is across Phase A and Phase B only

Instantaneous values

The meter measures instantaneous values that can be displayed.

Instantaneous values list
Voltage Phase A
Voltage Phase B
Voltage Phase C
Amps Phase A
Amps Phase B
Amps Phase C
Frequency
Power Factor Phase A

Power Factor Phase B
Power Factor Phase C
Power Factor Total
Active Power phase A
Active Power phase B
Active Power phase C
Active Power total
Apparent Power phase A
Apparent Power phase B
Apparent Power phase C
Apparent Power total
Reactive Power phase A
Reactive Power phase B
Reactive Power phase C
Reactive Power total

The display sequence of the instantaneous values is user programmable in the meter. See also section 12.2.11 “Normal Display”.

Power Factor

The meter determines the power factor from active and reactive power. It provides the power factor for each element or phase as well as total.

Power factor calculation method for element or phases (poly phase meter only)

$$PF_x = \frac{active\ power_x}{\sqrt{(active\ power_x)^2 + (reactive\ power_x)^2}}$$

x is a place holder for any of the following possible options: element 1, element 2, phase a, phase b, phase c.

Total power factor calculation method (including poly phase meter)

$$PF_{total} = \frac{active\ power_{total}}{\sqrt{(active\ power_{total})^2 + (reactive\ power_{total})^2}}$$

Format of power factor

The format for the power factor on display or in the report is as follow:

$$\boldsymbol{Q\ p.pp}$$

Where $\boldsymbol{Q}$ stands for the quadrant 1 to 4 (refer to section 4.1.6 Four Quadrant metering)

and $\boldsymbol{p.pp}$ stands for the power factor value from 1.00 to 0.00.

Instantaneous value averaging period

The instantaneous values are averaged over 500msec period except frequency which is averaged over 100msec period.

4.3 Data Profile

4.3.1 Load Profile

The meter supports up to 4 Load Profile groups which each of these groups supporting its own interval lengths of 10,15,30,60 minutes with F/W R4.9 and earlier. 5 minute intervals are also available with R5.0 F/W or greater.

Each Load Profile group on its own supports up to 4 channels that can be recorded when running R5.4 F/W or earlier and 24 channels when running 6.0 or greater.See section 12.2.9 “Load Profile ”.

The recorded quantity in each of the channels can be selected from any of the quantities as shown in section 4.2 “Measured Quantities” and also RSSI values as shown below.

The maximum recorded days across all sets of Load Profile are determined by the memory size. However, MeterDev allows the user to specify how much Load Profile memory to allocate to each of the 4 possible Load Profile groups. By default the memory is equally shared between the number of defined Load Profile groups. In MeterDev the memory allocated is shown as the approximate number of Load Profile days that will be recorded before wrap around of the Load Profile data occurs. If the default amount of memory is reduced for a particular Load Profile group then that same amount of memory can be allocated to another Load Profile group or groups. There is a minimum memory size that can be allocated to a Load Profile group. All the available Load Profile memory doesn’t need to be used but a warning will be displayed in MeterDev if there is any free space still available.

It is then possible to partially program one or more of the Load Profile groups that are currently running in the meter. Load Profile groups 1 to 4 have a priority where group 1 is the most important. If the memory allocated to each Load Profile group is unchanged then it is possible to change the type of Load Profile data or the number of channels or interval length without disturbing the lower priority Load Profile groups. Partial programming a particular Load Profile group will always clear all of the existing Load Profile data for that Load Profile group. If the memory size is changed in a higher priority Load Profile group then all lower priority Load Profile groups will also be cleared during partial programming of the higher priority group and will be disabled.

The Load Profile quantities can be read locally via the optical port or remotely via the communication module to the AMI Network Management System. All the profiles are organised as a circular buffer. The entries are stored in the same order as they occur, so the most recent entry is stored last. Once the buffer is full, the oldest entry will be overwritten by the most recent one. When reading the profile, the entries are given out the same way, i.e. the oldest entry first.

RSSI values profiling

If the meter is fitted with a transparent communication module it can profile RSSI value but these values will only be reported if a supported modem type is connected to the meter and the modem support options have been enabled in the meter program.

The RSSI values are read from the modem every 5 minutes.

RSSI Load Profile values can be saved as an Instantaneous value (most recent value from modem) at the end of the current Load Profile interval. A value of 65535 indicates that there is no modem connected or a recent power outage prevented a value from being obtained. If a 65535 occurs then the Load Profile interval will also be marked with a status named ”modem read failure”.

The minimum RSSI, average and maximum values can also be recorded in Load Profile.

Most modems will return an RSSI value of 99 if the modem cannot calculate a valid value. The value of 99 will be used in maximum RSSI Load Profile channels, not used for minimum RSSI (if a lower value had already been obtained) but it will be used in averaging. For this reason average RSSI values that are greater than 31 are an indication of at least some 99’s occurring during that particular Load Profile interval.

Value	*RSSI dBm*
0	-113dBm or less
1	-111dBm
2..30	-109dBm..-51dBm
31	-53dBm or greater
99	not known or not detectable
65535	no modem connected or a recent modem power outage

Reset

It is possible to reset Load Profile data. If the data structure of the Load Profile is modified (e.g. the number of energy registers captured or the length of the capture period) the Load Profile automatically resets to avoid any inconsistent data. At the same time a corresponding event is stored in the event log.

First Installation

The Load Profile recording will be factory disabled until the next power up event which will normally happen when the meter is first installed at the customer’s premises. This process will prevent the meter from filling with 0’s due the long initial power off period after the meter has left the factory.

The 0 filling process of the Load Profile in particular after a very long outage (many months) can block the remote as we optical communication port for several minutes depending on the number of Load Profile changes enabled.

Time change

> A Load Profile group will stop profiling data when a time change backwards has occurred that causes the time to pass back through an already completed Load Profile interval.

Care must be taken when the time is remotely adjusted through a transparent communication module which could stop the Load Profile from working without notification. Locally EMPWin+ tool box command prevents this by asking the user whether he wants to set the time and allowing the Load Profile to stop. It is recommended that remote auto time changes are not performed unless the dial-in software can determine if the time change will cause any Load Profile group to stop. Manual time changes could be performed provided the user ‘schedules’ the time change to occur at the most appropriate time within the shortest Load Profile interval that is being recorded.

Itron MV90 Load Profile configuration

If multiple Load Profile groups are to be used within MV90 then the Device ID name that is setup within MV90 needs to append #1 or #2 or #3 or #4 to the end of the meter’s Account ID. e.g. meter account ID is SAMPLE3277 and there are 4 Load Profile groups in the meter then in MV90 the four unique Device ID’s that need to be created will be SAMPLE3277#1, SAMPLE3277#2, SAMPLE3277#3 and SAMPLE3277#4. For additional details refer to Itron’s release documents for TIM_LGUA.

4.3.2 Event Log

The meter logs all enabled events with respect to meter functionality in the order of occurrence so as to keep a record for the meter. The events include the meter’s load control relay, disconnect relay, Load Profile, power supply, fraudulent events, and any relevant meter error or warning that might occur. These events and the time of occurrence are stored in the event log.

Sort order

All event logs are stored according to a "first in, first out" buffer arrangement. The entries are stored in the same order as they occur, so the most recent entry is stored last. When reading the log, the entries are given out the same way (oldest entry first).

Log size

The meter stores up to 500 events in the event log.

Events

The meter support several events such as:

- Changes of the clock
- Changes of the configuration
- Clearing of profiles
- Self check errors
- Activation of new parameters
- Activation of new time of use, etc.
- Power quality events
- Power failure events
- Fraud (Tamper) detection events

- Disconnect control event log

The complete list of possible events, which can be included in the event management can be found in Appendix B – Event Table”

Readout The event log can be read both remotely and locally.

4.4 TOU Registers

Time-of-Use (TOU) Tariffs

The meter supports 6 Time-of-Use tariff rates, which are user configurable. Time-Of-Use (TOU) Tariffs can be programmed in the meter. See also section 12.2.5 “TOU Switch Table”.

Maximum Demand

Maximum demand metering is available for each of the programmed rates. The cumulative demand can be recorded, available for most kW, kVA and kVAR channels (including the element quantities), and the date and time for which the maximum demand was recorded.

The demand interval can be set between 1,5,15,30 and 60 minutes. There can be up to six subintervals if rolling demand is required with a maximum of 15 minutes and a minimum of 1 minute for the subinterval.

A Billing read will report the Current Demand values and up to the last eight Self Read dates of Demand values. If Previous Billing periods are being created these will also report Demand values.

There are display options for TOU Demand metering. These are the maximum value, date and time when it occurred and Cumulative value. These are available per TOU rate or as an Any Time situation.

Current Billing Period

The current billing period is the accumulated energy totals and TOU registers at time of reading the meter.

Previous Billing Period

The previous billing period is a snapshot of the accumulated energy totals and TOU registers. The meter stores the last 14 billing periods. These reads are scheduled Weekly, Monthly, or on a particular month of the year via the recurring and non-recurring screens in EMPWin+. See also section 12.2.6 “TOU Recurring Dates” the parameter “Previous Period Read”.

Self Read The self-read is a snapshot of the accumulated energy totals and TOU registers at Midnight each day. The meter stores the last 8 self reads when operating FW R4.9 or earlier. The meter stores the last 35 self reads when operating FW R5.0.

4.5 Quality of Supply Monitoring

The meter records data which is useful for system monitoring or for investigation of local voltage and load problems. The meter can display, voltage, current, frequency, power and power factor on the LCD as part of the cyclic display or read both remotely and locally.

The complete list of recorded data can be found in section 4.2 “Measured Quantities”, Instantaneous values.

4.5.1 Under Voltage & Over Voltage recording (Sag and Swell)

Under voltage and over voltage events can be recorded when the voltage has passed outside programmed threshold limits. Each event will be recorded as a start time & date, end time & date and the worst case value that occurred during this period. The user can program the meter with the high and low threshold levels and the breach time. See also section 12.2.10 “Quality of Supply”.

4.6 Fraud Prevention and Detection

The meter records full or partial program changes, power outages and detects meter reverse energy flow (detailed below) to indicate the possibility of fraud. All tampering events will be stored in the event log.

The meter support the following fraud detection features:

- Reverse energy detection
- Detection of terminal cover removal in power and off power mode
- Load side voltage detection (in a future F/W release)
- Incorrect password login attempts.

4.6.1 Reverse Energy Detection

The meter can log reverse energy flow as an event and can issue a notification (alarm).

Reverse energy flow will need to be disabled for Co-generation applications.

The meter incorporates following features for detection and metering of reverse energy flow:

- The reverse energy (export) for reporting can be enabled or disabled in the meter’s program. Also, the point at which this energy is reported can be programmed. See section 12.2.3 “Energy Registers”.
- Once export energy has been detected (if detection is enabled) the meter will add the warning message to the event log and display list (after the last programmed entry). This display is retained whenever the export energy continues to flow and until the status of the meter is reset.

In addition a warning message is reported by the meter when a program read is made. This warning flag is retained in the meter and can only be reset by the EMPWin+ 'Reset Status' command or NMS software. Reverse (export) energy is accumulated in separate registers and is recorded with the same accuracy as forward (import) energy.

The meter's red LEDs do not flash for reverse energy consumption.

4.6.2 Detection of terminal cover removal (ON/OFF power)

The meter supports a detection of unauthorised meter cover removal via tamper switch situated under the terminal cover. The detection of that event enabled in power on mode as well as in off power mode. Once the meter detects such an event it will time stamp and event log.

The meter will use the time stamp of the time and date shortly after power restore for detection during power off.

4.6.3 Load side voltage detection (if supported by F/W)

The breaker disconnects all terminals on the load side. In case a disconnect command is issued to the meter and voltage still applies on load side, then the meter will time stamp and event log "voltage on load side". This event represents two potential conditions for the same log entry:

- Meter is bypassed (load side tamper detection)
- Malfunction of the disconnect relay

In both cases the tamper event is triggered and sends an alert message to the Network Management System.

Phase A needs to be connected in order to pick up this condition.

4.6.4 Password Protection

Correct passwords are required to access the meter via the optical port and communication module. Three consecutive incorrect attempts will lockout the user for one hour for that particular password level. The password is encrypted for additional password protection during communication.

The access levels are:

- Full read/write access.
- Two programmable intermediate levels with restricted read/write access.

Optical Passwords are defined by running a separate security program – EMSecure, part of the EMPWin+ suite of programs.

For more information see also section 12.4 Security

4.7 PredicTemp™

The meter supports a patented logic and algorithms to detect and alarm abnormal temperature rise with the meter installation. It uses temperature sensing inputs from the U-Series meter's metrology engine.

This feature reduces the risk of potential fire hazards due to

- Faulty installation (incorrect torque applied on terminal screws)
- Overload of the circuit

It can send alarms to the Network Management System if the meter is fitted with a communication module supporting this feature. Additionally it optionally can also disconnect the load downstream the meter. Please contact your local sales manager for further information.

4.8 Load Control Relay

The meter has an integrated 31.5A load control relay as an option which is linked to Phase C.

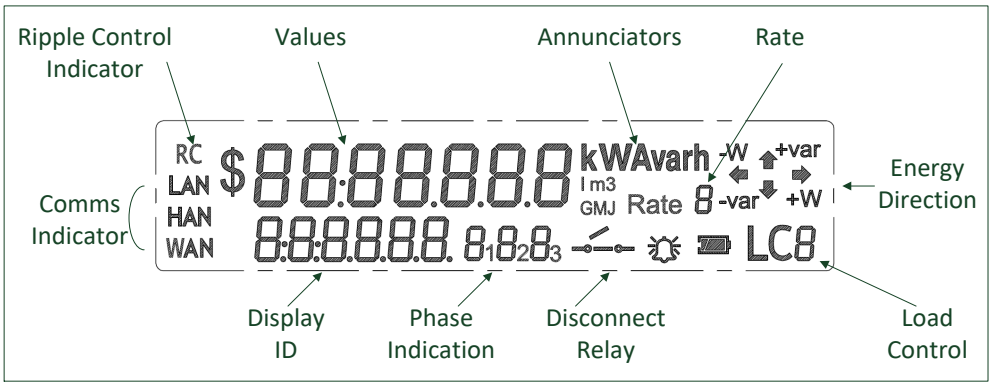
Phase C needs to be connected in order for LC to function. In case of a two phase installation, where Phase C is not connected, please ensure that the Phase C terminal screws are screwed down at recommended torque for LC to function.

4.8.1 Load Control Groups

Load control groups enable a variable number of meters to have their load control schedules overridden. The internal load control relay can be programmed either remotely or locally to respond to one primary group, one secondary group and one tertiary group. The commands will override or disable the internal load control command.

4.8.2 Load Control Relay Status

The LCD display shows a status LC1 to indicate the load control relay has closed then it will read LC 1.



4.8.3 Load Control Relay control Sources and Hierarchy

The meter acknowledges the following sources to request control of the load control relay.

- Loss of Supply or Disconnect
- Emergency Control
- Boost
- Priority override (override commands)
- Ripple control
- Normal Time Table (internal load control command)

Since multiple sources are available and may request control of the load control relay at the same time a clear hierarchy is mandatory to handle the different sources. In operation always the control source higher in the hierarchy gets priority to take control of the relay. The hierarchy is as follow:

1. Loss of Supply or Disconnect
2. Under Frequency (Emergency Control)
3. Boost (boost primacy enabled)
4. Priority override (override commands)
5. Demand Limit Load Control
6. Boost (boost primacy disabled)
7. Ripple control
8. Power restore
9. Normal Time Table (internal load control command)

4.8.4 Relay State Machine (RSM) for Load Control operations

4.8.5 RSM rules

The RSM follows three simple rules which are:

- At power up the RSM starts always from idle and newly initialised in power restore mode. No tasks from pre power down are carried over.
- Only one control source can control the LC relay at the time, therefore the control source higher in the hierarchy gets control.
- Each control source works independently and simultaneously to all other control sources.

As an example, if a priority override command is received to close the relay for the duration of 4 hours then the module instantly request for control but in the meantime it starts to count down the duration regardless whether it is in charge of the LC relay or not. This helps the user to exactly know how long this task will last for and by when this task has been completed.

With the control request by the priority override module, the RSM reassesses the situation and based on the given hierarchy it may give control to the priority override module. However, once the RSM gives control to the priority override module only then the module can close the LC relay. The priority override module will maintain the relay in close position for the reminder of the duration before it releases control. It could happen that during this time period another control source takes over the LC relay control. Then the relay state may change based on the programmed action of the control source in charge but regardless the priority override module is still counting down and once back in charge it will close the LC relay again for the reminder of the programmed duration.

Once the programmed duration has expired the priority override module releases control without changing the state of the LC relay. In this particular case the LC relay stays closed. The RSM reassesses the situation newly and gives control to the next requesting control source. Not switching the LC relay at the end of each task and leaving the switch operation with the next control source will eliminate any relay toggling and therewith eliminate short pulses to the appliances.

The LC relay does not have a predefined state per se. In case a defined state is required then we recommend to set the requested relay state in the normal time table module. This module has the lowest priority in the RSM and it will set the relay state accordingly once no other module is in charge.

4.8.6 Boost

The boost function will energise the controlled load for a pre-set duration, which is programmable from 1 to 8 hours in half hour increments. This functionality can be enabled or disabled both locally or remotely. Boost is found in different places in the RSM hierarchy one as boost primacy enabled the other as boost primacy disabled.

The following case is valid when boost primacy is enabled. While the load control relay is still under control by a priority override module operation the boost button can be pressed and will close the load control relay for the specified time. However, in this case the energy may be redirected to element 1 (see section 12.2.24 Controlled Load L1 – Boost)

A customer will need to look at the Load Control Modules Status section of the program read report to see how the Boost Primacy feature has been set.

4.8.7 Loss of Supply or Disconnect

This function performs a relay operation at the time a power outage occurs or on a disconnect command has been issued. This feature allows the utility to program the relay in a defined state while power off. Setting the relay to open prevents a load peak at power restore. See also section 12.2.20 “Controlled Load L1 – Loss of Supply ”.

4.8.8 Power restore

On power up the meter enters the power restore mode which provides a programmable duration in which the relay will remain in the status set in Loss of Supply. The RSM is not locked in to the power restore mode for the reminder of the programmable duration. All control sources higher in the hierarchy can take over control. However, if another control source takes over control during power restore is in charge then this will end the power restore mode immediately and power restore will not request control back regardless whether there is a reminder of the power restore duration.

Although the firmware distinguishes between the meter-power-restore-mode and the ripple-control-power-restore-mode. It will always enter the ripple-control-power-restore-mode if ripple control is selected as a control source otherwise it will enter the meter-power-restore-mode.

Refer to section 12.2.21 Controlled Load L1 – Power Restore. This is because ripple control requires a much more advanced feature set in power restore mode.

4.8.9 Priority override (override commands)

The meter receives override commands either via optical port or the optional communication module. The priority override command contains the relay action to be performed. The duration which the relay action shall last for is set in the meter program (refer to section 12.2.22 Controlled Load L1 – Priority Override).

4.8.10 Emergency Control

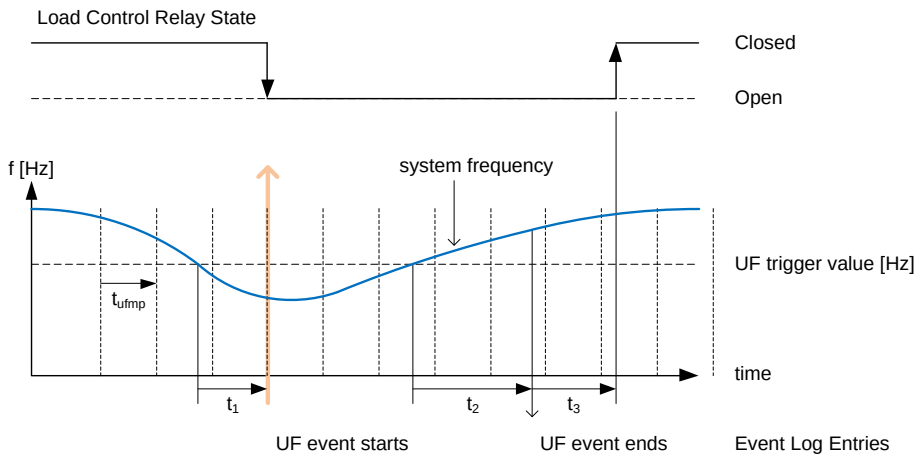
The meter provides the emergency control functions listed below. Depending on the control hierarchy level an emergency function may precede the actual control source and perform the programmed load control action. In any case the event is time stamped and logged in the event log.

4.8.10.1 Under Frequency

The meter monitors the mains frequency.

Phase A needs to be connected to enable this feature

If the frequency drops below a defined value and remains under for a defined breaching time then the selected relay operation will be performed. Usually an open relay command is performed to shed load.



t_{ufmp}: Under Frequency measurement period. This is the time duration in which the meter samples and averages the system frequency. At the end of this period a decision is made whether to open the LC or not.

t_1: Reaction time to perform the programmed LC action.

- best case reaction time between 0.5 x t_{ufmp} to 1 x t_{ufmp}
- otherwise maximal 1.5 x t_{ufmp}

t_2: Nominal frequency breach time

t_3: Random delay time after event ends

Event log entry including time stamp is done after the duration expired of either t_1 or t_2

See also section 12.2.25 Controlled Load L1 – Under Frequency Load Control.

4.8.10.2 Under Voltage (not enabled)

The meter monitors the mains voltage. If the voltage drops below a defined value and remains under for a defined breaching time then the selected relay operation will be performed. Usually an open relay command is performed to shed load.

4.8.10.3 Demand Limit

The meter monitors the total energy consumption. If the energy consumption exceeds a defined value in between a definite time period then the selected relay operation will be performed. Usually an open relay command is performed to shed load. Refer to Section Controlled Load L1 – Demand Limit Load Control for more details.

4.8.11 Ripple control

The E350 can be a ripple receiver which is another control source of the RSM. More detailed information about ripple control in section 9 Ripple Control.

Phase A needs to be connected to enable this feature

4.8.12 Normal Time Table

This control source switches the LC relay based on a defined Time Table (refer to section 12.2.26 Controlled Load L1 – Time Table).

It can be used to define a relay state in which the relay should stay in case no other control source is in charge of the LC relay.

4.9 Disconnect Breaker

The meter features an integrated disconnect breaker to connect or disconnect the power supply to the consumer under specific conditions. Max switching current of up to 100A at nominal line voltage and a maximum 10,000 cycles. All breaker operation are logged in to event log which are listed in Appendix B – Event Table.

4.9.1 Disconnect breaker operations

The disconnect breaker operations are categorised as below:

- Arm operation - Customer presses button to reconnect or disconnect.
- Local operation - reconnect or disconnect.
- Remote operation - reconnect or disconnect.

Arming

Arm operation mode is set by the network remotely or locally via the optical port to reconnect the disconnect relay. When the meter receives a reconnect command in this mode, the householder is required to press the push button “2” located on the meter for 3 seconds to reconnect the disconnected breaker. See also section 12.2.14 Relay Setting

Local

The disconnect breaker can be controlled locally via optical port.

Remote

The disconnect breaker can be controlled remotely via the communication network.

Disconnect Breaker Status

The contactor symbol is used to indicate the status of the disconnect relay.

The current state of the breaker is displayed as either

Breaker OPEN	Breaker Closed

LAN/HAN/WAN Indicators

4.9.2 Auto disconnect

If the Watts specified in the program exceed the limit for the length of the programmed breach period (during the monitoring period after a remote Breaker reconnect) then the Breaker will disconnect. See also section 12.2.17 “Auto Disconnect”.

4.9.3 Supply Capacity Control (SCC)

The meter has four supply capacity settings:

- Normal supply capacity limit (for both Import and Export of Active Demand)
- TOD import supply capacity limit
- TOD export supply capacity limit
- Emergency supply capacity limit.

These control settings are configured remotely or locally.

The normal, TOD import, TOD export or emergency supply capacity limit is configured to disconnect the Breaker and remain disconnected for the specified duration, where the load level exceeds the specified supply capacity limit over a specified period of time. The meter will log an event for all changes to the supply capacity settings and statuses.

See also section 12.2.16 “Supply Capacity Limit”.

It is possible to enable or disable the four supply capacity limitl options independently. For Normal, TOD import and TOD export supply capacity limits, the demand averaging is done for the defined trading intervals, and at each minute boundary; within the defined trading intervals; the limit will be checked and if the calculated demand exceeds the defined threshold then the relay operation will occur. If relayhas not been triggered for Normal, TOD import and TOD export supply capacity limits, then the Emergency supply capacity limit will be checked and relay will be triggered if necessary.

The Normal, TOD import and TOD export Supply Control limit triggering is checked every minute instead of `waiting for the end of the 30 minute Trading Period (synchronised to the meter’s hour and half hour intervals). It is possible the disconnect of the Breaker can occur part way through a Trading Period. For this reason it is recommended that the Breaker reclose period is at least 30 minutes otherwise multiple disconnect and reconnect operations can occur within a single Trading Period.

For Normal, TOD import and TOD export supply capacity limitmode single or multiple Trading Periods are still supported. For single Trading Period selection the energy accumulation and hence the demand calculation restarts at the beginning of the new Trading Period.

TOD import and TOD export supply capacity limits are introduced in R4.9 meter firmware. TOD import and TOD export allows user to specify a time of day (TOD) and monitoring duration (in minutes) for which the supply capacity control will be in effect. This is different from Normal supply capacity limit where it is not possible to define a time of day and monitoring duration for control.

Disconnect after 10 consecutive Breaker operations

After 10 consecutive Breaker disconnect operations due to SCC within a calendar day the Breaker will automatically (doesn't require the ARMed state to be defined in the meter program) enter the ARMed state that requires a button press to reclose the Breaker. A scrolling message will be displayed by the meter to indicate what needs to be done to reclose the Breaker. While in the ARMed state an external Breaker close commands will be ignored, a button press is required to reclose the Breaker or a midnight crossing. A power outage while in the ARMed state will not change the ARMed state.

As noted above if the Breaker has entered the ARMed state due to 10 consecutive disconnect operations due to SCC then the Breaker will automatically be reclosed at midnight. A midnight crossing will also reset the consecutive counter back to 0 as will a power outage.

Note: The Arming Timeout function (as described in relevant section) won't be triggered when the breaker is armed due to the 10 consecutive breaker open operations by the Supply Capacity Control (SCC) logic.

4.9.4 Loss of supply

The meter supports a Loss of Supply (LoS) option to switch the Breaker to the open position at power down. If this option is enabled the meter ensures the breaker is in the open position during meter loss of supply. Upon power restoration the Breaker reverts back to the state before meter loss of supply based on LoS settings. This option can be enabled or disabled locally and remotely. Refer to 12.2.18 Breaker Loss of Supply

Care must be taken when enabling this feature. A short-lived voltage dip could trigger the meter power down sequence. If the LoS feature is enabled then the breaker would interrupt the power supply for a longer period than the actual outage as determined by the LOS settings.

The time required to close the breaker after power restoration is within seconds and not immediately due to physical limitations.

4.9.5 Breaker Maintenance Cycle

Breaker Maintenance Cycle allows the user to specify approximately when the Breakers twitch exercise is performed. The twitch exercise makes three audible sounds once a day so the meter can be programmed to make these occur at a time that will not alarm the customer.

4.10 Communication

4.10.1 Overview

The meter features following communication interfaces:

- WAN/LAN Communications (Modular based communication)
- HAN (Modular based communication)
- Local - Optical Port

The meter can interface with different WAN/LAN/HAN communication modules such as any of the Landis+Gyr Gridstream modules, Silver Springs Networks RF mesh / ZigBee , WiMAX / ZigBee and 3G. The meter has been designed with the aim to physically separate the meter and WAN/LAN/HAN communication components.

The major functionality such as load profiling, Time-of-Use, other metering data and events are remotely accessible via the communication module. These values/information can also be locally read via the optical port using the Laptop or Hand Held Units, for the purpose of configuring, troubleshooting or when the endpoint itself is unavailable via WAN/LAN communications.

4.10.2 WAN/LAN Communication

The meter’s hardware has been designed to interface to compatible LAN communication module. Different LAN communication modules that are currently available to provide a flexible interface with the meter are:

- Landis+Gyr Gridstream RF module with ZigBee
- Landis+Gyr WiMAX module with ZigBee
- Silver Springs Networks RF mesh NIC with ZigBee

4.10.2.1 Transparent Communication

It is also possible to use the meter in an AMR application where a transparent modem is connected to the meter. The remote port supports the Landis+Gyr Kingfisher 2 protocol which is the same as the optical port. There are a few Head End Systems which support this protocol including Landis+Gyr EMPWin+, Landis+Gyr Advance Economic and Itron MV90.

4.10.3 Home Area Network (HAN)

The HAN allows for full connectivity and interaction between the distributor and customer appliances. The long-term view is that appliance manufacturers will build smart modules into their products that will be able to receive information from the grid or sense when it is appropriate to turn applications on or off.

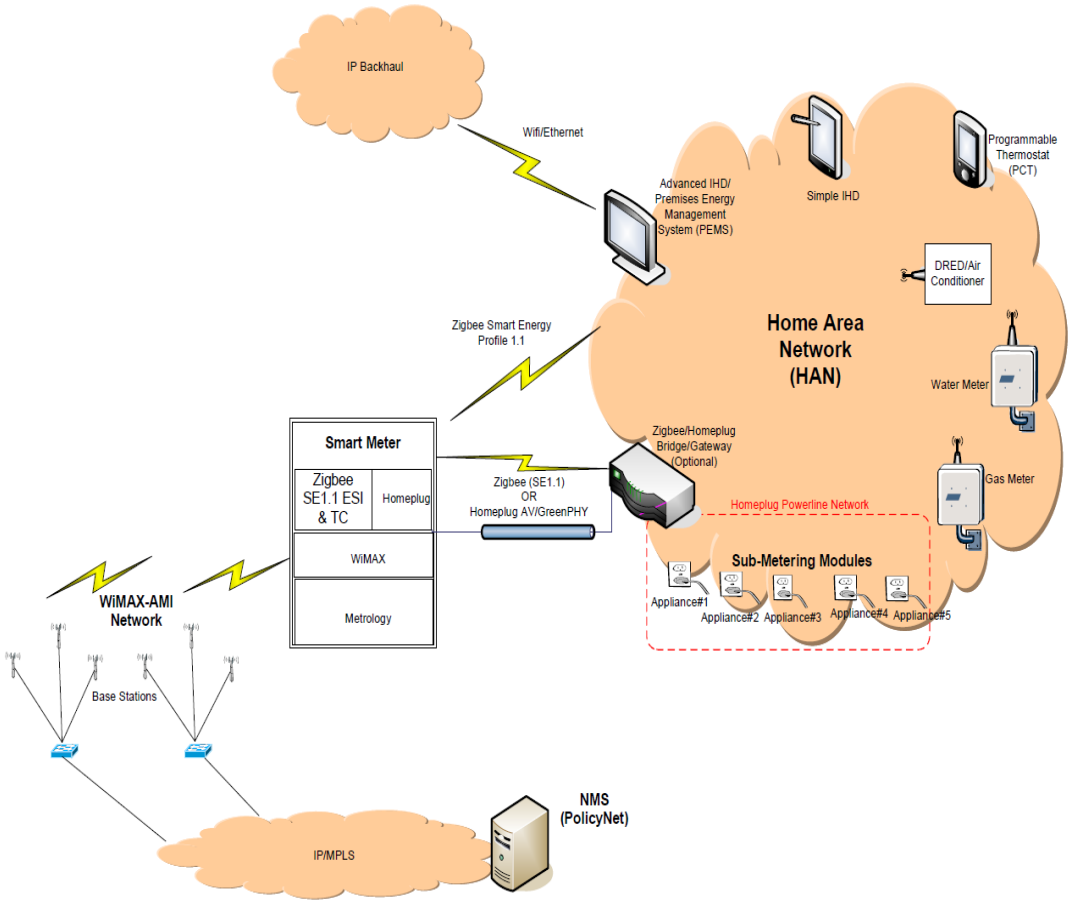


Figure 4.3 – Home Area Network Architecture

4.10.4 Modular communication

4.10.4.1 General

The meter is designed to allow field-upgradeable communication to address future communication needs.

The meter's modular design is unique in that it allows the existing communication module to be replaced with a new module without the need to replace the meter.

- When the time comes to replace the communications module this can be done without disconnecting customer supply.
- Communication module can be replaced without breaking main terminal cover seal.
- E350 supports the key ANSI (American National Standard for Utility Industry) communication standards.
    - ANSI C12.19 - ANSI standard for extracting and organizing meter data in a specific format.
    - ANSI C12.18 - ANSI standards that describe protocols used for two-way communications between meter and module.

These standards have enabled the U3400's interoperability with the Landis+Gyr Gridstream RF (RF Mesh), SSN (RF) and (4G, WiMAX) modules. These modules can be easily interfaced as your requirements change.

4.10.4.1 Signal strength detection at time of installation

This unique feature allows the installer to check the signal strength RSSI of a newly installed modem under normal operating conditions in order to determine whether the antenna gain is sufficient or not. As an example the meter as well as the modem antenna is inside a meter box. If the meter box is open to read the RSSI value off the display would not reflect the normal operating condition as the meter box door is normally closed. Therefore the meter supports the RSSI installation mode which allows the installer to check the RSSI while the door is closed.

The four available RSSI values are Instantaneous, minimum, maximum and average. These last three choices are re-calculated after the first RSSI value has been reached in the Alternate display list. The monitoring period for these calculations is determined in MeterDev by the Modem Installation Mode settings that offer from 20 to 240 seconds of monitoring. RSSI reads are attempted about every 4 seconds during the monitoring period. There is also an initial delay before the monitoring period starts; this is to allow for a meter box lid to be closed. It should be noted that the RSSI values that are read for Load Profile purposes are only read every 5 minutes and not every 4 seconds as they are in installation mode.

Modem Installation Mode display of RSSI values has four different stages. When the Alt button is pressed and the first available modem display item is shown it will be presented as C ----- (C followed by a space and 5 dashes). This will stay for the Initial Delay seconds (default is 10). The display will then change to C------ (C followed by 6 dashes) during the first part of the Monitoring period (user configurable in seconds). If during the Monitoring period the modem returns an RSSI value then the last five dashes will be replaced by the RSSI value padded with leading zeros e.g. C-00021. The first – after the C will remain until the Monitoring period expires at that time the last obtained RSSI value will be displayed (without the first dash) for up to 255 seconds or until the Alt button is pressed. If no RSSI value is obtained then the final displayed value will be C -----.

The four different RSSI values Instantaneous, minimum, maximum and average are all displayed as explained above but will have the users display ID to distinguish them.

4.10.5 ANSI Optical Port (Local Communication)

For onsite communications the meter uses an ANSI standard Type 2 optical port, which complies with AS1284.10.2 -1993. The ANSI standard port provides a fast and easy method of reading and programming the meter. High speed communication at the rate of up to 38400 baud is available through the optical probe.

The standard probe recommended is the Landis+Gyr AIP200 probe as detailed below.

Optical Port

To communicate through the optical port, a personal computer or a suitable hand-held device and an ANSI standard optical probe is required, together with a meter programming and/or reading software. A typical example is the **EMPWin+** software which supports the program for the U3400 meter.

AIP200 Optical Probe

The AIP200 probe is a dual standard probe useable for both ANSI and IEC1107 (Flag) standard ports. It is magnetically held to the port and locks into position for the ANSI type port. The U3400 meter is fitted with this ANSI standard port.

Figure 4.4 AIP200 Probe

AIP300 Optical Probes

The AIP300 ANSI optical probe is the perfect solution for data transfer with utility meters via Bluetooth or USB. The AIP300 Bluetooth probe gives the user greater mobility and provides a range of up to 10 metres.

For devices without Bluetooth, the USB function is available for use. The LED provides indication of data transfer and battery status.

Figure 4.5 AIP300 Probe

To set up the communication link, place the optical probe against the optical port. The D shape of the ANSI standard port assists to correctly align the probe to the port. When the magnet of the probe ‘clicks’ into contact with the steel face of the optical port, the magnet then holds the probe in place. A corresponding D-shaped depression in the surface ensures that this only occurs for the correct alignment. Once the connection is made the EMPWin+ software can handle the reading or programming of the meter.

5 Programmable Multifunction Outputs

The meter supports up to 4 pairs of potential free output contacts which are programmable to either pulsed outputs or threshold contacts.

The output contact specification is as:

- Four (4) sets of 2-wire potential free
- RJ45 connector
- Voltage Rating: 40VAC max and 12V- 24VDC
- Current Rating: 100 mA (DC or AC)
- Relay type: Solid State Relay

Pulsed Outputs are only supported with the depending HW configuration which needs to be ordered from factory.

5.1 Pulsed outputs

5.1.1 General requirement

The energy pulse weighting (quantity/pulse) and the pulse duration are programmable within defined limits. A Pulse divisor of 0 disables that particular energy pulse output otherwise 1 to 255 can be used to scale (slow down the pulse rate). The pulse width for energy channels can be set in 20m Seconds steps from 20 to 1000 mSeconds. A pulse width of 0 enables the recommended 50% duty cycle. The four pulse outputs can be configured into two different groups broadly known as Register or Threshold Control. The user can select how many of each type are required but is limited to a total of four.

5.1.2 Pulsed Output Registers

The meter provides the following programmable selection of quantities or values for each output contact that is defined as a Register.

- Same energy registers as listed in section 4.2
- End Of Interval (EOI)
- TOU Rate Alert (A to F) closed contact when rate is active

5.2 Threshold Control

There are three threshold conditions that can be used to trigger a relay operation if any of the enabled conditions are detected. It is the user’s choice to select an Open or Close pulse output operation and it would depend on what type of interfacing hardware is being used as there is no direct load control provided by the meter. A condition must exist for at least the Minimum Event Duration specified. All three conditions have an Event Duration period and a random delay after the condition has ended which is individually programmable.

5.2.1 Frequency

If the meter's supply frequency falls below this specified value then the selected relay operations will be performed (usually an open relay command to shed load). The frequency threshold can be set to a maximum of 49.9 Hz in steps of 0.1 Hz. 0 or a very low value will disable this option.

5.2.1.1 Low Voltage

If the meter's supply voltage falls below this specified value then the selected relay operations will be performed and the user's interfacing hardware can then perform the required load control or alarm process. The low voltage threshold can be set to any whole number value between 160V and 280 volts, 0 or a very low value will disable this option.

Also each individual phase can be selected to be monitored as part of the threshold control.

5.2.1.2 Demand Threshold (watts)

If the meter's measured load in watts exceeds this specified value then the selected relay operations will be performed and the user's interfacing hardware can then perform the required load control or alarm process. The watts threshold can be set to any whole number value (1 Unit = 1W) between 0 and 72'000 = (Un* Imax* 3phase) = (240V*100A*3) 0 will disable this option.

5.2.1.3 Relay Operation

For Threshold Control Outputs each relay can be programmed to Open, Close or be Disabled (Disabled means no change will occur on the physical relay however, an event shall still be logged). The initial relay state is the opposite of the relay action which will be performed when the event occurs. If the action is "disabled" then the relay output shall be in open state.

5.2.1.4 Power cycle or reprogram of the Threshold control outputs

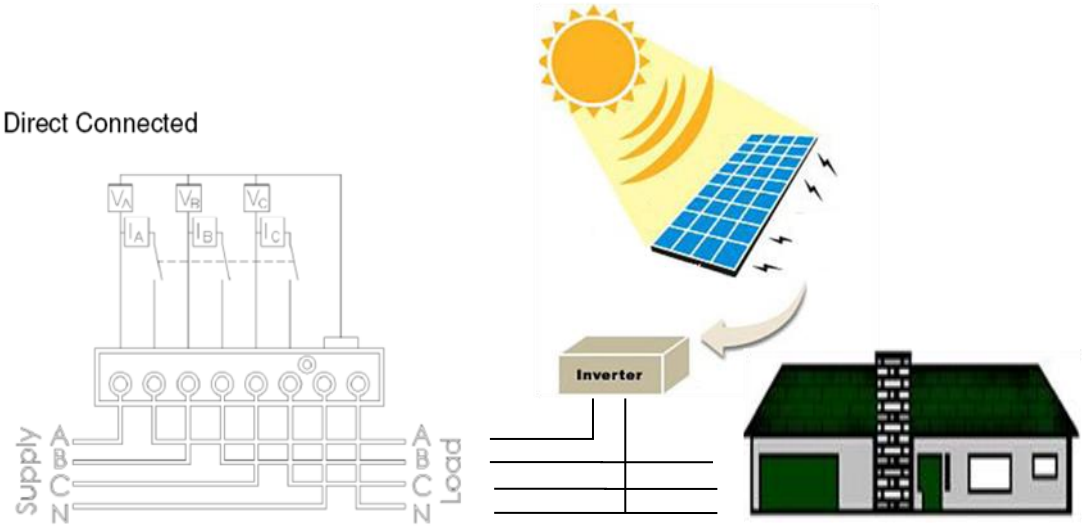
The threshold control outputs will be initiated at each power restore and after this function has been reprogrammed.

Meaning, all relays will be set into initial state and all delays will be reset.

6 Gross and Net Metering

6.1 Gross Metering

The meter supports gross metering for different applications such as PV or cogeneration. To meter gross energy the import and export energy flow on the load side of the meter must be completely isolated by connecting the wires to the appropriate meter terminals.



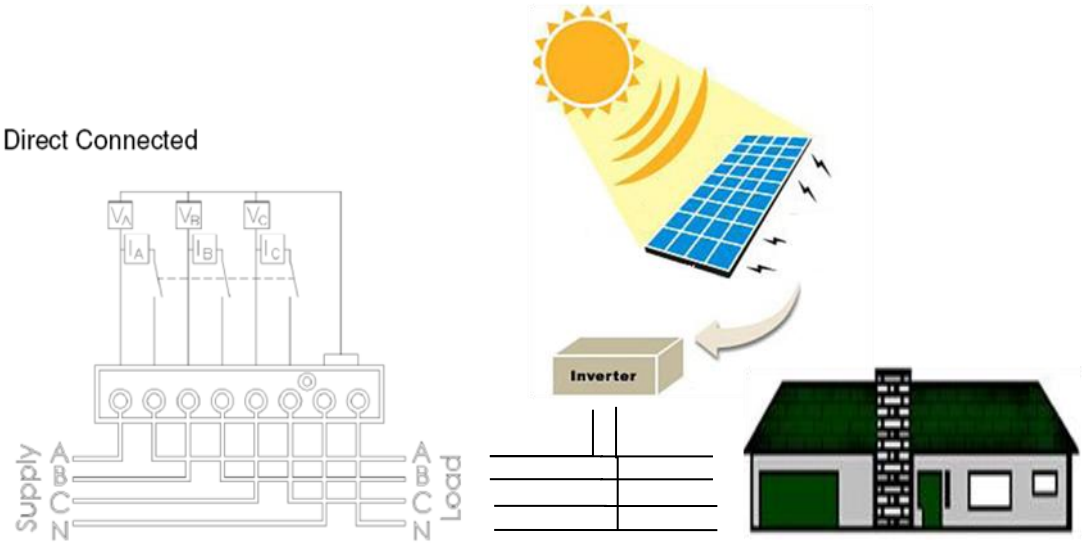
In gross metering applications with a 3 phase meter, the meter provides the gross values Import/Export on the phase level. Do not use the Total Import/Export registers for Gross Metering. Refer to 4.2 Measured Quantities

6.2 Net Metering

Net metering is normally achieved with the load and generation connected to the same meter terminal.

In case a net metering installation generates more power than consumed at the same time, then a disc meter turns backward and decrements the total energy register, while the electronic meter records the energy in the export register.

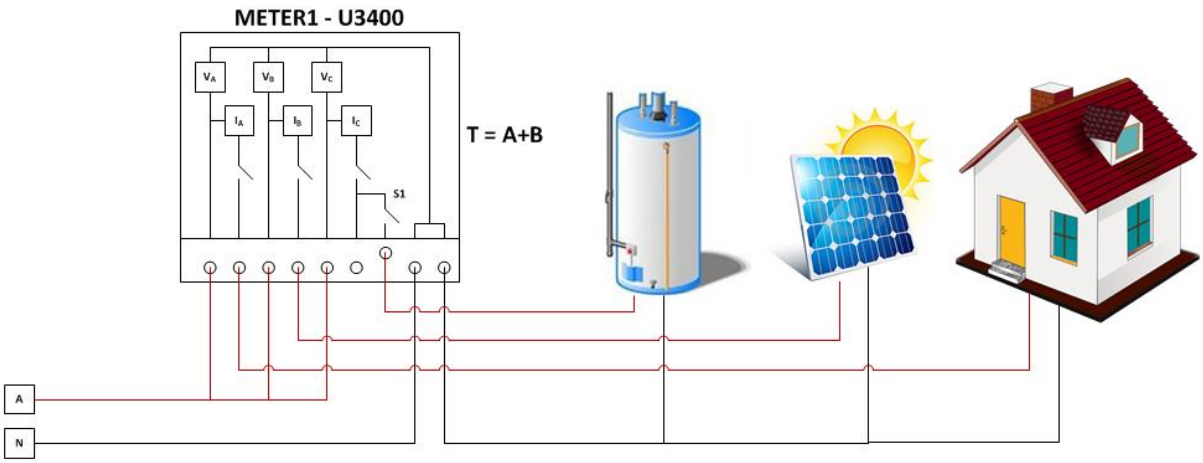
On the other hand if a net metering installation uses more power than produced, the disc meter turns forward and increments the total energy register while the electronic meter records the energy in the import register.



7 Single Phase Net Metering Application

7.1 U3400 single phase application

On the U3400 the three elements can be used to independently measure the Main Power, Solar and Hot Water circuits. The integrated ripple control functionality can be connected to the meter's internal relay to control the hot water system.



In the above example the following connections are made:

- Phase A – Main Power
- Phase B – Solar Panel
- Phase C – Hot Water

The total energy registers are configured to calculate the net total using the Phase A and Phase B registers so that any energy generated by the solar panel is subtracted the household usage.

8 Prepayment Mode

The meter is able to operate in a prepayment display mode which is managed by a smart prepayment solution platform.

For further details please refer to document:

E350 U-series Prepayment User Manual Appendix

9 Ripple Control

9.1 Overview

Ripple control is the technique of using amplitude modulation of power line supply in order to control the opening and closing of relays in remote equipment

Phase A needs to be connected to enable this feature

9.1.1 Telegram definition

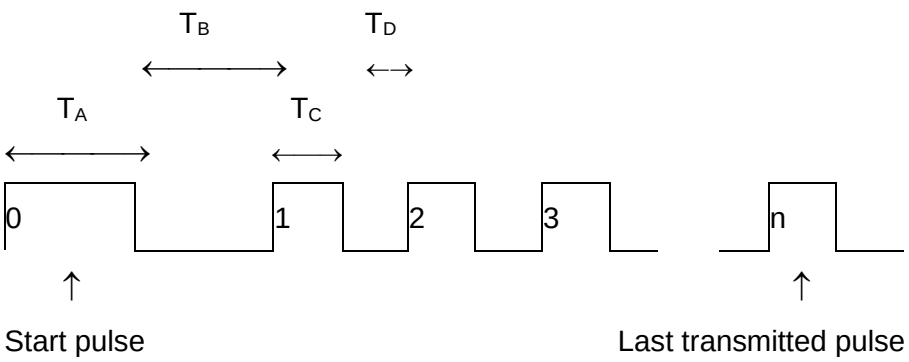
A telegram is defined as a ripple control transmission frame, and consists of a start pulse and a sequence of up to 50 pulses as shown in the figure below.

In general the ripple telegram is the carrier of the command to be executed in the meter.

9.1.1.1 Telegram frame

To define a telegram the following parameters shall be used with a nominal and minimum time:

- Start Bit Width
- Start Space Period
- Data Bit Width
- Data Space Period



	Description	Range of possible values
T_A	Start bit mark width	320 ms to 10 s
T_B	Start bit space width	320 ms to 10 s (in 10 ms intervals)
T_C	Data bit mark width	10 ms to 10 s
T_D	Data bit space width	10 ms to 10 s (in 10 ms intervals)
n	No. of command pulses	Maximum 50

9.1.1.1.1 Addresses and execute pulses

A telegram can consist of an address, an execute pulse, or both.

Each data pulse can be defined as follows:

0 No pulse (pulse absence must be verified)

1 Pulse present (pulse must be detected)

X Don't care (pulse may be present or absent)

To be classified as a valid pulse, the signal received must be within the filter parameters for a period of time greater than the pulse width.

9.1.2 Abort Pulses

The ripple control feature can be reset during a transmission by another Start pulse (abort pulse). The ripple control feature will operate in systems using a Reset/Start pulse to truncate telegrams. For this feature to work, the Reset/Start pulse must be more than double the width of the other pulses. The feature of resetting using the Start pulse can be enabled or disabled.

9.2 Specification

9.2.1 General

The U3400 has a new load control Relay State Machine (RSM) which handles the priority of the different relay control modules (see also 4.8 Load Control Relay).

The RSM is aligned to the NSMP section 7.6 Load Management through a controlled load contactor or relay.

The whole ripple control functionality is embraced in the ripple module. The ripple module is a control module of the RSM and runs independently and simultaneously to all other modules.

It will happen that another module which has the higher priority than the ripple module gets the control of the LC relay. In this case the ripple module has no longer control of the LC relay. Nevertheless the ripple module still runs in the back ground and receives ripple telegrams and actions the commands, as soon as the ripple module gets back in charge.

The U3400 ripple module has very wide range of user programmable features including ripple system, frequency, sensitivity, bandwidth and time delay. It can be software configured to operate in any of the conventional pulse-based ripple control systems. These include:

- Decabit (Zellweger)
- Semagyr (Landis + Gyr)
- Pulsadis (Enertec)
- Ricontic (Brown Boveri)

The ripple module is only supported if it is specified in the catalogue number. See also section 3.3 Catalogue Number.

9.2.2 Filter

Ripple protocol type (systems)	Decabit Pulsadis Semagyr Ricontic
Frequency	From 100Hz to 1600Hz in steps of 1Hz
Band Width	From 5Hz to 40Hz in steps of 5Hz
Operating Voltage[1]	From 0.30% to 4.00% in steps of 0.05%

9.2.3 Ripple Telegram settings

The U3400 has the capability of programming up to 16 different ripple telegrams.

Each telegram is set up as follow:

Telegram Name	A user definable name for that telegram
Code	Only supported with the use of Decabit. It is the short form of a dedicated bit pattern e.g. code +88 equals bit pattern 0011011100
Frame/Pattern	The bit pattern of a whole ripple telegram
Action	The command to be executed, see 9.3 Load Control functions
Time Delay	Applies to the command, see 9.3 Load Control functions
Re-Trigger	Flag allowing to re-trigger the same action, see 9.3 Load Control functions
Fail Safe	Flag to mark command specific fail safe, see 9.4.1.4 Command specific

The below functions are not supported by the U3400 via any ripple control:

Time setting via ripple telegram not supported

Tariff change via ripple telegram not supported

[1] Operating voltage is a percentage of the current incoming mains voltage.

9.3 Load Control functions

The U3400 ripple module provides a variety of load control functions to be executed once a ripple telegram has been successfully received. The meter distinguishes between relay actions, switch functions and control functions.

9.3.1 Relay actions

This is an action which is performed only once to switch the load control relay to the programmed state. The actions are:

- Close
- Open

Normally the action is performed instantly without any time delay on receipt of a valid telegram. However for both actions an individual time delay can be programmed. The selectable delays are:

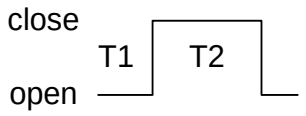
- Fixed delay (from 0h to 12h in steps of seconds)
  The relay switch action is performed at the time when the expired duration from time of receipt of a valid telegram exactly matches the fixed delay.
- Random delay
  The programmable settings include a minimum time delay and a maximum time delay from 0h to 12h in steps of seconds. The meter calculates and records a new random delay time which is within the range of the programmable settings at each power up.

9.3.2 Switch Functions

The switch functions are different from the relay actions described above because they perform multiple load control relay state changes based on one single valid received telegram.

9.3.2.1 Pulse

The pulse is defined as a relay state change from open to close and back to open.



A pulse telegram starts the pulse delay (T1) without switching the relay. The relay closes when time (T1) has elapsed and reverts back to open, at the end of (T2).

The programmable settings for pulse delay (T1) include a minimum time delay and a maximum time delay from 0h to 18h in steps of seconds.

In case a fixed delay is requested then the programmed minimum value must be equal to the maximum value. Otherwise the meter calculates and records a new random delay time which is within the range of the programmable settings at each power up.

The programmable settings for pulse duration (T2) are in the range of 0h to 18h in steps of seconds.

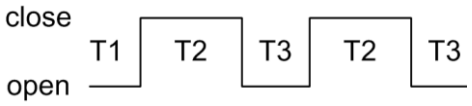
9.3.2.1.1 Retrigger

The meter allows optionally re-triggering the pulse. If the re-trigger option is enabled then the following rule applies.

- If the pulse telegram is actuated during the pulse delay (T1), it starts the pulse again without switching the relay.
- If the pulse telegram is carried out during (T2) relay closed, the pulse duration time (T2) starts again without switching the relay. The relay does not open until the time (T2) has elapsed.

9.3.2.2 Cycler

The cycler causes continuous repetition of the close time (T2) and open time (T3) after the initial delay (T1) has elapsed.



The programmable settings for the initial delay (T1) include a minimum time delay and a maximum time delay from 0h to 18h in steps of seconds.

In case a fixed delay is requested then the programmed minimum value must be equal to the maximum value. Otherwise the meter calculates and records a new random delay time which is within the range of the programmable settings at each power up.

The programmable settings for close duration (T2) and open duration (T3) are in the range of 0h to 18h in steps of seconds.

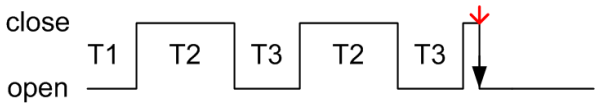
A “cycler start” telegram starts the cycler with the initial delay (T1) without switching the load control relay.

- If the initial position of the relay is open, then after T1 it will close and will commence cycling (open and close) in accordance with the cycle time T2 and T3.
- If the initial position of the relay is in closed, then after T1 it will open and will commence cycling (close and open) in accordance with the cycle time T3 and T2.

The U3400 supports two kinds of cyclers. These are the standard cycler and the air conditioning cycler. The only difference between these cyclers is how they act on a “cycler end” telegram. While the standard cycler stops immediately and opens the relay, the air conditioning cycler checks first in which position the relay is. In case it is in close position it will not open the relay until T2 has expired. This is to protect appliances from short pulses.

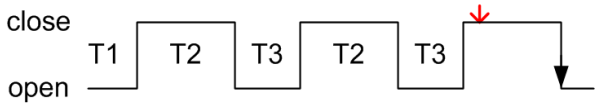
9.3.2.2.1 Standard cycler

Once a “cycler end” telegram is received the standard cycler stops with immediate effect and switches the relay to the open position.



9.3.2.2.2 Air conditioning cycler

Once a “cycler end” telegram is received the air conditioning cycler checks first in which position the relay is. In case the relay is in open state then the cycler stops. If the relay is in the closed position, T2 has not expired yet, it will not open the relay until T2 has expired.



9.3.2.2.3 Retrigger

The meter allows optionally retriggering the “cycler start” command. If the re-trigger option is enabled then the following rule applies.

- If the “cycler start” command is received whilst T1, T2 or T3 are running then T1 shall start without switching the load control relay.

9.3.3 Control functions

The ripple module supports the different control functions as below, which are in relation to the load control relay only.

9.3.3.1 Time Table

If the ripple module is set to the Time Table mode then the load control relay will operate according the programmed scheduler.

Time Table mode provides the same features and the same configuration as in the main meter program. Refer to section 12.2.26 Controlled Load L1 – Time Table.

To command the meter into the time table module, the action “Time Table Start” must be selected when setting up the ripple telegram.

To exit the time table mode, the action “Time Table End” must be selected when setting up the ripple telegram.

Without a “Time Table End” ripple telegram programmed, the meter will remain in this mode. It is possible for another control module higher in the RSM hierarchy will get control but after releasing control it will fall back into time table mode again.

A random delay time option applies to the switch time if programmed. E.g. the relay is currently in the open position. The telegram “Time Table Start” is received at 8 pm and the time table requires the relay to be in the closed position between 7 pm and 7 am with a random time of 20 minutes. In this case the relay will wait a random time of up to 20 minutes before switching to the close position.

The time table mode is used for two reasons

- Fail safe, selectable as an option
- For maintenance on the ripple injection plant

9.3.3.2 Emergency Load Control Shedding

The emergency load control shedding functions as defined in 4.8.10 Emergency Control can be temporarily enabled or disabled by a ripple telegram. The following options are available:

- Enable or Disable Under Frequency
- Enable or Disable Under Voltage (not released)
- Enable or Disable Demand Limit (not released)

Further it is feasible to simulate the emergency load control shedding using a ripple telegram. The intention is to measuring the amount of controllable load in a system.

The emergency test only operates on receipt of the following telegrams actions below. Once it is activated it will remain in this mode until the programmed delay time has expired.

- Test Under Frequency
- Test Under Voltage (not released)
- Test Demand Limit (not released)

9.4 Fail safe

The fail safe function serves to bridge the absence of ripple control pulses/telegrams. This absence may be as a result of a transmitter failure or a network signal issue.

The fail safe is activated once the fail safe time has expired.

The fail safe timer is reset on the successful decoding of a telegram plus additional requirements given in the different fail safe types below.

The fail safe timer can be programmed for time intervals between 0 hours and 72hours in steps of 1 minute.

9.4.1 Types of fail safe

There are four fail safe types where each of them has another object to maintain the network. The technical difference between them is under which condition the fail safe time is reset.

9.4.1.1 Disabled

Fail safe is not enabled at all.

9.4.1.2 Track specific

The fail safe timer is reset once a start bit plus the defined number of valid data bits are received.

If none of the received telegrams matches above specification, a global fail safe event is triggered.

Track specific fail safe checks whether the ripple injection plant is operating rather than checking for a specified ripple telegram.

9.4.1.2.1 Valid Data Bits

To allow the track specific fail safe function to operate, the number of data bits within a valid telegram is considered. With Decabit there are exactly 5 data bits in a valid telegram. Other protocols are user definable therefore this option is defined as the minimum number of valid data bits in a telegram.

9.4.1.3 Device specific

The fail safe timer is reset once any of the ripple telegrams programmed in meter is received. If none of them has been receive within the fail safe time, a device specific fail safe event is triggered.

9.4.1.4 Command specific

If command specific fail safe is selected then the program requests to mark at least one or more individual ripple telegrams as command specific.

The fail safe timer is reset once it successfully receives one of the specific marked ripple telegrams.

If none of the marked telegrams has been received within the fail safe time a command specific fail safe event is triggered.

A scenario for command specific fail safe is when doing air conditioning cycling. When the customer agrees to this type of control on their appliance, there is typically a maximum period that an event will last for. The risk to the Utility is that the “cycler stop” telegram is not received by the receiver. To reduce this risk a command specific fail safe time is used. During the control period the meter may have heard other telegrams for hot water control etc. The fail safe time must not be effected by these signals.

9.4.2 Fail safe actions

Once fail safe is active the following actions can be programmed:

- Open load control relay
- Close load control relay
- Go into time table module

The load control relay action is performed immediately.

9.4.3 Fail safe recovery

Recovery from fail safe can be programmed either manually or automatically. The meter recovers from fail safe immediately on receipt of any ripple telegrams programmed in the meter only.

However there is one exception where the fail safe action is set to “Time Table” mode and the recovery mode is set to manual. In this special case the meter must receive a “Time Table End” telegram to exit fail safe.

This recovery mode is useful to protect the meter switching the relay in cases where the utility injects ripple telegrams for test purposes, after a ripple injection plant has broken down. It is also intended to prevent the meter from exiting fail safe when receiving ripple telegrams spilled over from other ripple plants.

<table>
<tr><th>Fail Safe Action</th><th>Manual Recovery</th><th>Automatic Recovery</th></tr>
<tr><td>Open</td><td rowspan="2">The meter exits fail safe immediately on receipt of any ripple telegrams programmed in the meter</td><td rowspan="3">The meter exits fail safe immediately on receipt of any ripple telegrams programmed in the meter</td></tr>
<tr><td>Close</td></tr>
<tr><td>Time Table</td><td>The meter will not exit fail safe unit it receives a “Time Table End” telegram. Once it receives this telegram it exits fail safe immediately.</td></tr>
</table>

Manual recovery fail safe in combination with the “Time Table” as the fail safe action requires a dedicated telegram “Time Table End” to exit fail safe mode. If this telegram is not programmed the meter will remain in “Time Table” mode forever.

In the scenario the air conditioning cycling is programmed in fail safe time table, then meter honours the air conditioning operation and waits for the reminder of the close duration to elapse before exit fail safe mode.

9.5 Power Restore

The ripple module has its own independent power restore module. Even so it has exactly the same priority as the power restore module for non ripple meters in the overall RSM.

The RSM allows having both power restore modules enabled concurrently. In this case and if the ripple module is enabled then the ripple power restore module settings apply.

The power restore module as part of the ripple module is controlled by the RSM. This means a module higher in the hierarchy gets control at any time during power restore. If the Ripple Power Restore module is pre-empted by a higher priority module and when that module gives up control and ripple is next in line, then ripple will action the relay to a state dictated by the power restore module, unless failsafe is triggered or a telegram was received. It should be noted that the power restore module is nothing but an action of the ripple module.

9.5.1 Power restore actions

The below power restore actions are available

Restore actions	Description
Unchanged	Load control relay remains in the state it was during power off.
Open	Load control relay opens
Close	Load control relay closes
Time Table	The load control relay will operate according the programmed scheduler.
Memory	The memory function executes the last received ripple telegram prior to when the Power Down occurred.

9.5.2 Power restore delay

The programmable settings for a power restore delay include a minimum time delay and a maximum time delay from 0h to 18h in steps of seconds.

In case a fixed delay is requested then the programmed minimum value must be equal to the maximum value. Otherwise the meter calculates and records a new random delay time which is within the range of the programmable settings at each power up.

9.5.3 Power restore sequence

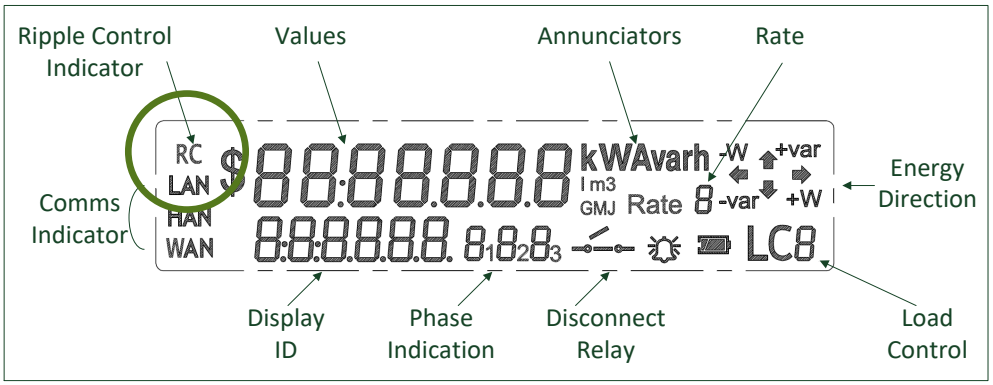
The RSM enters the power restore module immediately after power is applied to the meter. The power restore module task never voluntarily releases control and the power restore is not killed when pre-empted refer to 9.5 Power Restore. The ripple power restore is an action of the ripple module.

> ⓘ The meter that was in “fail safe” mode prior to the power outage remains in “fail safe” mode.

9.6 Display

9.6.1 Definition of the Ripple Control Indicator (RCI)

The RCI expresses the functionality of the ripple control module as a whole independent system and does not describe the Load control relay status.



9.6.2 Ripple Control Indicator behaviour

In general the U3400 acknowledges five different ripple control states. These states are displayed with four different ripple indicator behaviours because state 1 and state 2 can be consolidate.

State	Meter Type	Description	RCI behaviour on display
1	Non Ripple Meter	Ripple module permanently disabled	Permanent OFF
2	Ripple Meter	Ripple module temporarily disabled via EMPWin tool box or remote AMI command	Permanent OFF
3	Ripple Meter	Ripple is receiving a ripple telegram which had a valid start bit plus matches the filter values.	Fast flashing ON/OFF t_{on}= t_{off} = 0.5sec

4	Ripple Meter	Ripple module is in fail safe.	Slow flashing ON/OFF t_{on}= t_{off} = 2sec
5	Ripple Meter	Ripple module is working correctly	Permanent ON

The ripple indicator will maintain the ripple state status regardless whether the ripple module is in charge of the load control relay or not.

Use case: While the LC is controlled by under frequency load shedding the fail safe timer expires and fail safe gets activated. The RC indicator immediately reflects the fail safe state.

9.7 Ripple event log and status

The meter records two sets of information, the ripple event log and the ripple status log.

Each relay action performed by ripple is optionally recorded in the meter event log.

9.7.1 Event log

The event log records a time stamped event including various information for each recognised ripple telegram that was programmed into the meter.

The log will record the last 150 ripple commands received.

Event log frame

Time	**Description**					
Date and Time	Overall message	Message content				
1	2	3	4	5	6	7

1. Date and time
2. Telegram ID plus the relay assignment (LC1 stands for load control relay one)
3. Telegram ID
4. Signal strength
5. Delay Timer
6. Telegram action
7. Retrigger information

9.7.2 Status log

The ripple status log records various information’s as shown below.

Ripple configuration status	True: Ripple is enabled and has a valid configuration False: Ripple has not been configured	
Last known telegrams	Telegram ID	
Last known signal strength	Voltage	
Last unknown signal strength	Voltage	
Last unknown number of bits	Number	
Last unknown received telegram	Bit pattern e.g. Decabit 1010101010	
LC No	Last Action	
Always 1 (only one LC relay)	Enumerator	
LC No	Telegram No	Telegram Count
Always 1 (only one LC relay)	1	how many times the telegram has been received since last reprogramming
Always 1 (only one LC relay)	16	how many times the telegram has been received since last reprogramming

Definition of an unknown telegram: Telegram matches the filter value settings but the telegram doesn't match any of the pre-programmed telegrams in the meter.

9.8 Program Summary

It contains the program ID plus the version number. It also provides the option of a free text note field.

The notes are not written to the meter but stored on the program data base.

10 Installation

This chapter describes the mechanical/ physical configurations and installation procedure of the U3400 meter.

10.1 Storage and Transport

The meter should be stored and transported in its original packing case or an equivalent level of packaging. The meter should be checked for physical damage after transportation

The meters are packaged 6 in a pack for transporting. Larger quantities are generally supplied on pallets with shrink plastic cover.

10.2 Safety Considerations

Dangerous Voltage
Disconnect the supply when installing or servicing a meter.The connecting conductors at the point of installation must be voltage-free for installation of the meter. Contact with live components is dangerous to life. The relevant supply fuses should therefore be removed and kept in a safe place until finishing work, so that they cannot be re-inserted by other persons unnoticed.

To avoid possible damage to the meter, at no time, any voltage in excess of 300 volts should be applied to this meter.

The meter is sealed to maintain the security and prevent unauthorised access or opening of the meter. As per NMI regulation, our approval number has been provided on the seal.

10.2.1 Presence of touch current during installation

Landis+Gyr reminds installers to be aware that touch currents may be present during the installation of a U3400 meter.

The U3400 meter is double-insulated and designed to operate without a direct earth connection. When double-insulated equipment is connected to mains power, voltage will be present on some accessible conductive parts such as antennas. The presence of this voltage is not abnormal.

Due to the presence of this voltage, a meter installer touching an accessible conductive part may feel a very low current that flows from the conductive part, to the installer and then to ground. This is known as a touch current. The threshold at which an installer can detect a touch current will vary between individuals and according to environmental conditions; in most cases there will be no perception of current.

The U3400 meter is in compliance with the touch current limits of IEC60950-1 and it is therefore verified that it does not present a safety hazard to the installer, nor does it indicate an internal equipment failure.

In order to measure the voltage present on an accessible conductive part, it is important to use a low-impedance multi-meter as a high-impedance multi-meter will provide an incorrect reading. The voltage present on an antenna, and its associated touch current, can be removed by earthing the antenna, such as by installing it on an earthed metal meter box.

10.3 Mechanical Installation

The spacing and layout of main terminals is as per the SD and SDM meters and in accordance with AS1284.11. The normal position for mounting a meter is vertical with the LCD display at approximately head height. The overall meter dimensions are given in Figure 7.1, 7.2 below and in “Appendix A – Specifications”. The physical position of the 2 bottom screws is identical to the popular SD, SDM, Q3,Q4, EM3000 series and EM5100 meter range, minimising, necessary modifications to the metering panel when replacing these meter with the U3400 meter.

10.3.1 Recommended Terminal Screw Torque

Landis+Gyr have conducted intensive test to define nominal values for correct installation torque to securely hold a conductor in both steel and brass terminal cages for the U-Series meters.

Based on torque settings across a range of conductors, the minimum recommended tightening torque of 2.5 Nm is required to ensure a correct installation with the cage terminals. The maximum limit is 3.0 Nm.

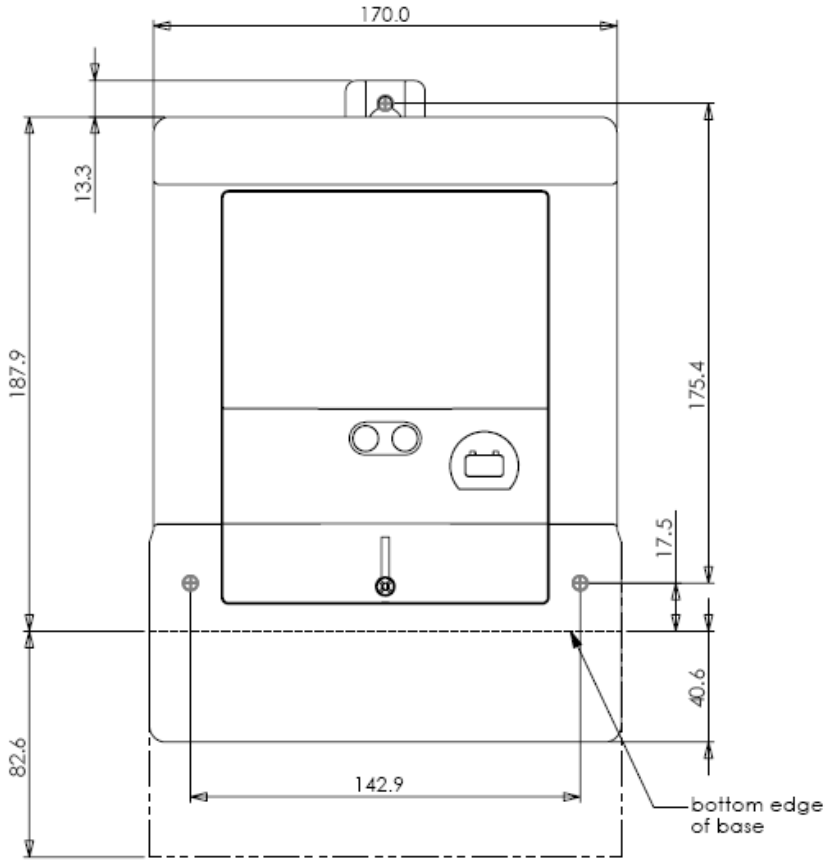
The recommended torque for Load Control terminals is 2.0 Nm +/- 10%.

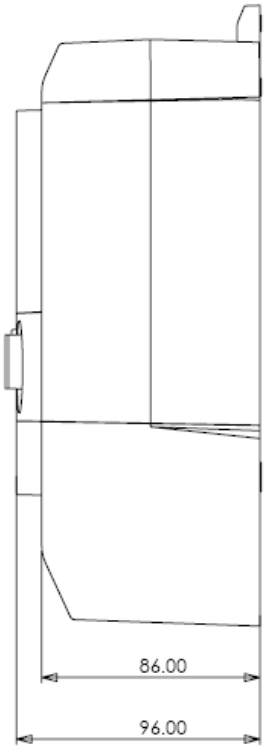
It is recommended inspecting all terminal screws after tightening for deformation.

It is recommended to use uninsulated ferrules on cables smaller than 25mm².

Battery operated screwdrivers should not be used for final torqueing

10.4 Meter Dimensions

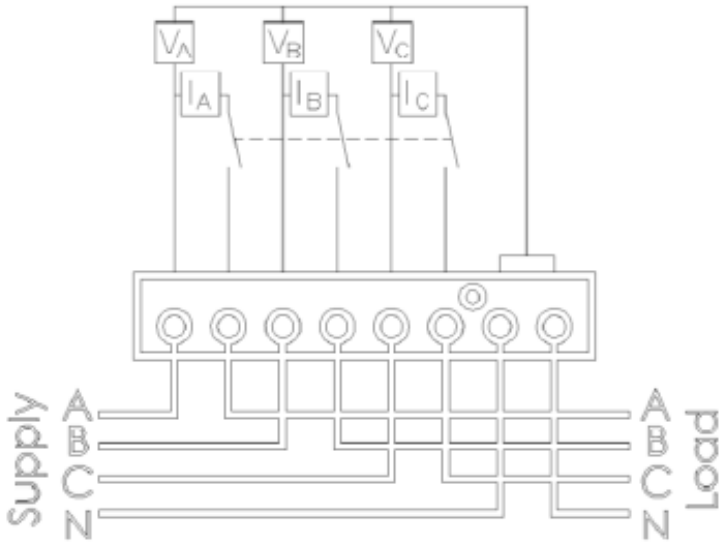




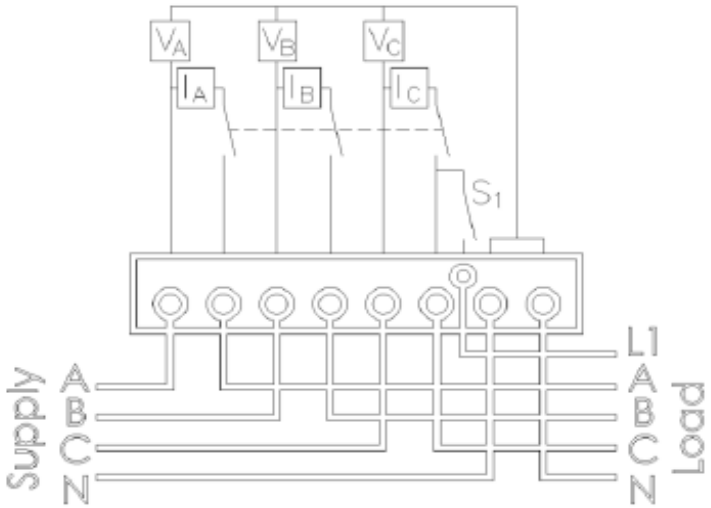
10.5 Configurations

The U3400 meter is available in three configurations.

10.5.1 U3400 (no load control relay, no Multifunction Output Contacts)

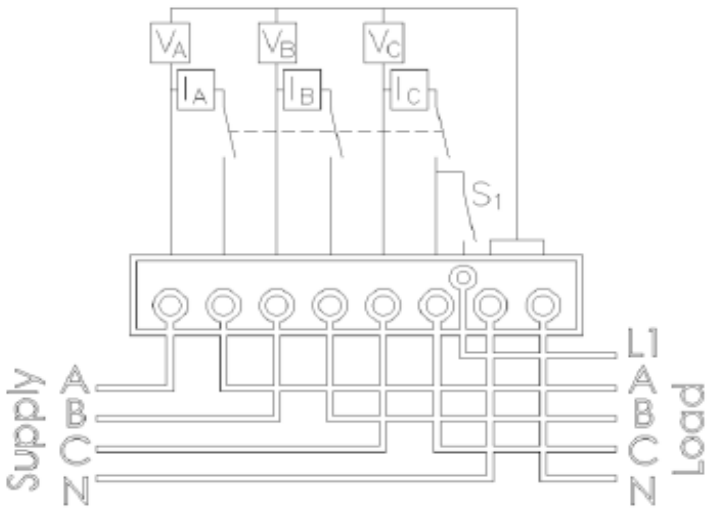


10.5.2 U3401 (with load control relay, no Multifunction Output Contacts)



10.5.3 U3401 (with load control relay, with Multifunction Output Contacts)

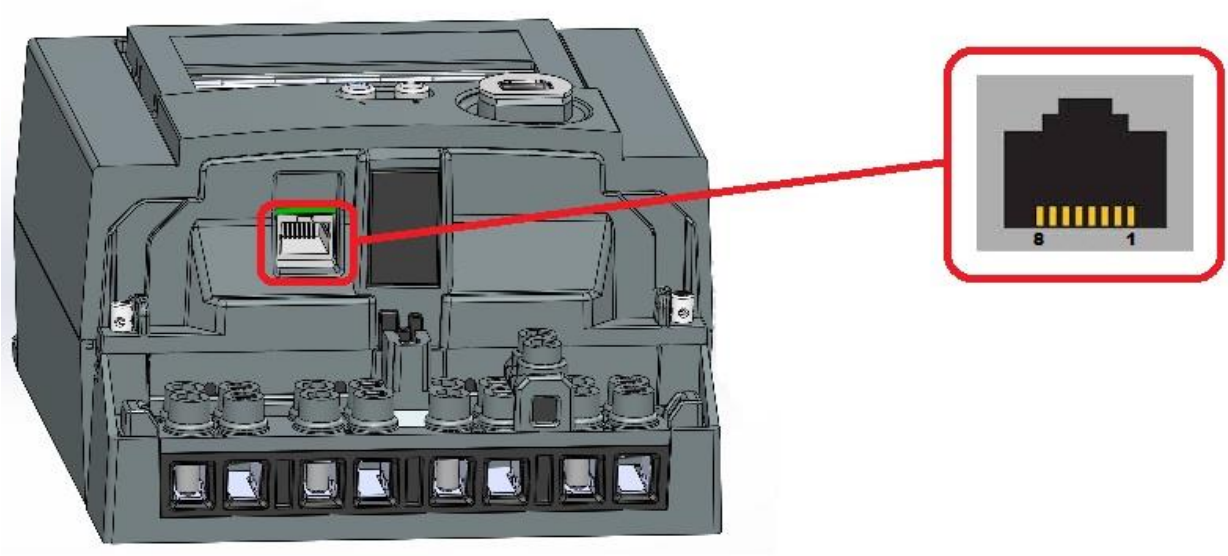
This configuration is only available with a minimum order quantity. Please contact your sales representative for further information.


VA
VB
VC
IA
IB
IC
S1
Supply
A
B
C
N
L1
A
B
C
N
Load

10.6 Multifunction Output Pin assignment

The Multifunction Output pins assignment of the RJ45 connector is as shown below

Pin	Assignment
1	PO1
2	PO1-COM
3	PO2
4	PO2-COM
5	PO3
6	PO3-COM
7	PO4
8	PO4-COM



10.7 Module installation

The communication module is plugged into the top section of the meter and covered with the module cover. The module cover has the provision to be sealed to prevent any unauthorised access. Communications modules are hot swappable allowing the utility to insert or remove modules while the meter is powered up. The communications module and compartment is designed to be intrinsically safe and is free of mains voltages. This is intended to protect the technician's safety while installing communications modules to the meter.

If an external antenna is required to be connected to the communication module, the breakout on the top left of the module cover is removed to allow the antenna cable to run through.

The pictorial installation procedure of the module is shown below:

Figure 7.6 Module Installation

11 Operation

This chapter describes how to read the meter on-site and how to use the buttons and LED (Light Emitting Diodes) indicators. This chapter details status codes on the meter display.

11.1 Layout

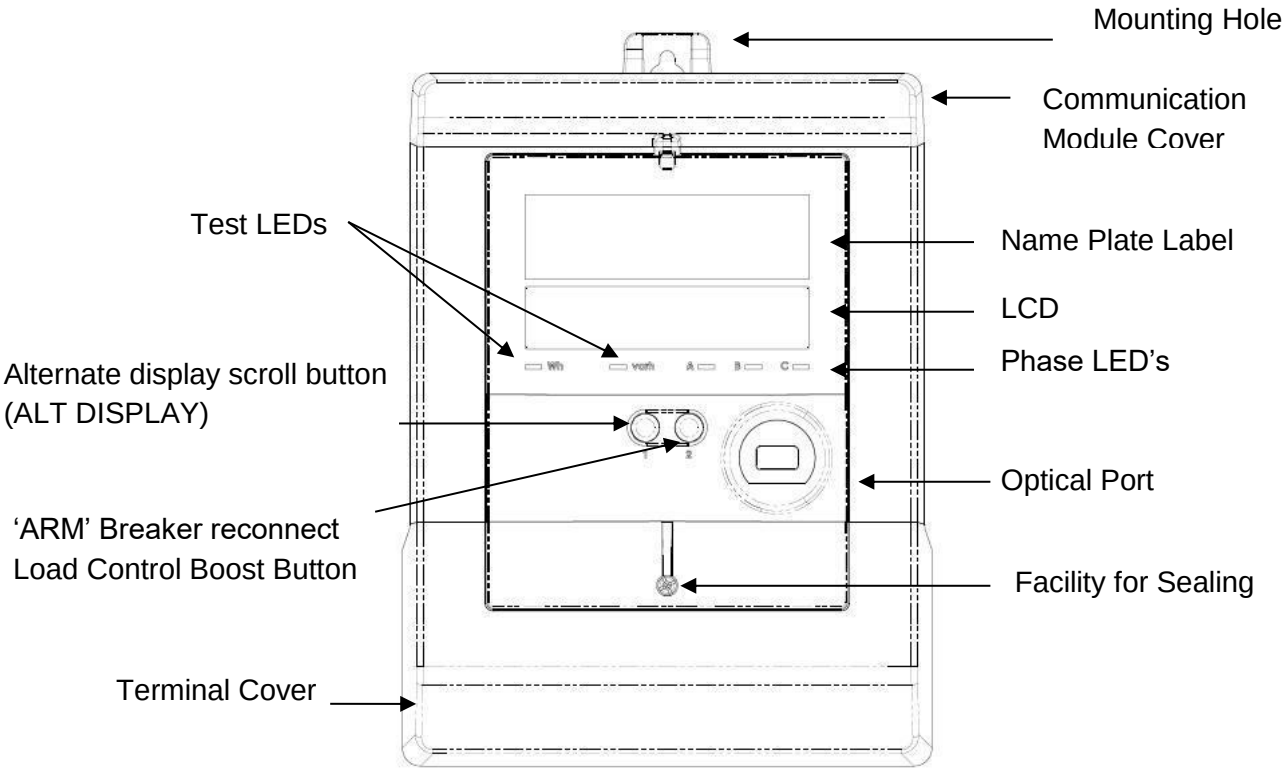


Figure 11.1 Layout

Name plate Label

Provides all required meter ratings and can include the utility or meter supplier's registered name and logo. The variable data such as meter property numbers is inserted in the area provided (bar-coded) with a clear plastic 'non-removable' clip-in cover.

LCD Display

A highly contrast wide angle liquid crystal display (LCD). The custom designed LCD can be read in a wide range of lighting conditions. It has been designed to provide optimum viewing with the meter mounted at eye level and above, in accordance with normal utility practices. However, excellent viewing is available for large variations in angle, both indoors or in full sunlight

Consumption Indicators (Test LEDs)

The visible red colour LEDs are used for showing either active (kWh) or reactive energy pulses (kvarh). The two red LEDs are pulsing at a rate proportional to the measured load. The meter constant is set at 1 Wh and 1 varh per pulse.

Phase LED

Phase indicator LEDs indicate the meters voltage elements are energised. All 3 indicators should be lit for correct 3phase operation.

Button 1 This button allows the reader to scroll through the available display items. To move along the sequence quickly, the button should be pressed multiple times until the required display is located. To select the alternate displays the button must be held for 2 seconds.

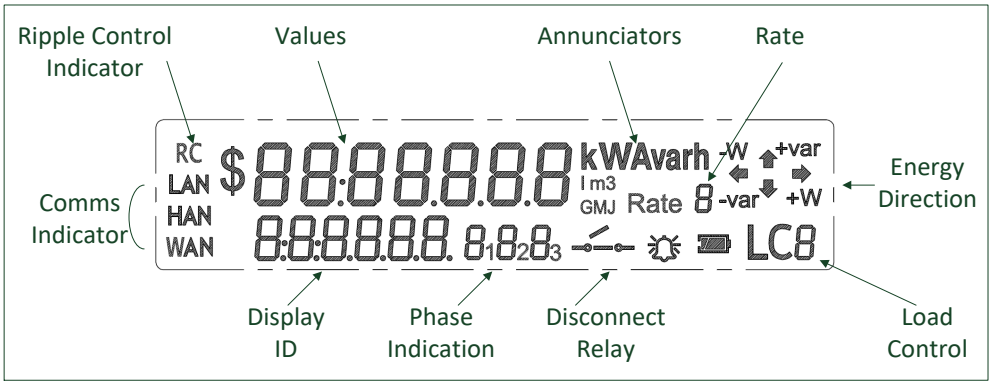
Button 2 This button allows the user to reconnect the supply when in an armed state or activate the boost function (where enabled).

Optical Port ANSI Type-2 standard for two way communication with the metering device, for programming and the collecting meter data, using a laptop or handheld fitted with an ANSI optical probe. For more information see also section 4.10.5 “ANSI Optical Port (Local Communication)"

Meter Terminal Cover

Mechanical security and sealing device. Sealing of the meter is required for security and to prevent unauthorised access or opening of the meter. A verification marking has been provided on these seals as it is required by NMI. There is a sealing provision of Φ2.4mm hole for main cover, terminal cover and a Φ1.0mm for communications cover. (Maun seals are supplied as a standard on the U3400 meter).

11.2 Reading the Display



Display ID The display could cater for a 6 digit-number however, it currently supports only a two-digit number that identifies the display value.

Values The value on the display, gives the energy values consumed during the current billing period and other information like date, time, program ID, Revision numbers, etc. Sections of the Value are sometimes used to convey extra information such “PF” used in front of the power factor value.

L+G meters display energy on the LCD display in units of kilowatt hours (1000's of watt hours). When displaying energy L+G meters light up an annotator that indicates the value displayed is in units of kWh's (1 kilowatt hour = 1000 watt hours).

The displayed energy values can be customised to:

- Hide or show leading zeros
- Display between 0 to 3 decimal places.
- The energy registers are displayed with 7 digits, with up to 3 decimal places depending on the parameter setting.

The meter can sequentially display up to 20 items in the normal display list and 10 items in each of the Alternate display lists. The actual items to be displayed and the order of display are selected during programming the meter and can only be altered by reprogramming the meter
The display can automatically scroll through the configured displayable items, where the hold time for each item can be selected in the range 1 to 8 seconds. More information about programming the display starts at section 12.2.11 "Normal Display"

Example leading zeros:

Customers can enable or disable the 'display of leading zeroes' on the LCD. When this feature is enabled, the display shows the character '0' in front of the energy register. This is most easily demonstrated with an example, below.
If the energy register contains 12.345kWh, and the display is configured for 3 decimal places and no leading zeroes, then the LCD will show
12.345 kWh
However if leading zeroes are enabled (and the display has 7 digits available), the LCD will show
0012.345 kWh
As can be seen above - the leading zeroes feature simply fills digits to the left of the 'most significant digit' with zeroes, to emulate the mechanical register on an induction disc meter.

Example Number of Decimal Points:

Customers may select the number of decimal points to be displayed on the LCD. These decimal points allow the customer to see the amount of fractional kWh's accumulated by the meter. The customer may select anywhere from 0 to 3 decimal points.

When 0 decimal points are selected, the meter only displays the number of 'whole' kWh's on the LCD. Remember that the U-Series whole current meters have a rollover limit of 1,000,000kWh – so this number will be at most 6 significant digits in length before rollover.

This may be confusing to customers as the number of energy digits available on the screen is 7.

In the example below, the energy register has a value of 999,999.999 kWh – and the display is configured to show 0 decimal points and no leading zeroes.

999,999 kWh

When the meter is configured to display 3 decimal points, then a decimal point annotator is lit up on the display, and 3 trailing digits to the right of the decimal point are visible on the LCD. This allows the customer to see energy down to the finest ‘resolution’ available on the meter, i.e. watt hours.

In the example below, the energy register has a value of 99,999.999 kWh – and the display is configured to show 2 decimal points and no leading zeroes.

99,999.99 kWh

If the example above was changed such that 3 decimal places were configured, then the most significant digit would be ‘clipped’ from the LCD.

9,999.999 kWh.

Notice that there is a ‘9’ which cannot be displayed in the above example

However the electronic readout of the register would still be equal to 99,999.999 kWh.

Rate

The present TOU Rate A, B, C, D, E, F being used to record energy usage.

Annunciators

Various annunciators provide information about the quantity displayed with a unique identification number and with information about the energy units like kWh, kVAh and kvarh.

Energy Direction Arrows

The direction of the power for kW, kVA and kvar are displayed on the panel with the direction arrows. These arrows are instantaneously updated and will display the direction of the power at every instance of time.

Energy Direction Arrows	
+W	Active energy import (Energy Received from the Utility).
-W	Active energy export (Energy Received from the Customer).
+var	Reactive energy (kvarh Imported by customer)
-var	Reactive energy (kvarh Exported by customer)
+va	Apparent energy import (Energy Received from the Utility).
-va	Apparent energy export (Energy Received from the Customer).

Energy Direction Arrows start at 100 mA or 20 W.

LC

Enunciates the status for the load control relay. The numbers after ‘LC’ indicates whether it is the primary or other relays in operation, numbers are shown to indicate the closed state for that particular relay.

Disconnect Relay

The contactor symbol is used to indicate the status of the disconnect relay.

Breaker OPEN	Breaker Closed

LAN/HAN/WAN Indicators

This indicates whether the meter is connected to a network. This needs to be configured to turn on the appropriate icon. Commands can be sent to the meter (usually via the NIC) to enable or disable these indicators. They can be programmed to flash slow (2 seconds on and 2 seconds off) or fast (every 300mseconds) or to be permanently On or Off.

Ripple Control Indicator

Visible indication of the operation of ripple control is provided by the Ripple annunciator on the meter display.

The Ripple annunciator will appear when the meter has a valid Ripple program. The RC icon will flash when receiving the correct frequency.

11.3 Meter Status codes

The meter features Standard Meter Status codes and Manufacturer Meter Status codes. The Status code information can be programmed for display.

11.3.1 Display Status codes

If the Status information is programmed on the alternate display, the corresponding alternate display can be selected by holding Button 1 for 2 seconds. Then as part of the scroll sequence, the Status code will be displayed

11.3.2 Status Handling

The meter can display Status codes that alert the user to various problems. If the meaning of the Status code is unclear, we suggest that one generates a readout from the meter with EMPWin+ and contacts Landis+Gyr for clarification.

When a critical status occurs, the meter continues to operate and does not lock or freeze the display cycle. These Status codes are then cleared only via software with a reset command.

Standard Status codes (S1 on Display)	
Code	**Description**
S1 0x0001	Cyclic Redundancy Check Sum (CRC)
S1 0x0040	Clock Error
S1 0x0080	Measurement warning detection[2]
S1 0x0100	Low Battery
S1 0x1000	Tamper Switch Activated
S1 0x2000	Export Energy Detected

Manufactures Status codes S2	
Code	**Description**
S2 0x0001	Flat Battery
S2 0x0002	Power Fail Counter Mismatch
S2 0x0004	Export Energy Detected on Phase A
S2 0x0008	Export Energy Detected on Phase B
S2 0x0010	Export Energy Detected on Phase C
S2 0x0020	Neutral Integrity Related
S2 0x0040	Phase A Voltage Lost
S2 0x0080	Phase B Voltage Lost
S2 0x0100	Phase C Voltage Lost
S2 0x2000	Phase Rotation Error
S2 0x4000	Prepayment Mode Enabled
S2 0x8000	Prepayment Unoccupied Status

[2] This flag may indicate a measurement error in the past. Please contact your Landis+Gyr representative for further diagnosis.

It is possible to have more than one Status reported at one time. Care must be taken when deciphering status codes as they are reported in hexadecimal (base 16)

Example 1 If the meter reported **S1 1100.** The meter would have a Low Battery and a Tamper Error.

Example 2 If the meter reported **S2 000C.** The meter would have lost Phase A and Phase B Voltages

12 Software configuration Tool

AMI meters have multiple functions allowing the meter to be utilised for a large number of applications. During manufacture of the U3400 meter, the meter must be programmed (configured) with the utility/ customer configured program. This configures metering information and other operating parameters to be displayed, load control, Load Profile, time-of-use setup (if required), disconnect breaker functionality and security arrangements (access passwords etc.). This configuration procedure and programming details are explained in this chapter.

In general, the meter is supplied from the factory as pre-programmed to customer requirements, so a detailed knowledge of meter programming is not necessary. Landis+Gyr local offices can provide assistance for most programming matters.

The actual programming of the meter is performed using L+G's commissioning and programming tool called EMPWin+. For those who wish to develop programs themselves, this chapter and the on-screen help available in EMPWin+, will provide all the necessary details.

12.1 EMPWIN+

The meter is configured with EMPWin+, a Windows based software package. The Landis+Gyr Part number for this software is **CD-EMPWin+.** This suite of programs covers all of Landis+Gyr's AMI meters.

EMPWin+ is a software support application developed by Landis+Gyr, Australia

The EMPWin+ software versions 2.4.4 or higher supports U3400 electronic meter.

Please refer to the firmware release notes and the EMPWin+ release notes to see which meter firmware version is supported by which EMPWin+ version.

The various modules that make up the full EMPWin+ software package are listed below.

Under current development practices, new technologies and requirements are ever evolving, and from time to time software may be updated to introduce new features, provide service updates and performance improvements. Software updates are implemented at the discretion of asset owners and Landis+Gyr will advise of software updates as and when they are released.

Please refer to the EMPWin+ User Manual for detailed information.

Figure 0.1 AIP200 Probe connected to the optical port

12.1.1 MeterDev - Program Developer

Meter program development tool, can operate as a standalone program for the creation of new or the editing of existing meter programs.

The basic characteristics of the meter are specified which will determine how the meter is to function. This occupies the majority of the screens in the EMPWin+ Program Developer.

The programmer then needs to specify the display options to appear on the meter display. For the meter this is very simple with a maximum of 40 displayable items.

When the programmer is satisfied with the choices made, the EMPWin+ Program Developer will generate a *program file* (*.mcd).

12.1.2 MeterComm - Communications

All communications to the meters, is carried out from this program. It forms the hub of the EMPWin+ software package.

The Communications program allows the user to write this program to the meter. It can also perform readings and tool box commands.

To open the communications program a user name / Password combination is required only if EMSecure is installed.

Once into the communications program, the meter-specific information, such as the meter serial number and the customer’s account ID must be entered.

Download (Write) the meter-specific configuration file to the meter using the optical port.

12.1.2.1 MeterComm Settings

MeterComm has a Settings menu option that is shown before a connection is established to a meter. The Settings option has three sublevels being Properties, Retry Settings and EM Comm settings.

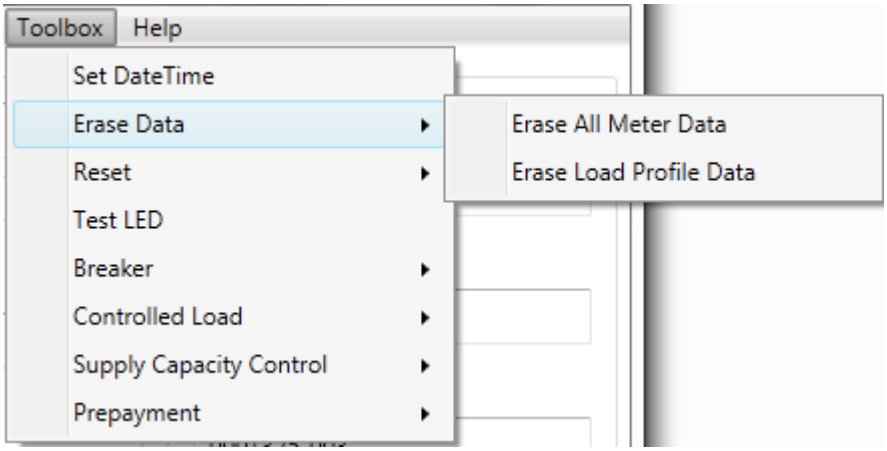
The Properties option can be used to configure the PC's Programming Station number (from 0 to 9900) and the Communication Timeout extension units from 0 to 8 (that are used during remote communication sessions – typically set to 4 units). Only the first user in EMSecure in Aphabetical order can modify this setting.

The Retry Settings options can be used to set the number of retries for logging into the meter and the number of seconds to delay between retries. These values are typically set to 1.

EM Comm settings (not used with U series meters) are additional features used with the EM5400.

12.1.2.2 Toolbox commands

Toolbox commands are dynamically configured based on the type of meter being used and what features have been enabled by the meter's program. A typical U3400 Toolbox list is shown below.



Set DateTime option actually only allows the time within the current date to be reset not the date (this is for security and event logging reasons). If the date needs to be changed then the meter must be fully programmed.

Erase Data can allow all Billing and Load Profile data to be erased or if Erase Load Profile Data option is used then just the LP. If the program has multiple Load Profile groups then specific Load Profile groups can be erased without disturbing others.

Reset option supports the subtypes of Event Log, Ripple Log (if ripple is enabled), Status (clear current error and warning flgas) and Passwords (write new optical passwords to the meter).

Test LED allows either the Wh or VARh LED to be configured to pulse for import or export direction and also specifies for how long to stay in that position.

Breaker option can Disconnect or Reconnect the 100 Amp Breaker.

Controlled Load option can be used to open or close the meter's LC relay (depends on meters configuration options) and also to control specific LC functionality such as Under Frequency or Ripple Control of the LC relay. Following Priority Override Commands are also provided:

- Open: no parameter
- Close: no parameter
- Instant Open: StartDateTime, Duration
- Instant Close: StartDateTime, Duration
- Delayed Open: StartDateTime, Duration, Random Delay Min(00:01-12:00), Random Delay Max (00:01 - 12:00)
- Delayed Close: StartDateTime, Duration, Random Delay Min(00:01-12:00), Random Delay Max (00:01 - 12:00)
- Start Load Cycling: StartDateTime, Duration
- Terminate priority override: Random Delay(1-120) (StartDateTime 0, Min 0)

Supply Capacity Control enables (or disables) ESCC, Normal supply Capacity Control mode, TOD Import, TOD Export supply capacity control limits independently.

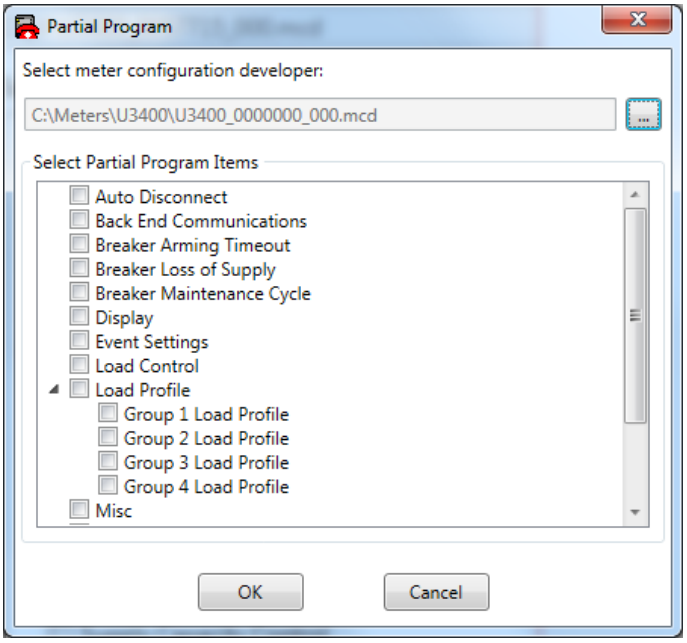
Prepayment option allows the meter to be switched between normal Credit mode and Prepayment modes. There is another option that allows the meter to be placed in Occupied or Unoccupied state but only if the meter is in Prepayment mode.

12.1.2.3 Remote connect with MeterComm

MeterComm can now be used to remote connect to U series meter that are fitted with a suitable modem or TCP/IP connection board or modem. In MeterComm click on the Remote Communication box and enter either the meter's CSD phone number or TCP/IP address. The family type U series must be selected. The Site Description field can be used to save a name for this specific communication session. Previously saved entries are available on the Phone Book tab.

12.1.2.4 Partial Programming

MeterComm supports Partial programming of many of the U3400 features as shown below.



Auto Disconnect relates to all options shown on the AutoDisconnect GUI page of MeterDev.

Back End Communications is for Transparent Modem configuration options.

Breaker Arming Timeout is for configuring the Breaker Arming Timeout setting.

Breaker Loss of Supply is for configuring the Breaker behaviour during and after a power outage.

Breaker Maintenance Cycle allows the User to specify when the Breakers twitch operation will be performed so that any noise produced doesn't disturb the customer.

Display relates to Normal and Alternate 1 and 2 options

Event Settings as specified in the Events Settings GUI

Load Control can be used to change all the Load Control Relay State Machines features and at the same time modify the LC switch times.

Load Profile can be used to change a single Load Profile group or a combination of LP groups.

Misc is used to change the Use Crystal for Time Keeping; Reset Password Lockout on a Power outage option; the Optical Priority option; and the Export Energy warning levels for each element

Pulsed Outputs allows these features to be enabled/disabled or modified.

Quality Of Supply changes the settings shown on the QOS GUI

Relay Settings changes the 'Breaker/Relay Present' and 'Enable Arm for' settings on the Relay Settings GUI but not the 'Breaker Arming Timeout' setting or the 'Load Control Group' assignment numbers.

Supply Capacity Control can change all settings shown on the SCC GUI and also enables the mode set in the program.

TOU partial programming can be used to change all TOU related items including the number of TOU rates and registers for both energy and demand; TOU rate switch times; Recurring and Non-recurring special days.

Due to the scope of these changes and the structure of existing TOU values it will be necessary to erase all previous and current TOU values from the meter if this feature is used.

12.1.3 MeterViewer

Program to view the Load Profile data, billing information including Energy and Demand values, meter program data read produced.

The detailed reporting of the meter's functioning and non-billing data can be used for system monitoring of supply quality, for security checks and for fraud investigation.

12.1.4 EMSecure - Security

Meter access security system development program used for controlling access to the meter. The security systems are detailed at the end of this chapter.

EMSecure is not part of the EMPWin+ package. It is a standalone package.

12.1.5 Meter-specific parameters

The 2 parameters described below are generally different for each individual meter. They are therefore not set up as part of the stored meter program but are entered at the time the meter program is downloaded (part of the Write/Full Program screen entries). They do not appear in the EMPWin+ Program Developer screens (see EMPWin+ user manual – section on Write a Program to a Meter).

Serial Number

Description	The serial number of the meter. (The meter also has a manufacturing serial number, which cannot be changed).
Values	Up to 20 alphanumeric characters.

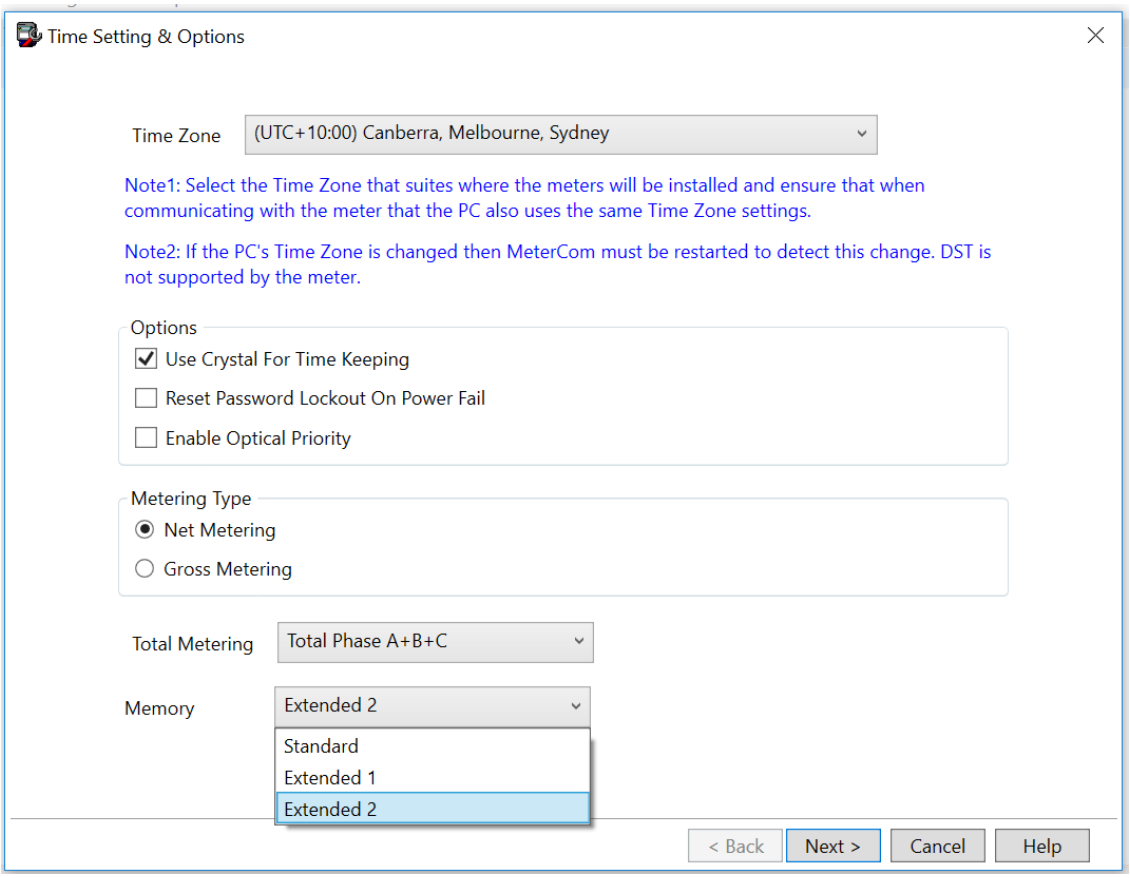
Account ID

Description	The customer's account ID or other location identifier.
Values	Up to 20 alphanumeric characters.

12.2 Configuration of the Meter

12.2.1 Time Setting & Options

The time zone selection is for the purpose of selecting the respective geographical time location for where the meter will be installed. This enables the collection software to apply time zone offsets if they are collecting data across different time zones.



Time zone

Description The drop down list is pulled down to select the appropriate time zone with respect to meter site of installation.

Default selection: (GMT+10.00) Canberra, Melbourne, Sydney.

The meter doesn't support DST

Use Crystal For Time Keeping

If unticked then the meter clock will be synchronised with the mains frequency.

Reset Password Lockout on Power Fail

Description The meter can be programmed to reset the lockout by power cycling the meter. See also section 12.3.3 “

Enable Optical Priority

Description It gives priority to the optical port in case a remote access to the meter happens concurrently. It also kicks off the remote port at "safe" points in the remote communication cycle when a local communication is establishing. Otherwise the local communication port is locked out until the remote session is finished.

Metering Type

Description This provides the option of configuring the totalling registers to be calculated in either Net or Gross Metering fashion. The functionality is described in sections 4.2 and 6. It shall be noted that Instantaneous Power Total display and load profile parameters and Pulse LED operation will remain as Net irrespective of the Metering Type chosen.

Total Metering

Description This provides the option of configuring the total registers to be calculated across A+B+C (default), or across any other combination of registers including A+B, A+C, or B+C.

Memory

Description The Memory selection gives the choice between:

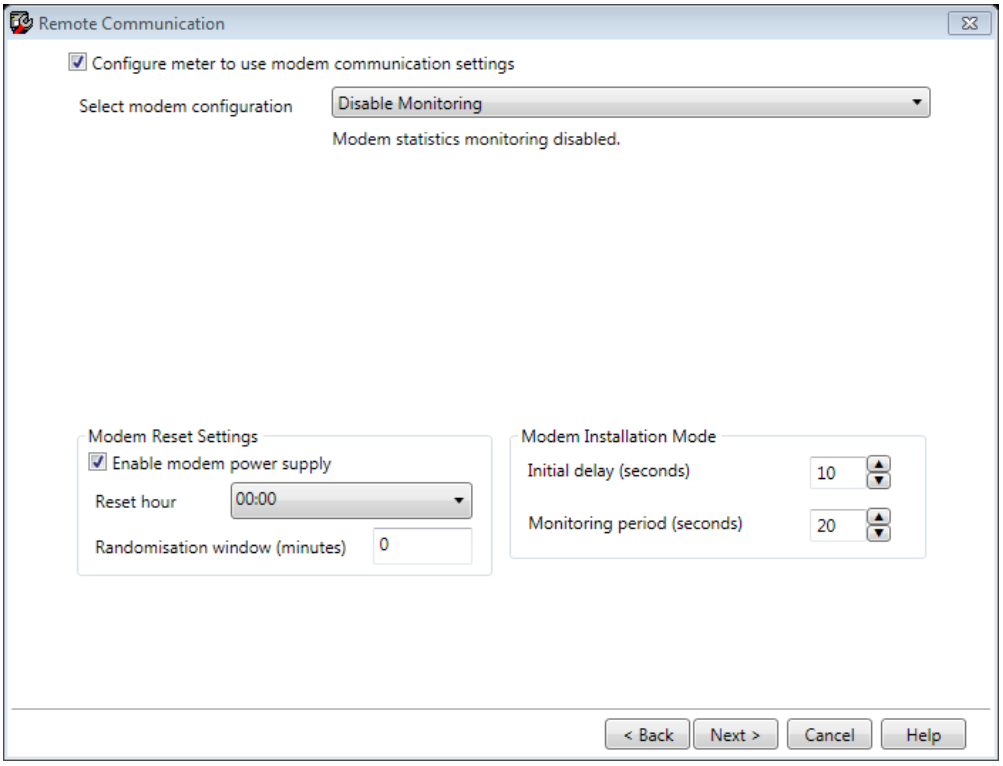
- **Standard** (4.9 F/W or earlier)
- **Extended 1** (5.0 F/W or greater)
- **Extended 2** (6.0 F/W or greater for 64Mb memory)

12.2.2 Remote Communication

This configuration is available for all U series meters that have fw version R4.0 or later fw version and is used to define the behaviour of a transparent modem (thereafter only the modem).

Communication Modules which do not have their own operating system are not considered being a modem this includes the Landis+Gyr Gridstream RF and the SSN NIC's.

Do not enable this feature when the meter is used with a communication module which has its own operating system. By doing so the specific communication module settings in the meter will be ereased and the module will no longer be able to communicate to the meter.



Configure meter to use modem communication settings

Description If the meter doesn't have a modem then the option should not be selected

If the modem option is enabled then additional RSSI display and Load Profile monitoring options can be used.

Select modem configuration

Description

There are currently three modem types that have been tested and although the default settings is the same for each meter type they have been setup as separate types just in case timing parameters need to be changed in the future. The main modem feature that is currently available is for the support of RSSI values that can be returned by the modem.

Modem Reset Settings

Description

This option is to enable the meter's port to power the modem or not and to specify a time of day when the modem's power supply can be optionally reset for 30 seconds, there is also a random number of minutes after the specified hour so that multiple modems don't all power cycle at the same time.

Modem installation Mode

Description

The modem Installation mode settings have been made user configurable so that this feature can match the meter/modem installer's normal operating practises.

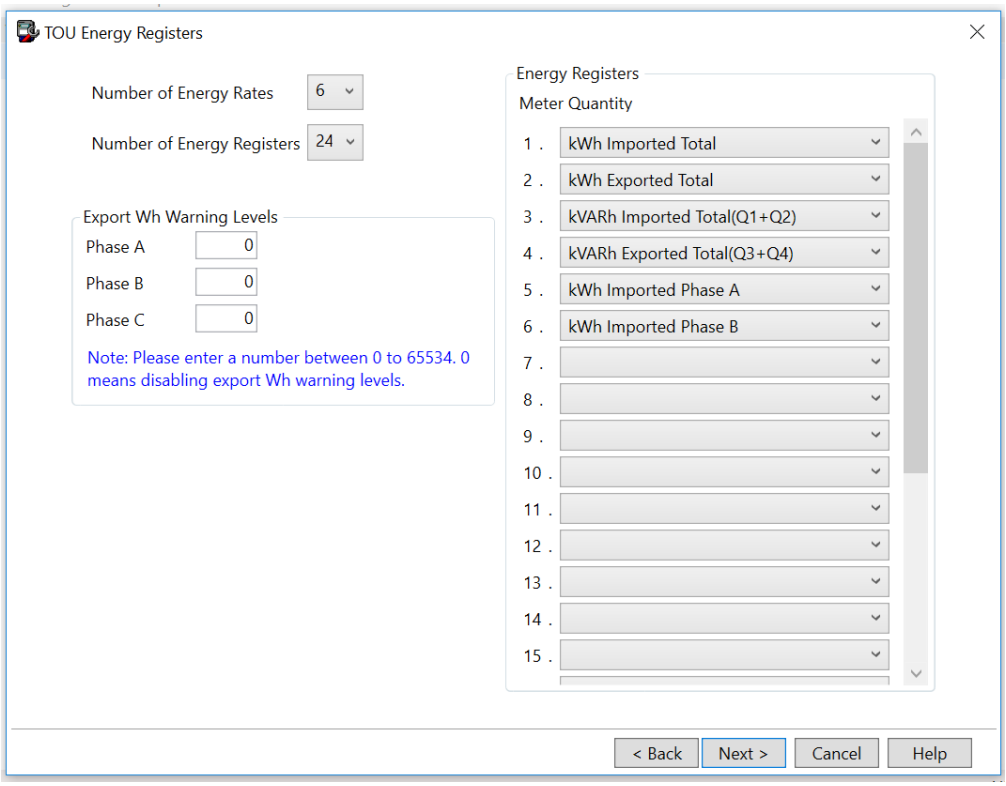
Signal strength detection at time of installation

The meter is capable to display RSSI values in one of the Alternate Displays. The intention is to allow the meter/modem installer to check the signal strength at a new location. The four available RSSI values are Instantaneous, minimum, maximum and average. These last three choices are re-calculated after the first RSSI value has been reached in the Alternate display list. The monitoring period for these calculations is determined in MeterDev by the Modem Installation Mode settings that offer from 20 to 240 seconds of monitoring. RSSI reads are attempted about every 4 seconds during the monitoring period. There is also an initial delay before the monitoring period starts; this is to allow for a meter box lid to be closed. It should be noted that the RSSI values that are read for Load Profile purposes are only read every 5 minutes and not every 4 seconds as they are in installation mode.

Modem Installation Mode display of RSSI values has four different stages. When the Alt button is pressed and the first available modem display item is shown it will be presented as C ----- (C followed by a space and 5 dashes). This will stay for the Initial Delay seconds (default is 10). The display will then change to C------ (C followed by 6 dashes) during the first part of the Monitoring period (user configurable in seconds). If during the Monitoring period the modem returns an RSSI value then the last five dashes will be replaced by the RSSI value padded with leading zeros e.g. C-00021. The first – after the C will remain until the Monitoring period expires at that time the last obtained RSSI value will be displayed (without the first dash) for up to 255 seconds or until the Alt button is pressed. If no RSSI value is obtained then the final displayed value will be C -----.

The four different RSSI values Instantaneous, minimum, maximum and average are all displayed as explained above but will have the users display id to distinguish them.

12.2.3 Energy Registers



Energy Register / Meter Quantity

Description This section of the screen allows for the selection of up to six TOU metering quantities (energy registers) for Standard and Extended 1 memory setting and 24 for Extended 2 memory setting.

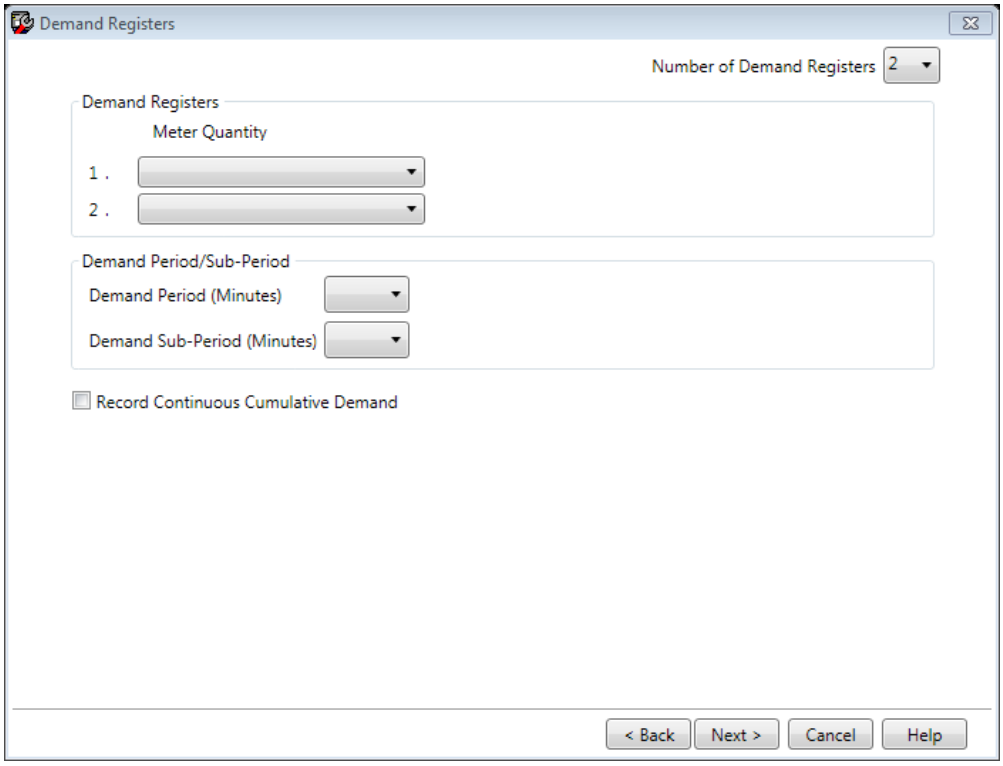
Total energy registers are always recording and are available.

Export Wh Warning Levels

Description The level at which this warning is activated is programmable.

Values 0 to 65534 (a value of 0 will disable the export warning.)

12.2.4 Demand Registers



Number of Demand Registers

Description This enable a customer to program up to two TOU demand registers in the meter.

Demand Register / Meter Quantity

Description This section of the screen allows for the selection of the metering quantities (energy registers) listed in the pull down window initiated by clicking the down arrowhead on the right hand side of each box. The Demand registers are the same as offered for TOU energy measurements.

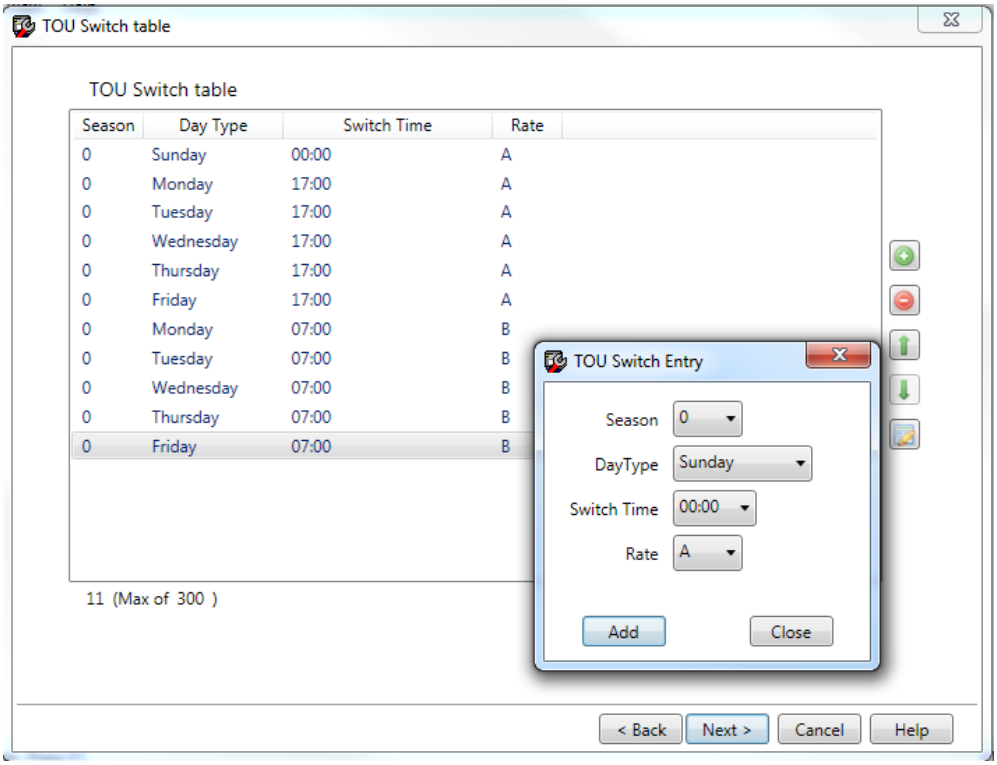
Demand period/sub-Period

Description The level at which this warning is activated is programmable.

Values The Demand measurement period can be 1, 5,10,15,30 or 60 minutes with a maximum of six sub periods e.g. 30 min with 5 minute sub periods. The Demand TOU rate switch times are the same as used for TOU Energy.

12.2.5 TOU Switch Table

Rate switch table allow for the specification of the times at which rates are switched.



Rate Switch Points

Description

Rate switch points are the times at which the tariff rates change for each combination of day type and season type.

Rate switching is controlled according to the combination of day and season type - there are thirteen day types and up to four season types (0 = same season all year).

Switching times can be specified in fifteen-minute intervals. There are six independent tariff rates at which energy consumption can be metered. A maximum number 300 entries are available.

Values

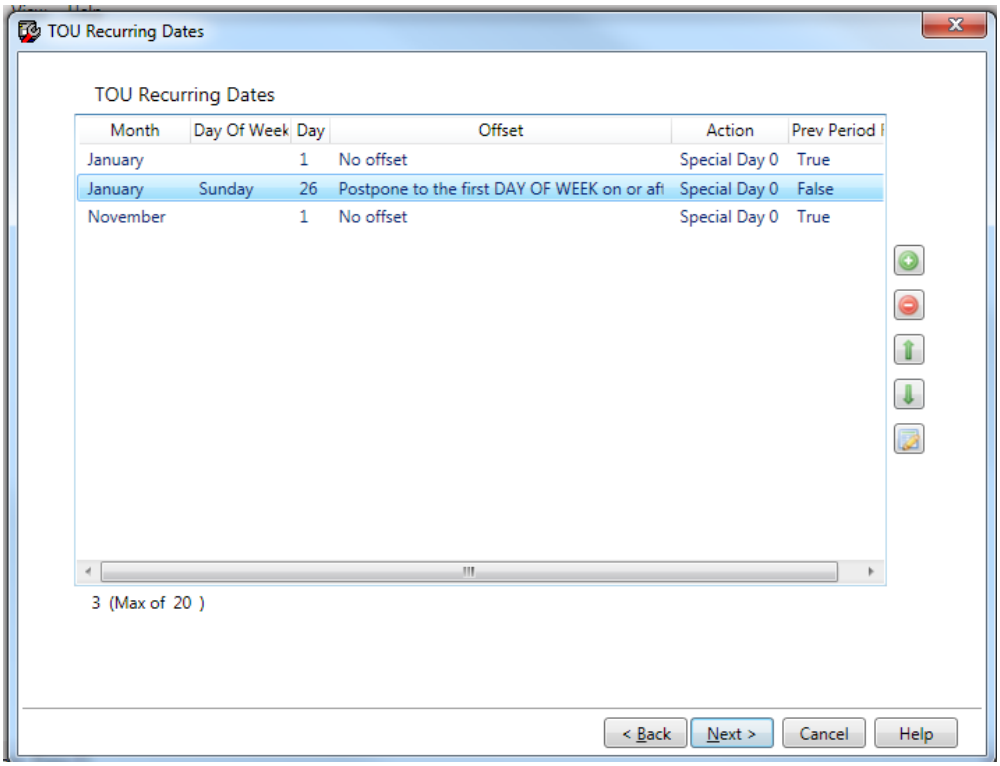
The table below details the allowable values in the rate switch points table.

Season	0, 1, 2, 3
Day Type	All Weekday, All Weekend, Saturday, Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Special 0, Special 1, Special 2, Special 3.
Switch Time	0:00 to 23:45 in 15 minute increments
Energy Rate	A, B, C, D, E, F

The meter will stay in the current rate unless specifically changed. There is no default state for midnight. Therefore, it is important to specify an appropriate rate and relay status at midnight (00:00) for each day type to ensure that the meter will start up in the correct rate.

12.2.6 TOU Recurring Dates

These are dates that occur regularly each year. The meter performs the specified action every year on the specified date (up to 20 entries can be listed).



Description

The selections in this table specify the date, the action, and if a previous period read is required.

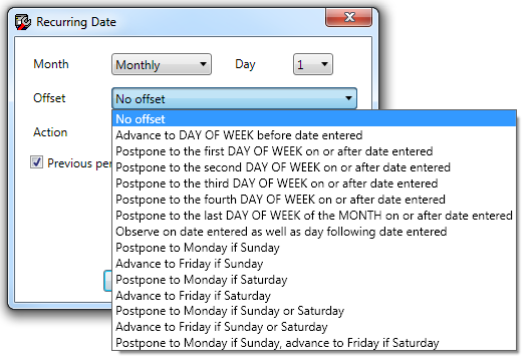
Values

The table below details the allowable values in the recurring dates list.

Recurring Dates List Values	
Month	Jan - Dec, monthly, weekly
Day	1 -31
Offset	Refer to Figure below
Action	Season change 0 to 3, Special day 0 to 3
Previous Period Read	True or False, Saves a snapshot of all current billing registers as a previous period read

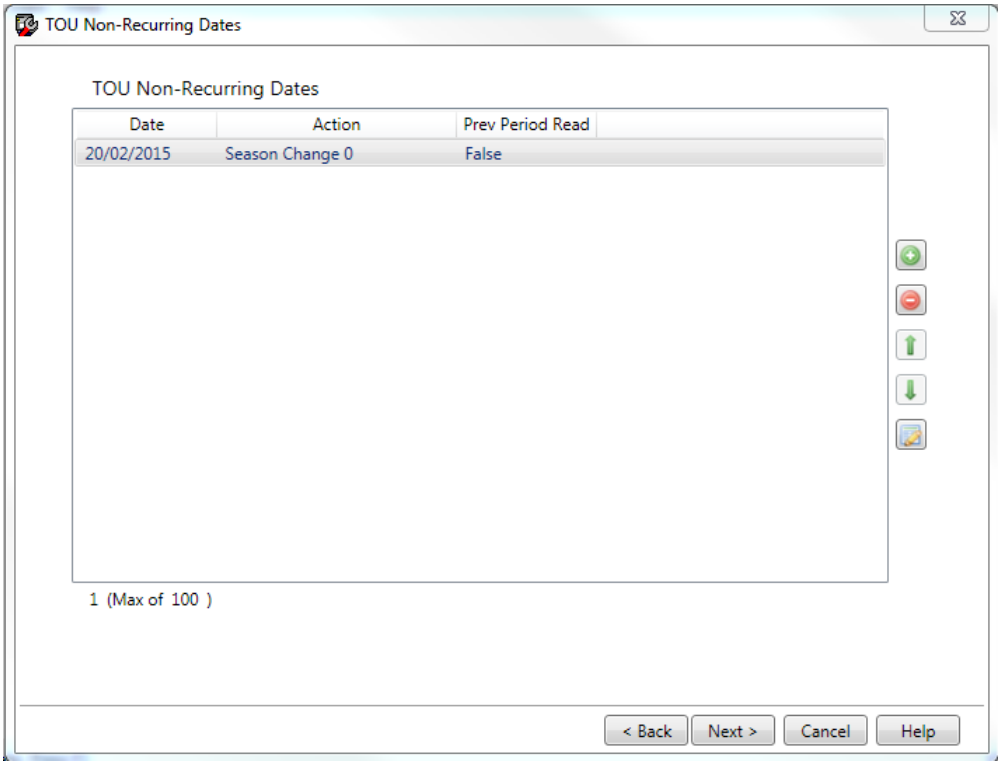
Any Special Days created should have appropriate TOU switch table entries.

Offset



12.2.7 TOU Non Recurring Dates

These are dates that do not occur regularly (for example, Good Friday). Each irregular date must be separately specified for each year (up to 100 entries can be listed). The meter performs the specified action on the specified date.



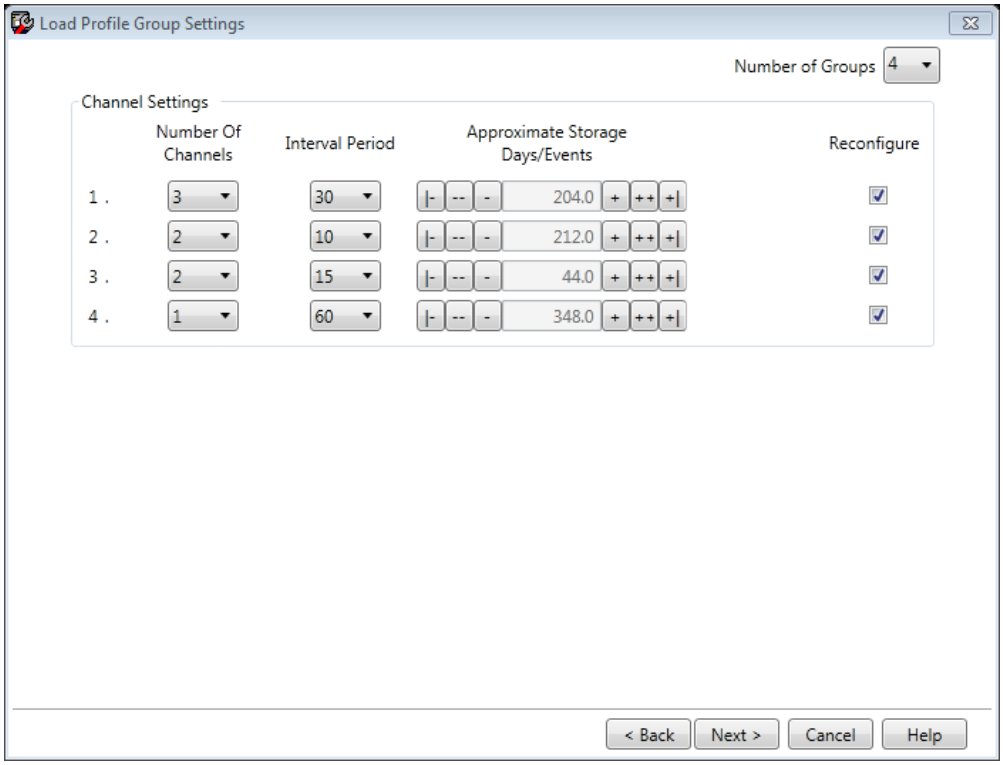
Description — The selections in this table specify the date, the action and if a previous period read is required.

Values — The values are the same as for cyclic dates except that the Year must be entered with the dates entered.

12.2.8 Load Profile Group Settings

Load Profile groups 1 to 4 have a priority where group 1 is the most important. If the memory allocated to each Load Profile group is unchanged then it is possible to change the type of Load Profile data or the number of channels or interval length without disturbing the higher priority Load Profile groups.

Partial programming a particular Load Profile group will always clear all of the existing Load Profile data for that Load Profile group. If the memory size is changed in a higher priority Load Profile group (lower group number) then all lower priority Load Profile groups (higher group numbers) will also be cleared during partial programming of the higher priority group (lower group number).



Number of Groups

Description Up to 4 Load Profile groups can be selected from the pull down list

Channel settings

Description This allows customising the individual Load Profile sets with regards to number of channels, interval periods and even the number of storage days per sets.

Number of Channels

Description This defines the number of channels for each individual Load Profile set.

- Up to 4 channels are supported for Standard and Extended 1.
- Up to 24 channels are supported for Extended 2.

Interval Period

Description

The meter supports 10, 15, 30 and 60 minute interval periods for F/W 4.9 and earlier and the addition of 5 minute intervals for F/W 5.0 and greater.

Approximate Storage Days/Events

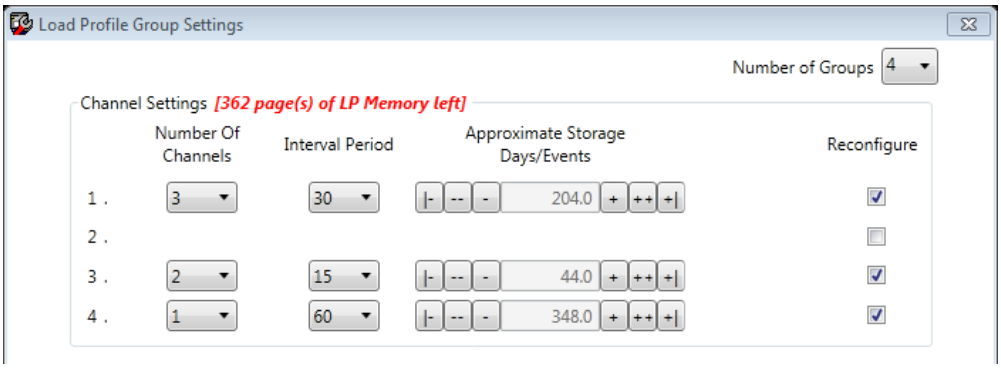
Description

Based on the settings the tool will calculate the number od storage days per Load Profile set based on the maximum available memory capacity. It is possible to adjust the Approximate Storage Days based on the principle the maximum available memory is fix. Hence to increase the Approximate Storage Days in one set another set must decreased the Approximate Storage Days first.

Reconfigure

Description

This is used to preserve a set of Load Profile for future use. As shown in the example above 4 Load Profile groups have been defined, each with a certain amount of Load Profile memory space allocated to them. Now, by un-ticking the reconfigure box on group 2 as shown below, if this meter program (*.mcd file) is used to fully program a meter then the meter will only have 3 active Load Profile groups but space (362 pages of LP memory) will be available in the meter to add group 2 at a later date (provided it has the same memory size as was originally reserved).

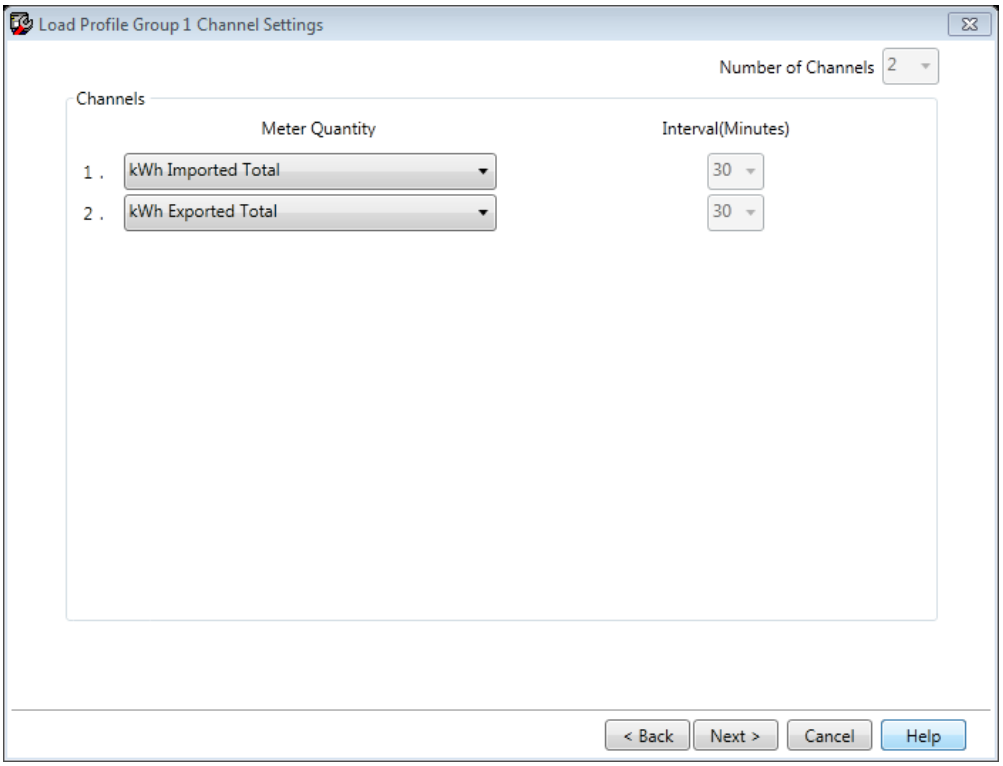


A typical use case might be to define 4 Load Profile groups but when the meter is initially programmed don't select groups 3 & 4 to be configured (un-tick Reconfigure). This program could then be modified to enable groups 3 & 4 and disable groups 1 & 2 and saved as a new program that is only used for partial programming of Load Profile groups 3 & 4. Groups 1 & 2 could be used for energy recording and groups 3 & 4 for QOS items that can be turned on or off by partially programming groups 3 and/ or 4 at appropriate times without ever running the risk of accidently restarting groups 1 or 2.

The meter will always check for Load Profile group memory requirements and use the hierarchy rules if the customer makes a mistake.

12.2.9 Load Profile Group 1 … 4 Channel Settings

The below meter program definition for GUI ‘Load Profile Group 1 Channel Settings’ will be the same for all Load Profile Group Channel Settings.



Number of channels

Description The greyed out value indicates that this has been defined on the previous GUI.

Channels / Meter Quantity

Description Each channel can record one quantity selected from the pull down list (duplication is permitted if required). A detailed list is show in section 4.2 Measured Quantities Registers plus the first eleven Instantaneous Quality Of Supply items plus RSSI.

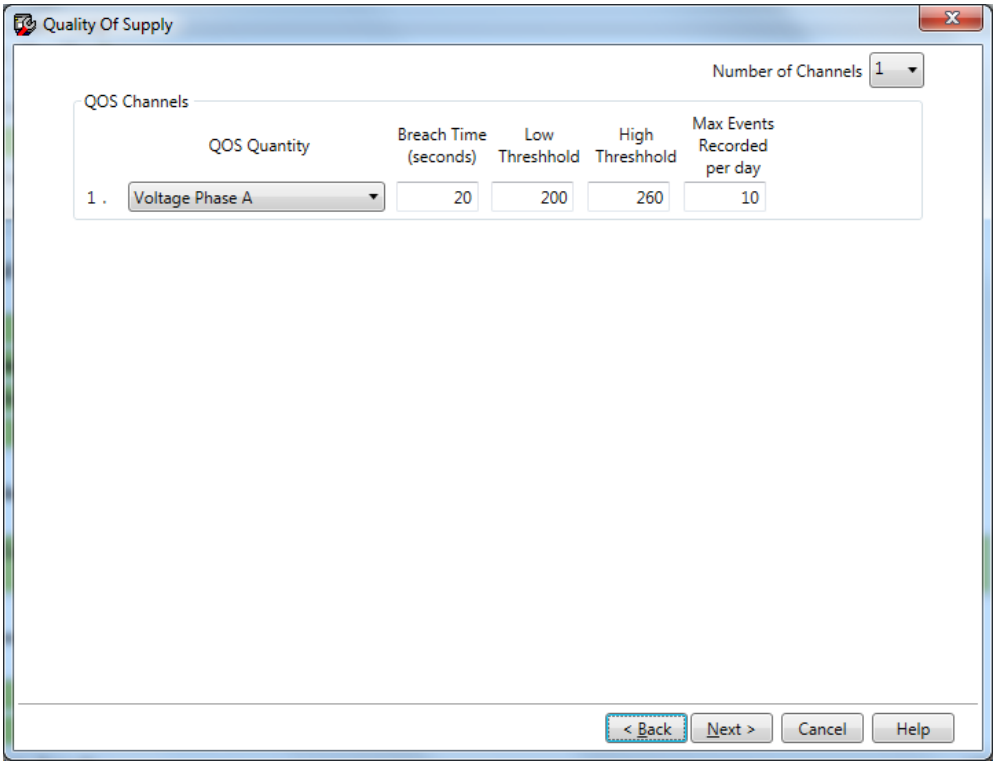
Channels / Interval (Minutes)

Description The greyed out value indicates that this has been defined on the previous GUI for this particular Load Profile set.

There is an overflow bit that gets set if 65535 is exceeded and this can be used to double the unit reporting value. This is handled automatically be EMPWin+

12.2.10 Quality of Supply

Recording for Voltage Sag and Swell, three channels are available



Minimum Duration (Breach time)

Description This value represents the minimum time in seconds that the quantity will be outside of the specified limits before it is recorded as an event. The time limit is in seconds from 0 to 255 (4.25 minutes).Typically this might be set to several seconds to avoid picking up surges.

Values 0 to 255 seconds

Low Threshold

Description Low threshold values can be entered for event recording to be triggered for excessively low voltage values. Typically this might be set to 220. If 0 is entered then the low limit will not be checked. Always set to 0 when not in use to avoid unnecessary data overwrites.

Values 0 to 300 volts

High Threshold

Description High threshold values can be entered for event recording to be triggered for excessively high voltage values. Typically this might be set to 255. If 0 is entered then the High limit will not be checked. Always set to 0 when not in use to avoid unnecessary data overwrites

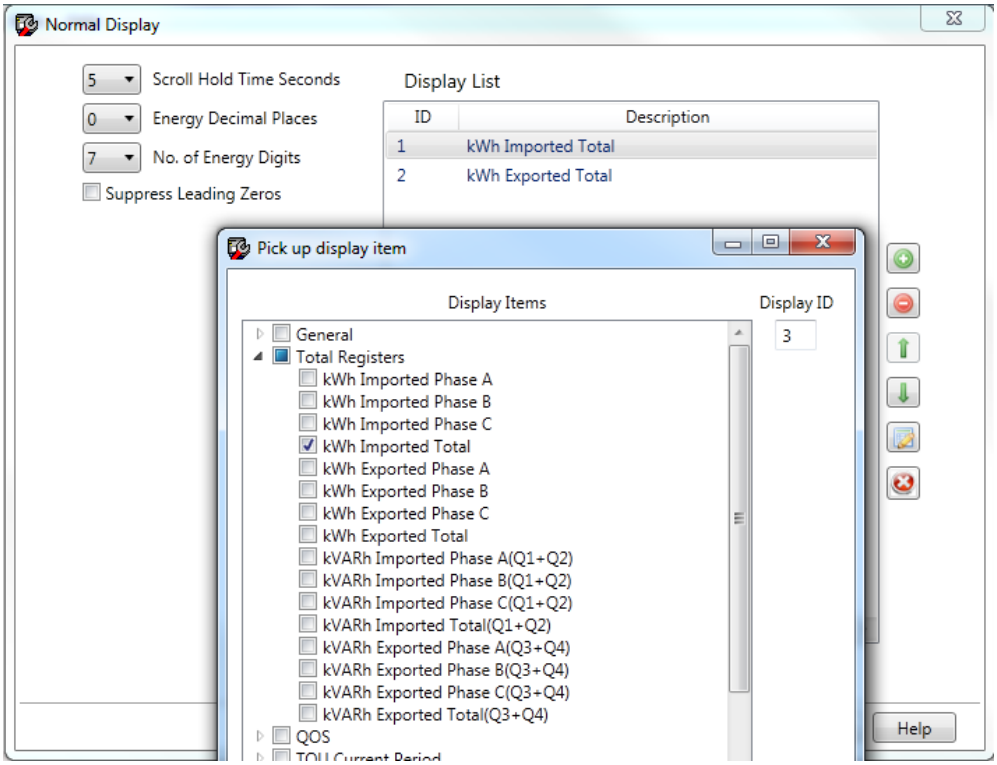
Values 0 to 300 volts

Max event recording per day

Description	Max event recording lets the user limit the number of Under and Over voltage events stored in a particular day. Typically this might be set to less that 200 to avoid over flowing the total event log of 500 in less than a day.
Values	0 to 300 Events

12.2.11 Normal Display

The programmable selections on the 'U3400 display' screen allow the user to program the LCD to display the items required in any order.



The items to include are available from the **Green Plus** icon button (bottom-most) which pops up the pick up display item. The Red minus icon button will delete the item highlighted on the display list. The up and down arrow icon are used to move the items up and down the display list. The edit button will allow for the item configuration to be edited. The red cross item will delete all items on the display list. The pickup list contains the available items while the display list contains those items added for display.

The order of the display items in the meter display cycle can be altered according to the user's choice. The display sequence of the items shown on the LCD will be as per the order in 'Display List'.

SCC Re-close Timer is only selectable under the Normal Display options.

Scroll hold time

Description The time, in seconds, for which each display option is shown on the meter display during the scroll sequence when the meter is set to automatically scroll. Typically this is set to five or six seconds.

Values Integer between 1 and 8, default is 5

Energy & Demand Decimal Places

Description The number of decimal places displayed when an energy value is displayed. The higher the number of decimal places the quicker the register readings wrap around. From EMPWin+ 2.8 onwards a check is made for a valid combination of energy digits and decimal places.

Values 0, 1, 2, 3; default is 0

No. of Energy & Demand Digits

Description The meter can display up to 7 energy digits (includes the decimal places). It is recommended that all 7 digits be used to maintain the longest period before the register rollover occurs.

It is important that customer understand that the display register settings (No. of Energy Digits plus Energy Decimal Places) is only valid for the display register and is not valid for the meter internal energy register when it comes to a register overflow. The meter internal energy register has a rollover limit of 1,000,000kWh with a resolution of 3 decimal points. As an example if the display register is programmed with 6 energy digits and one decimal point in the format 12345.6 kWh then the display register will overflow at 99,999.9kWh. However, the electronic read out of the same register will not overflow and register 100,000.0kWh. This register will overflow at 999,999.999kWh instead.

Values 5 to 7; default is 7

Suppress Leading Zeros

Description The display can be set to remove any leading zeros when an energy value is displayed.

Values Checked or unchecked; default is not checked

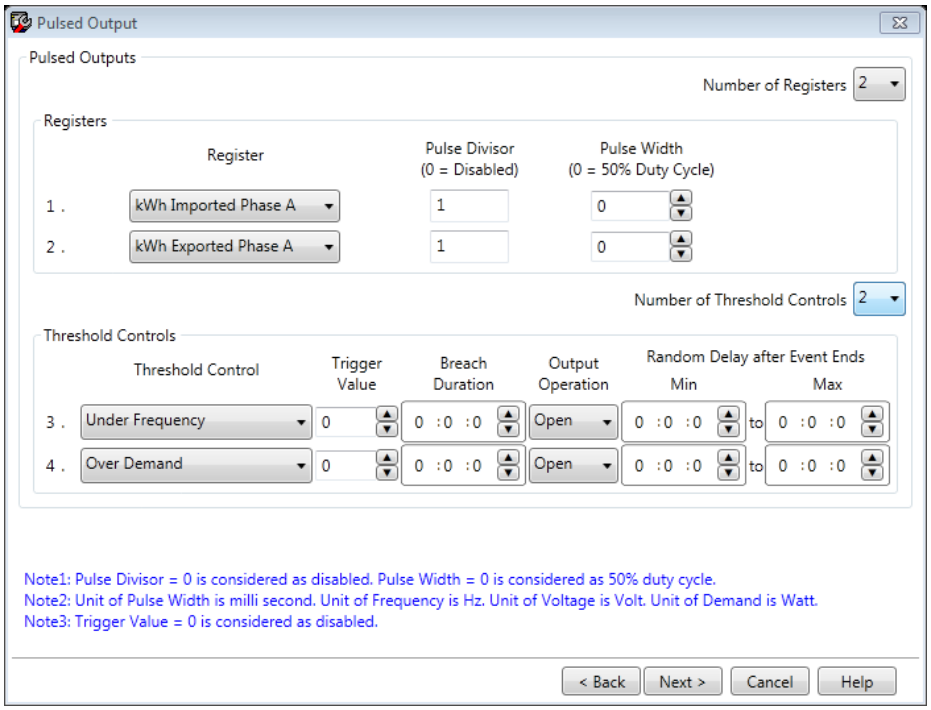
12.2.12 Alternative 1 and 2 Display

Alternative 1 and 2 display are 2 additional display screens which, when selected, are similar to the ‘Normal Display List’ with the name of the list changed to ‘Alternate 1 Display ’ or ‘Alternate 2 Display’. The pick list and the following operations are the same for both the alternative display options.The lists allow up to 20 items in the normal display list and 10 items in each of the alternate display lists.The Alternate Displays have an additional Modem group available for selection of modem related values such as RSSI values.

The Alternate 2 Display list also offers the Comms Display item that can be used by communication modules to display information on the meter's display.

12.2.13 Pulsed Outputs (Multifunction Outputs)

The meter has maximum 4 output contacts which can be allocated to Registers and Threshold Controls. The registers have the higher priority meaning whenever the Number of Registers is changed it will immediately change the number of Threshold controls accordingly.



Number of Registers

Description: Up to 4 registers can be allocated, the reminder of the 4 registers will be available for Threshold Controls.

Registers

Description: These are the register which will drive the output contact.

Values: All measured energy quantities, all End of Interval from 1 minute to 60 minute and the Tariff Rates A..F

> The End Of interval is linked to the meter internal clock with the full hour as the reference a matching Load Profile interval group does not need to exist.

Pulse Divisor

Description: Used to define the number LED pulses to the output contacts.

Values: Value 0 to 225

Where 0 means pulse output is disabled
Where 1 means 1 output pulse per 1 LED pulse

Where 255 means 1 output pulse per 255 LED pulses

Pulse Width

Description Used to define the pulse width in milli seconds

Values Value 0 to 1000msec in steps of 20msec

Where 0 means 50% duty cycle

Number of Threshold Controls

There are three threshold conditions that can be used to trigger an output contact operation if any of the enabled condition are detected. A condition must exist for at least the minimum breach duration specified. All three conditions have a random delay period which is applied after the condition has ended before the output contact is switched back to is previous state.

Description Up to 4 Threshold Controls can be allocated depending on the number of registers already used.

Threshold Controls

Description These are the instantaneous quantities which will be monitored and which will trigger the output contact if the programmed threshold as well as the breach duration is expired.

Values Under Frequency, Under Voltage for each phase and Over Demand.

Trigger Values

Description If this value is exceeded either over or under depending on the instantaneous quantity selected then the breach duration will start counting down. In case the monitored quantity gets back to normal before the breach duration has counted out then the breach duration is reset too.

Values 0 = disabled

Quantity	Range	Steps
Under Frequency	0 – 49.9Hz	0.1Hz
Under Voltage	0 – 280V	0.1V
Over Demand	0 – 72000W	1W

Breach Duration

Description This is the duration for which the monitored quantity must constantly be either over or under the programmed threshold before the output contact will change its status.

Values From 0 seconds to 1hour.

Output Operation

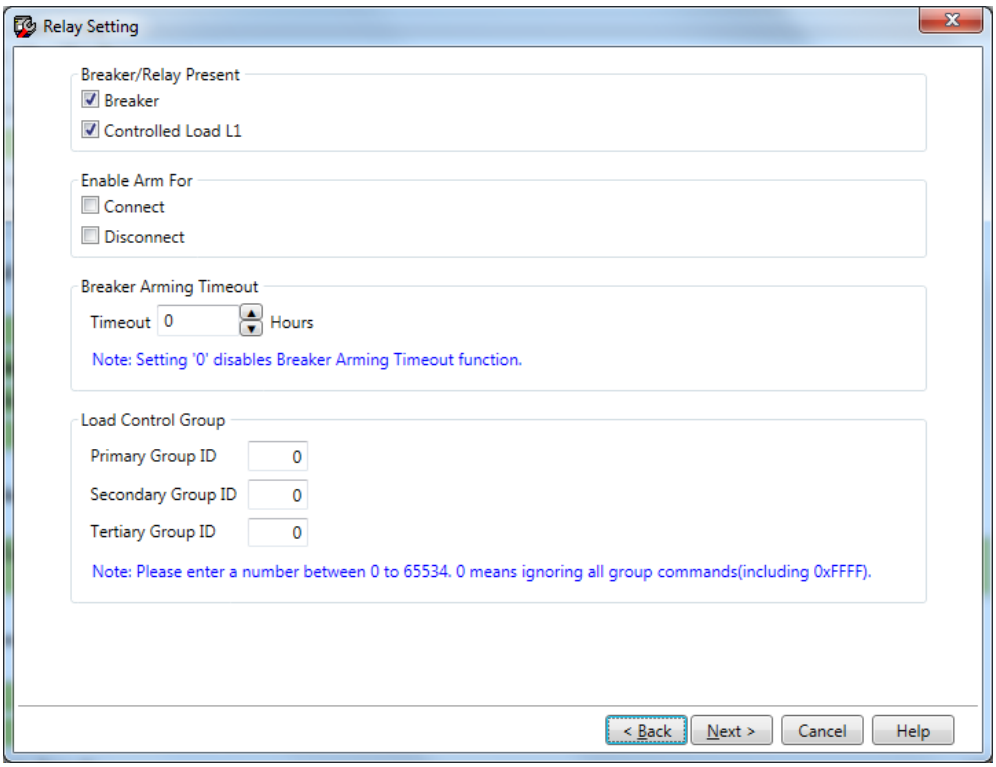
Description	Each relay can be programmed to Open, Close or be Disabled (Disabled means no change will occur on the physical relay however, an event will be logged). Normally relays would be programmed to Open during an Emergency Event. The initial relay state is the opposite of the relay action which will be performed when the event occurs. If the action is “disabled” then the relay output is in open state.
Values	Open, Closed, Disabled.

Random Delay after Event Ends

Description	When the condition has ended (eg. Frequency returned back to normal) then the relay will stay in the position until this randomly calculated delay period has expired. In case the condition is still present at the time the random delay duration expires, then the random delay timer is reset and starts again.
Values	The random duration is from 0 seconds to 18h in steps of 1 second.

The threshold control outputs will be initiated at each power restore and after this function is been reprogrammed. Meaning, all relays will be set into initial state and all delays will be reset.

12.2.14 Relay Setting



Breaker/Relay Present

Description Enables breaker and load control relay functionality.

The meter will ignore all breaker or relay commands, both internal & external, if the box is not ticked.

When changing these settings the breaker or relay will remain in the position they were at the time of the program download.

Enable Arm

Description Enable arming of the disconnect breaker. If arming is enabled, pressing and holding button 2 on the meter is required to complete the remote operation of the disconnect breaker.

Breaker Arming Timeout

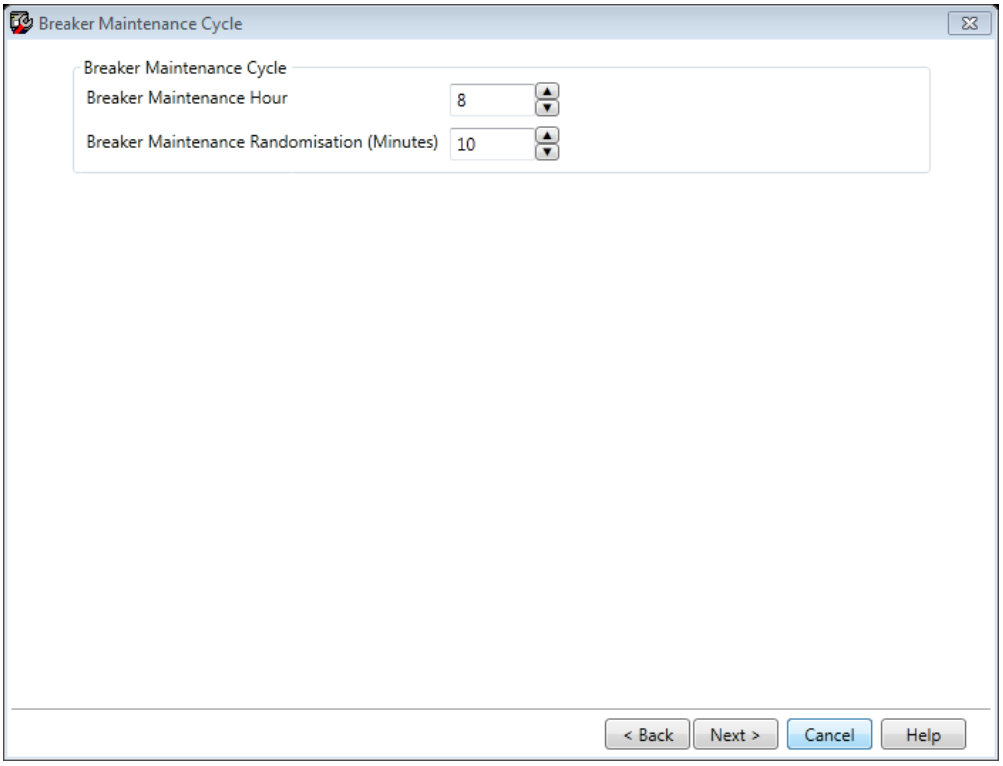
Description There has been an option added in this F/W release to allow the Breaker ARMed state to be automatically terminated if the Alt2 button has not been used to complete the Breaker operation within a specified time period. The Breaker Arming Timeout value is in the Relay Settings page of MeterDev. If 0 is selected then Breaker Arming operation will be the same as in all previous F/W builds which means the Alt2 button must be pressed to complete the Breaker operation. The timeout can be set in hours from 0 to 48. If a power outage occurs during the Arming timeout period then on power up the remaining time will be reduced by the outage duration period (provided the outage was for at least one minute). If the ARMing timeout period expires then an event will be logged and the meter display. returns to the Normal Display list in the meter**. The Breaker will remain in the same state as it was prior to the ARMing command being sent to the meter.**

Sending the same Breaker operation command while in the ARMed state, will cause the ARMing timeout period to restart again. If the opposite Breaker command is sent then ARMed state is terminated without the Breaker switching. If the Breaker ARMing timeout value is Partially Programmed then any currently active ARMed Breaker action will be terminated without the Breaker switching.

Load Control Group

Description This will assign the meter to an initial control group (Load Control Groups are used for broadcast commands to control the LC relays in the grouped meters with a single remote communication request).

12.2.15 Breaker Maintenance Cycle



Description Breaker Maintenance Cycle allows the user to specify approximately when the Breakers twitch exercise is performed. The twitch exercise makes three audible sounds once a day so the meter can be programmed to make these occur at a time that will not alarm the customer.

Breaker Maintenance Hour

This is the offset of hour counted from midnight when the meter will perform the Maintenance Cycle.

Values 0 to 24 in hours

Breaker Maintenance Randomisation (Minutes)

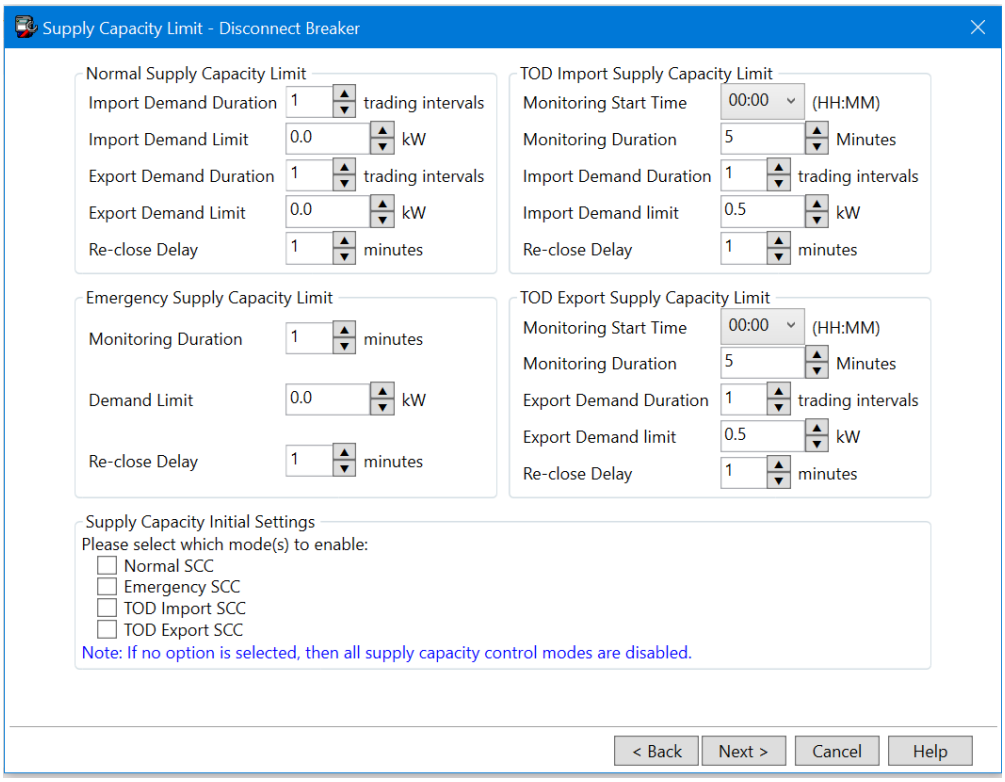
Within this random delay starting from the Maintenance Hour meter will perform the Maintenance Cycle.

Values 2 to 59 minutes in steps of 1minute

12.2.16 Supply Capacity Limit

Supply Capacity Control

The meter can configure four supply capacity limit settings – a normal limit (includes imported and exported energy), TOD import, TOD export and an emergency limit.



Any program change (full or partial) will reset all timer values for trading intervals and demand duration in order to maintain correct operation after program has been changed.

Normal supply capacity limit

The normal supply capacity limit includes limits for exported energy and imported energy. The parameters included for each of the energy are demand limit, demand duration in 30 minute trading periods, reclose time.

Import Demand Duration

Description — The number of 30 minute trading periods, synchronised to the clock hour and half hour, during which the average imported KW demand is calculated.

Values — 1 to 10, increments of 1 trading period synchronised to the hour and half hour

Import Demand Limit

Description — The imported energy demand limit above which the supply contactor (disconnect breaker) opens. The Demand limit is checked every minute within the current 30 minute Trading period.

Values — 0 to 99 kW in increments of 0.5 kW

Export Demand Duration

Description — The number of 30 minute trading periods during which the average exported KW demand is calculated.

Values 1 to 10, increments of 1 trading period synchronised to the hour and half hour

Export Demand Limit

Description **I**f the meter's total measured load in watts exceeds this specified value then the disconnect breaker will open.

Values 0 to 99 kW in increments of 0.5 kW

Reclose Time

Description On exceeding the normal supply capacity demand limit for either import or export load limits, the (disconnect breaker) opens for the specified time period (in minutes) and then closes again. This time period during which the supply contactor (disconnect breaker) remains open before it recloses is referred as the reclose time.

Note: There is a new display option that has been made available to support the SCC Re-close Timer countdown value. It is only made available in the Normal Display list. It is shown as either 'SCC t' or with the remaining Breaker reclose time in hh.mm.ss format.

When the Breaker has been opened due to a Supply Capacity Control operation the meter will now display how long before the Breaker will reclose. The displayed item is a clock countdown value and when it reaches 00.00.00 the Breaker reclose operation will start.

The SCC t (SCC Remaining Time display) item must be manually selected to be part of the Normal Display list. If it is selected to be part of the Normal Display list, when it is not activated it will be displayed as 'SCC t' in the Normal Display.

If SCC opens the Breaker then the next time the Normal Display reaches the SCC t, the display will lock on this and the remaining time before the Breaker recloses will be continually displayed. Manual scrolling through the other display items is still supported even while the countdown is running.

Values 1 to 60, in 1 minute increments

TOD Import supply capacity limit

The TOD import supply capacity limit includes limits for imported energy. The parameters included for this function are monitoring start time, monitoring duration, demand limit, demand duration in 30 minute trading periods, reclose time.

Monitoring Start Time

Description The time of day (TOD) when supply capacity control monitoring will start.

Values in HH:MM in with 30 minute steps.

Monitoring Duration

Description	The duration starting from the time of day (TOD), for which supply capacity control monitoring is active.
Values	in minute (5 to 1440) in steps of 5 minutes.

Import Demand Duration

Description	The number of 30 minute trading periods, synchronised to the clock hour and half hour, during which the average imported KW demand is calculated.
Values	1 to 10, increments of 1 trading period synchronised to the hour and half hour.

Import Demand Limit

Description	The imported energy demand limit above which the supply contactor (disconnect breaker) opens. For F/W version R3.0 onwards the Demand limit is checked every minute within the current 30 minute trading period.
Values	0 to 99 kW in increments of 0.5 kW.

Reclose Time

Description	On exceeding the normal supply capacity demand limit for either import or export load limits, the supply contactor (disconnect breaker) opens for the specified time period (in minutes) and then closes again. This time period during which the supply contactor (disconnect breaker) remains open before it recloses is referred as the reclose time.

TOD Export supply capacity limit

The TOD export supply capacity limit includes limits for exported energy. The parameters included for each function are monitoring start time, monitoring duration, demand limit, demand duration in 30 minute trading periods, reclose time.

Monitoring Start Time

Description	The time of day (TOD) when supply capacity control monitoring will start.
Values	in HH:MM in with 30 minute steps.

Monitoring Duration

Description	The duration starting from the time of day (TOD), for which supply capacity control monitoring is active.
Values	in minute (5 to 1440) in steps of 5 minutes.

Export Demand Duration

Description	The number of 30 minute trading periods during which the average exported KW demand is calculated.
Values	1 to 10, increments of 1 trading period synchronised to the hour and half hour.

Export Demand Limit

Description	**I**f the meter's total measured load in watts exceeds this specified value then the disconnect breaker will open.
Values	0 to 99 kW in increments of 0.5 kW

Reclose Time

Description	On exceeding the normal supply capacity demand limit for either import or export load limits, the supply contactor (disconnect breaker) opens for the specified time period (in minutes) and then closes again. This time period during which the supply contactor (disconnect breaker) remains open before it recloses is referred as the reclose time.

Emergency Supply Capacity Limit

The emergency supply capacity limit for Import energy includes parameters such as demand limit, demand duration, reclose time.

Import Demand Duration

Description	The time period (in minutes) across which the average kW of imported energy is calculated.
Values	1 to 60, increments of 1 minute

Imported Demand Limit

Description	The emergency supply capacity limit required to calculate the average kW of energy imported from the network across a certain time period, above which the supply contactor (disconnect breaker) opens.
Values	0 to 99 kW in increments of 0.5 kW

Reclose Time

Description	On exceeding the emergency supply capacity demand limit, the supply contactor (disconnect breaker) opens for the duration of the specified time period (in minutes) and then recloses again. This time period during which the supply contactor remains open before it closes again is referred to Reclose Time.

Note: There is a new display option that has been made available to support the SCC Re-close Timer countdown value. It is only made available in the Normal Display list. It is shown as either ‘SCC t’ or with the remaining Breaker reclose time in hh.mm.ss format.

When the Breaker has been opened due to a Supply Capacity Control operation the meter will now display how long before the Breaker will reclose. The displayed item is a clock countdown value and when it reaches 00.00.00 the Breaker reclose operation will start.

The SCC t (SCC Remaining Time display) item must be manually selected to be part of the Normal Display list. If it is selected to be part of the Normal Display list, when it is not activated it will be displayed as 'SCC t' in the Normal Display.

If SCC opens the Breaker then the next time the Normal Display reaches the SCC t, the display will lock on this and the remaining time before the Breaker recloses will be continually displayed. Manual scrolling through the other display items is still supported even while the countdown is running.

Values 1 to 60, in 1 minute increments

Supply Capacity Initial Setting

Any of the four settings can be enabled or disabled independently.

.

Supply Capacity Initial Settings
Please select which mode(s) to enable:
- [ ] Normal SCC
- [ ] Emergency SCC
- [ ] TOD Import SCC
- [ ] TOD Export SCC

Note: If no option is selected, then all supply capacity control modes are disabled.

12.2.17 Auto Disconnect

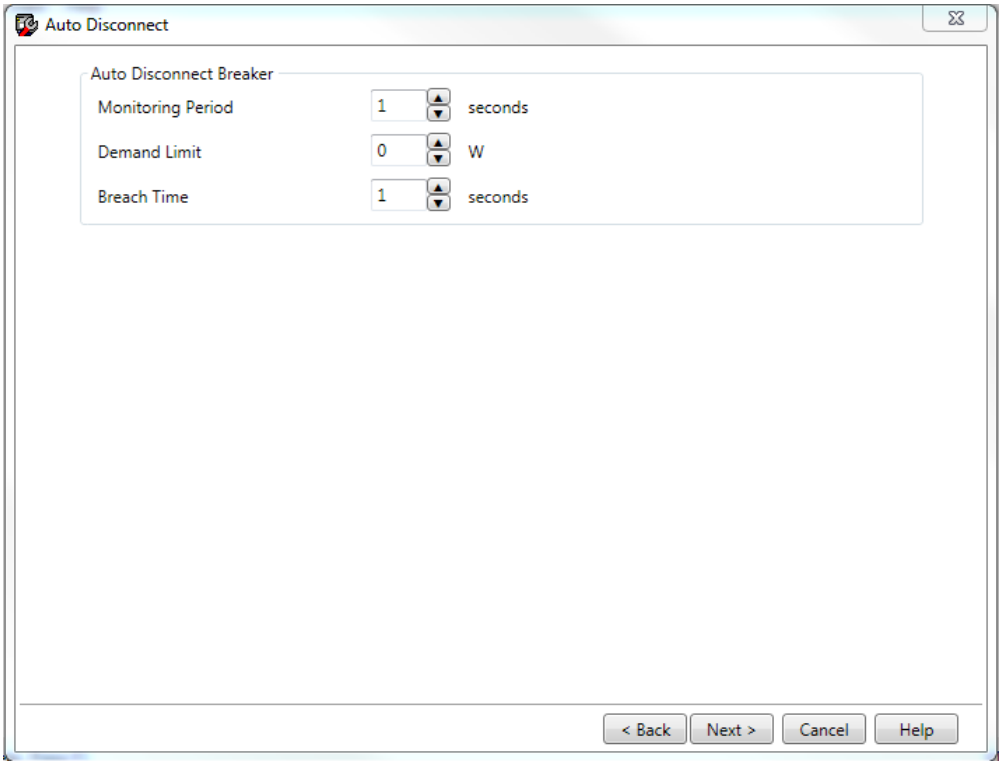
The auto-disconnect (AD) functionality of the breaker is activated as a conditional sequence of an external reconnection command. This function requires a remote breaker reconnect to start the Watts import load versus limit checking.

If the Watts specified in the program exceed the limit for the length of the programmed breach period (during the monitoring period after a remote Breaker reconnect) then the breaker will disconnect, otherwise the breaker will remain connected and the process will terminate at the end of the monitoring period.

If a power outage occurs during the monitoring period and the breaker is still connected then on power restore the process will start again as if the reconnect command was just received.

If the breaker is disconnected due to the auto-disconnect trigger and then an outage occurs the breaker will remain disconnected when the power is restored and a new external reconnect command will need to be sent to restart the process.

The auto-disconnect process when initiated, takes precedence over the Supply Capacity Control processes defined above.



Monitoring Period

Description	The monitoring period is the time after a remote reconnection has occurred.
Values	1 to 3600 seconds, increments of 1

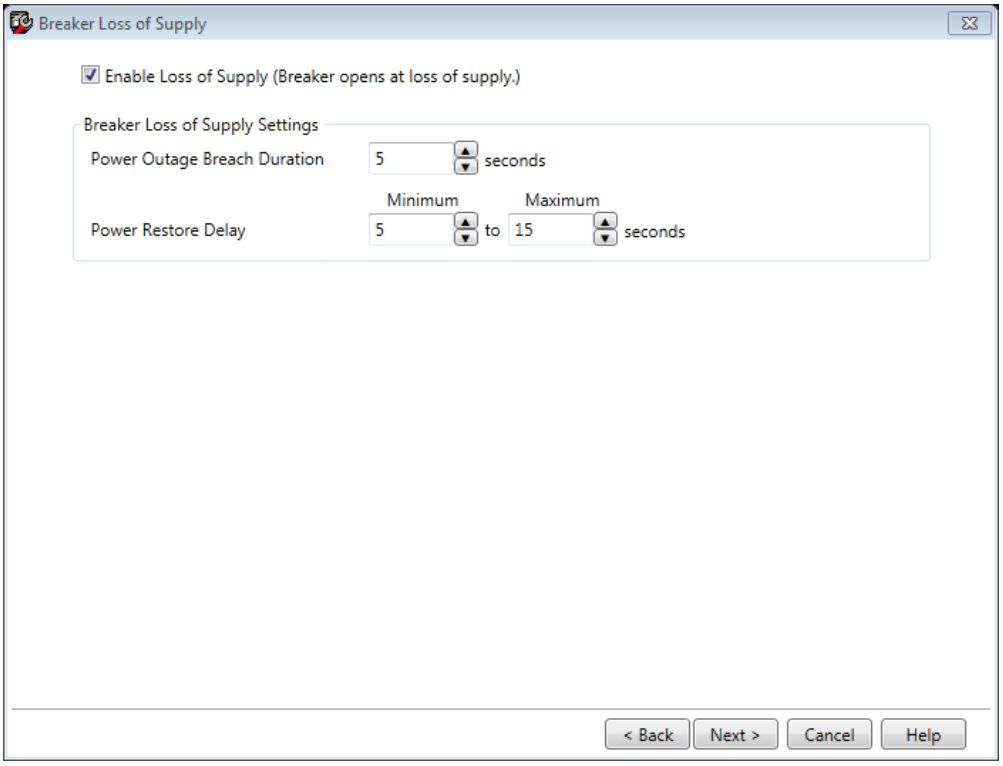
Demand Limit

Description	Demand limit (Watts). If the watts exceed this specified value the Breaker will be disconnected.
Values	0 to 2500 W in increments of 20, if set to 0W this disables the function

Breach Time

Description	This value represents the minimum time in seconds that the Quantity will be outside of the specified limits before the disconnect occurs.
Values	1 to 3600, in 1 second increments

12.2.18 Breaker Loss of Supply



Enable Loss of Supply

Description The meter supports a Loss of Supply (LoS) option to switch the Breaker to the open position at power down if this box is ticked.

Power Outage Breach Duration

Description If the power outage exceeds this breach duration then the Power Restore delay will apply otherwise the Breaker will close at power restore without using the programmed delay.

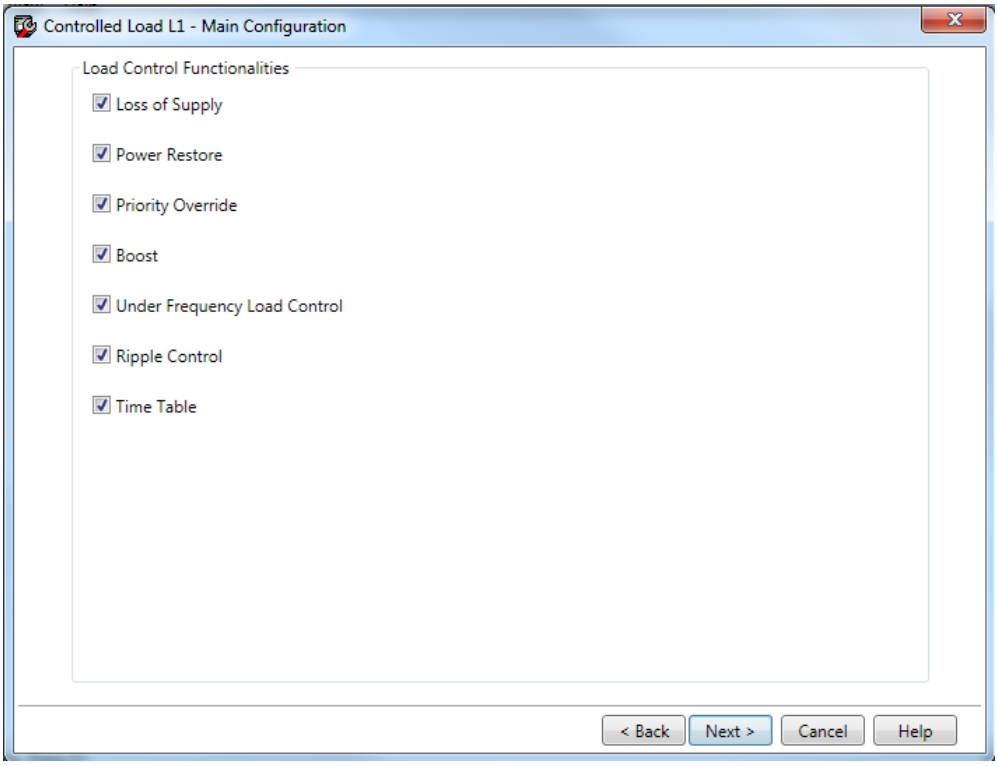
Power restore delay

Description The meter calculates a random delay between the min and max value at each power restore which is then applied to the power outage breach duration before the meter closes the breaker.

A fixed delay is added to the override period by programming the same delay for minimum and maximum.

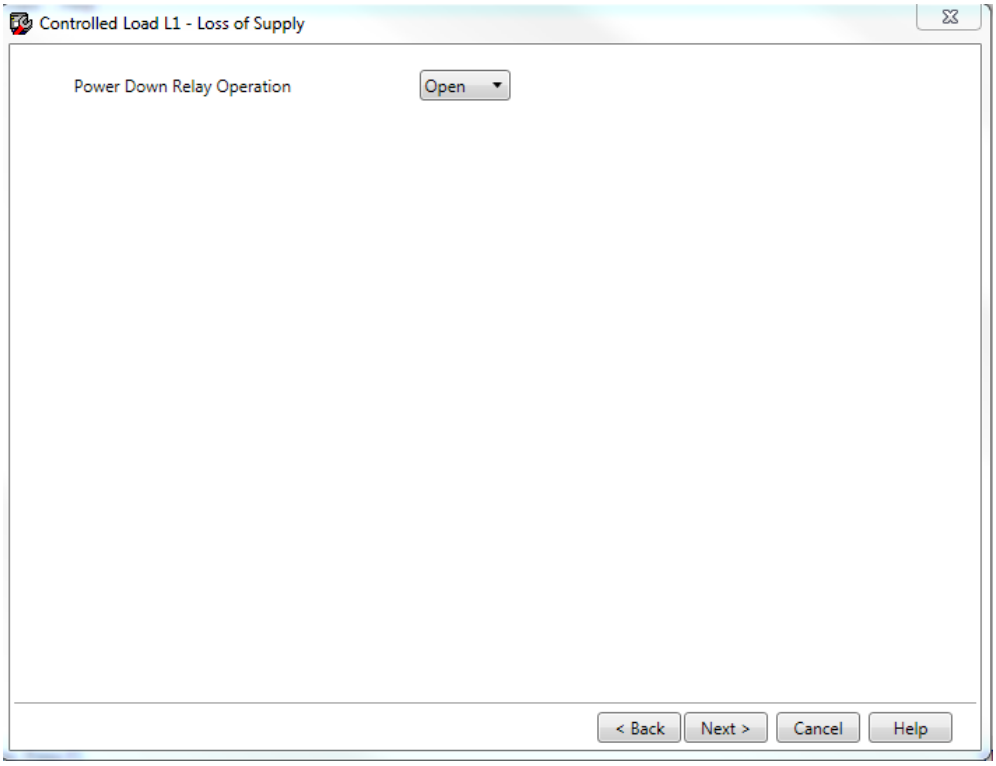
Setting both values to zero (0) results in the Breaker closing as quickly as it can after the power is restored.

12.2.19 Controlled Load L1 – Main Configuration



Description	This is the main screen of the Load Control Relay State Machine as described in 4.8 Load Control Relay It is used to program the RSM. These settings are only valid for the 1st load control relay.
Values	✓ (enabled) or blank (disabled), if a module is disabled it will not appear in the program sequence.

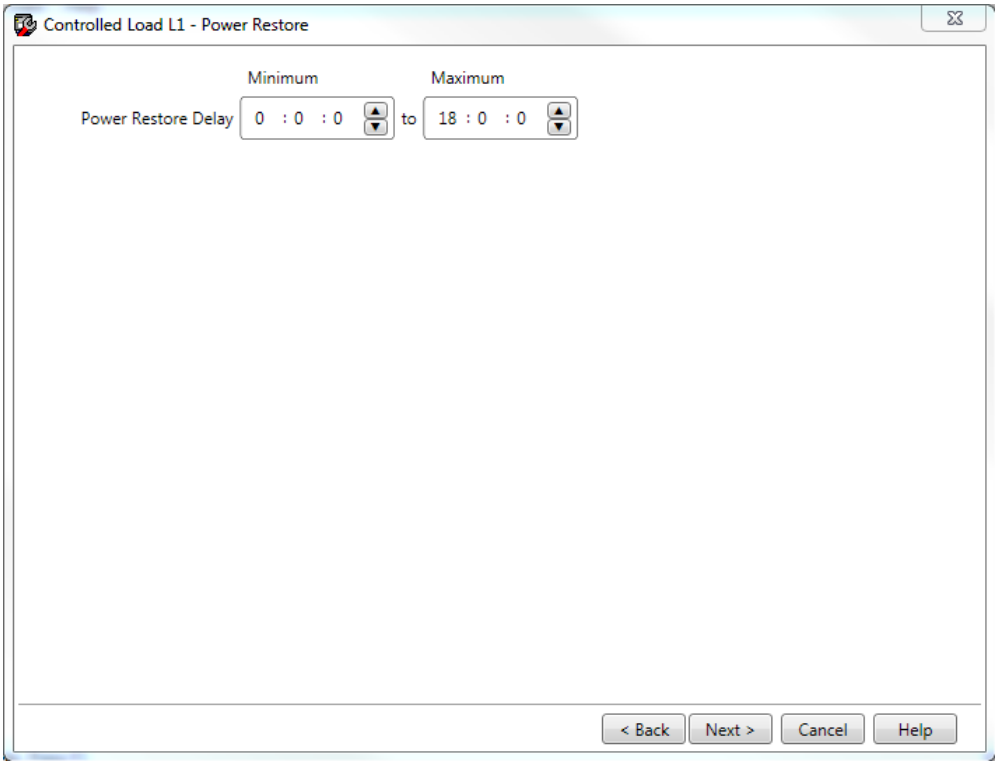
12.2.20 Controlled Load L1 – Loss of Supply



Power Down Relay Operation

Description	If a power outage occurs then the behaviour of the load control relay can be set as below. Refer to 4.8.7 Loss of Supply or Disconnect
Values	open or closed

12.2.21 Controlled Load L1 – Power Restore



Minimum Power Restore Delay

Description	This is the minimum delayed duration in which the relay maintains its programmed position.
Values	hh:mm:ss in increments of 1 unit (max value 18:00:00)

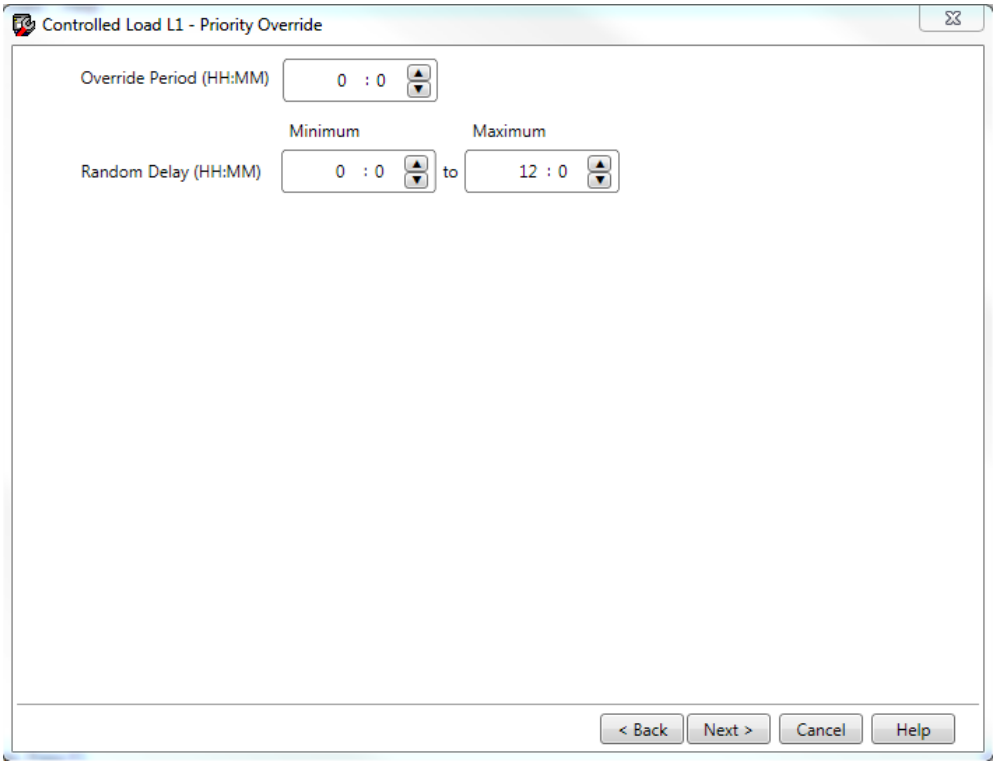
Maximum Power Restore Delay

Description	The meter calculates a random duration between the min and max value at each power restore. Once that random duration expires, the module will release control of load control relay immediately. However, this is the maximum duration the in which the relay could maintains its programmed position.
Values	hh:mm:ss in increments of 1 unit (max value 18:00:00)

A fixed delay is set by programming the same delay for Minimum and Maximum.

Setting both values to zero (0) results in an immediate action, no delay.

12.2.22 Controlled Load L1 – Priority Override



Override Period

Description	The duration the relay will remain in the commanded state. If 0 is selected then the relay cannot be switched by the override command.
Values	hh:mm in increments of 1 minute (max value 48:00)

Minimum Random Delay

Description	This is the minimum delayed duration which is added to the override period in which the relay maintains its programmed position.
Values	hh:mm in increments of 1 minutes (max value 12:00)

Maximum Random Delay

Description	The meter calculates a random duration between the min and max value at each power restore. This random delay is added to the programmed position. Once that random duration expires, the module will release control of load control relay immediately. However, this is the maximum duration which could be added to the override period in which the relay could maintain its programmed position.
Values	hh:mm in increments of 1 minutes (max value 12:00)

A fixed delay is added to the override period by programming the same delay for minimum and maximum.

Setting both values to zero (0) results in an immediate action after the override period expired.

Priority Override Command Support in R4.9

Definition

'Priority Overide' is a command, that the remote head end system (HES) can send to the meter. Hence NSM would need to have the Priority Override command integrated so that it can be used with L+G E350 meters. EMPWin+ can also be used to send these commands to the meter when connected to it locally or remotely.

Implementation

In addition to the Turn On and Turn Off commands, R4.9 firmware supports following new commands for Priority Override. The following table shows the legacy supported command and the new commands supported by R4.9 firmware, along with the supported parameters that can be sent with the command.

S.No.	Command	No Parameters	Start Date/Time	Duration	Min Randon Delay	Max Random Delay	Cycle On time	Cycle Off time
A.	Legay Turn ON	√						
B.	Legay Turn OFF	√						
1.	Turn On Immediately		√ (Specify date and time. Option available to do it immidiately)	√ (Specify from 1 to 2880 minutes)	No delay	No delay		
2.	Turn Off Immediately		√ (Specify date and time. Option available to do it immidiately)	√ (Specify from 1 to 2880 minutes)	No delay	No delay		
3.	Delayed Turn On (using Switch Time Randomisation given in command)		√ (Specify date and time. Option available to do it immidiately)	√ (Specify from 1 to 2880 minutes)	√ (Specify Min delay)	√ (Specify Min delay)		
4.	Delayed Turn Off (using Switch Time Randomisation given in command)		√ (Specify date and time. Option available to do it immidiately)	√ (Specify from 1 to 2880 minutes)	√ (Specify Min delay)	√ (Specify Min delay)		
5.	Start load cycling		√ (Specify date and time. Option available to do it immidiately)	√ (Specify from 1 to 2880 minutes)	Meter Timetable to be used for min randon delay.	Meter Timetable to be used for max randon delay.	Meter Timetable to be used for cycel On time.	Meter Timetable to be used for cycel Off time.
6.	Terminate priority override (to release control by command 1,2,3,4,5 above)					√ (between 0 and 120 min)		

Empwin+ (EMComm.exe) supports the operation of these commands from the Tool Box Menu. Refer to **Error! Reference source not found.** f or more details.

New Priority Override commands do not refer to the the meter settings for duration and random delay. These parameters have to be sent along with the Priority Override command as per the command table shown above.

There is no command for Stop Load Cycling, because stop load cycling is synonymous to 'putting relay in one state (On or Off) for the duration of priority override'. Hence Command S.No. 1,2,3,4 in the above table can be used to stop load cycle but retain control of the RSM. Command S.No. 6 (Terminate Priority Override) can also be used to stop load cycle and release control of the RSM.

The difference in Command S.No. 1 & 2 with Command S.No. 3 & 4 is that S.No. 1 & 2 commands do not use any random delay, whereas S.No. 3 & 4 commands use random delay.

Command S.No. 5 (Start Load Cycle) will refer to the values defined in the meter cycler (in Load Control Time Table of the meter program) for Min / Max Delay (T1) and Cycle On / Off duration (T2/T3). Hence in R4.9 firmware time table module of the RSM cannot be disabled. Having dependency on the meter's cycler, allows the user to broadcast a single command for different cycle requirement (as set in meter's Load Control Time Table).

The meter will not retain any information of the Priority Override command, if a power cycle happens before, or during the priority override start date/time.

If the meter is in U1200 compatibility mode (Section 12.2.19), then the new priority override commands S.No. 1 to 6 will not be supported by the meter, however it will retain the behaviour for the legacy commands as per R4.5 firmware version.

After expiry of Override duration specified in the Priority Override command, meter shall release control of the CLC/R, and let the RSM take over the switch operation.

Operation of the Terminate Priority Override command

a) If the Terminate Piority Override command is received by the meter before the specified start time and date of the Priority Override (including legacy Priority Override) command, then the Priority Override command shall be cancelled and the CLC/R shall not commence priority override.

b) If the Terminate Piority Override command is received during the Priority Override (including legacy Priority Override) command, then the Priority Override command shall be cancelled and CLC/R control is decided by the RSM (next highest operational load control module) either after a random delay or end immediately as specified in the Terminate Priority Override command.

c) If the terminate priority override command is received when no priority override is programmed (including legacy Priority override), then no changes to the CLC/R state will be made, and the RSM will decide the control of CLC/R.

Priority Command Rejection

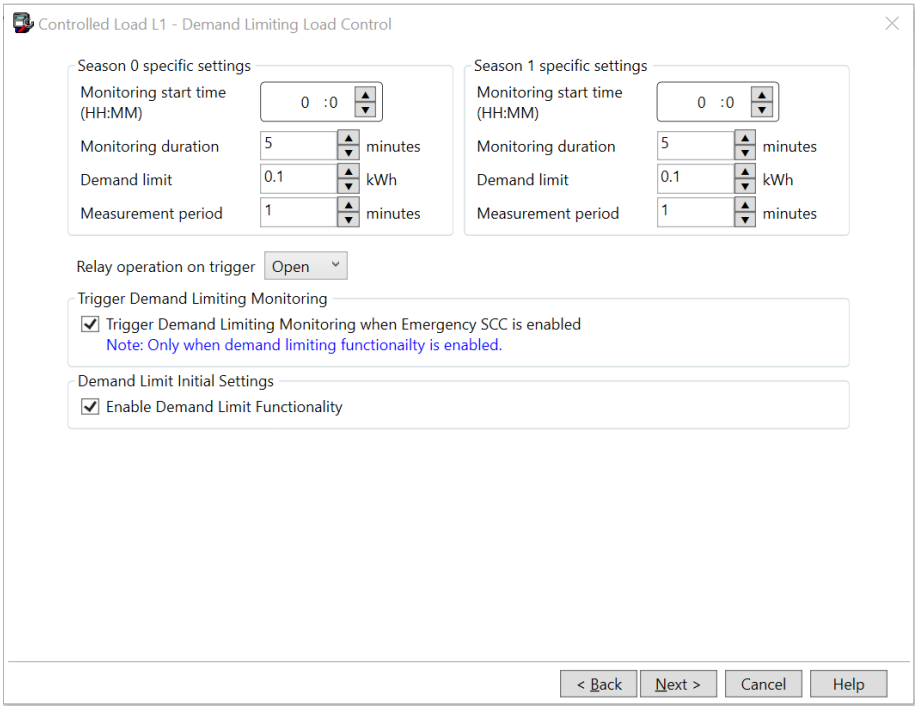
a. Priority Override command will be rejected if the meter receives the command after “start time+min delay” is past.

b. If Priority Override command is received between “start time” and “start time + min delay”, then the min delay will get reduced, however the max delay and the Priority Override duration will remain same.

Proirity Command Collusion

a. If a new Priority Override is received by the meter, when another Priority Override command is already running, the new command will override the old command, if the new command is valid.

12.2.23 Controlled Load L1 – Demand Limit Load Control



Demand Limit Load Control Support in R4.9

DLLC (Demand Limit Load Control) feature provides a programmable setting in the meter that allows LC Relay to opeate when the demand exceeds the set limit in the monitoring period.

Monitoring Start Time

Description	Start of monitoring period when demand is assessed by meter.
Values	minutes in increments of 1 minute (5 to 1440).

Demand Limit

Description	This is the Value of demand (as a function of time) in kWh. If the total kWh consumed between the monitoring period exceeds this value, the meter operates the LC Relay as defined in ‘Relay Operation on Trigger’ settings.
Values	in kWh in increments of 0.1 kWh (0.1 to 99.0).

Measurement Period

Description	This is the measurement period within the measurement duration, where the kWh measurement happens. kWh value of each measurement period is summated in the measurement duration and is assessed with the set limit in the monitoring period.
Values	minutes in increment of 1 minute (1 to 60).

ⓘ Each season has independent settings for DLLC.

Relay Operation on Trigger

Description	This selection determines whether the relay will OPEN or CLOSE when demand limit exceeds.
Values	Open or Close.

Trigger Demand Limit Monitoring

Definition	This feature allows user to link or de-link Demand Limit Load Control functionality with Emergency Supply Capacity Control (ESCC) when ESCC is enabled.

Trigger Demand Limiting Monitoring when Emergency SCC is enabled

Description	As described above for ‘Trigger Demand Limit Monitoring’.
Values	✓ (ESCC linked with DLCC) or blank (ESCC de-linked with DLCC).

Enable Demand Limit Functionality

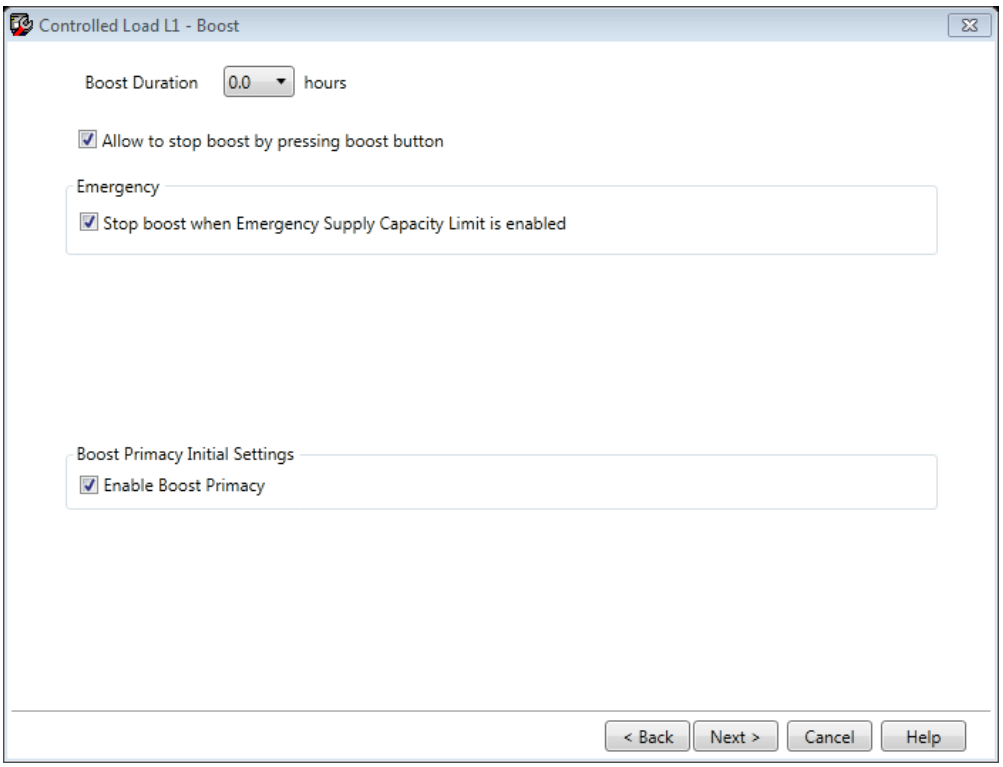
Description	This setting is provided to enable the DLCC feature and save the initial settings to the meters.
Values	✓ (enabled) or blank (disabled)

Refer to Appendix E – DLLC Use Cases, for examples of DLLC operation.

EMPWin+ 2.12 version supports R4.9 and will allow setting of the Demand Limit Load Control (DLLC) feature in a R4.9 firmware based meter. A meter with any other firmware version (lower than R4.9) shall not be configured with EMPWin+ 2.12 for DLLC features, as this could cause problems with meter functionality.

12.2.24 Controlled Load L1 – Boost

The boost functionality will energise the controlled load for a specified time, from 1 to 8 hours in half hour interval.



Boost Duration

Description	The time required to energise the controlled load for a requisite time which is selectable in 0.5 hours increments. To disable the boost button set the duration to 0.
Values	0 to 8 hours

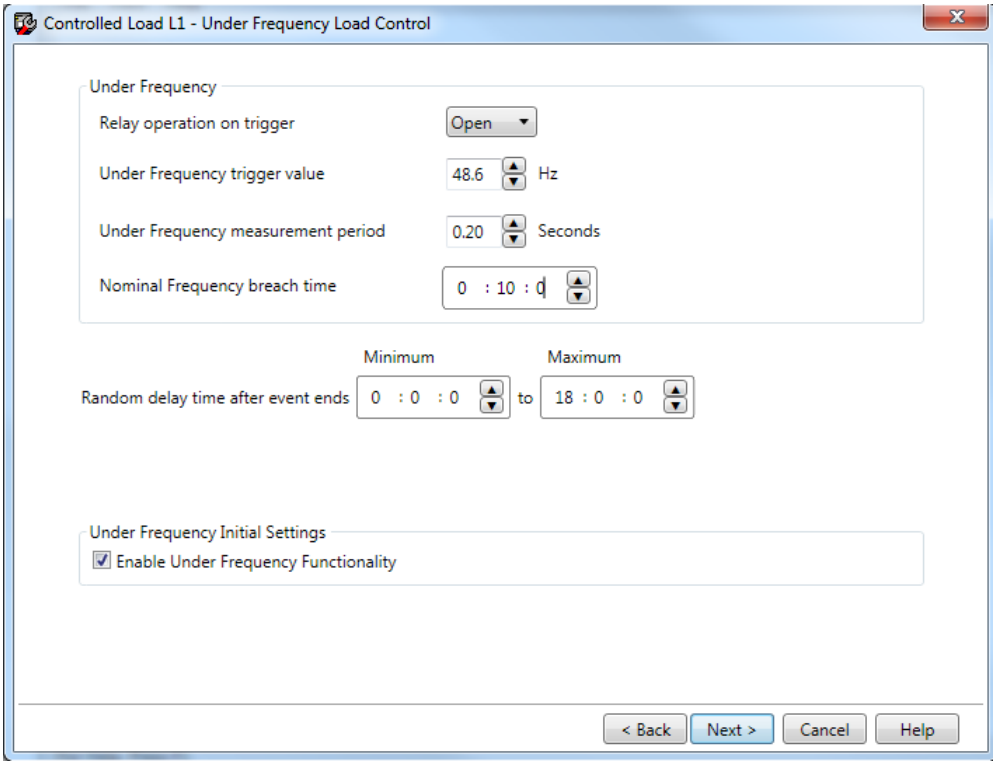
Allow to stop boost

Description	Boost can be stopped by pressing the boost button.
Values	✓ (enabled) or blank (disabled)

Enable Boost Primacy

Description	Enables boost primacy and therewith sets the boost module higher in the Load control RSM hierarchy.
Values	✓ (enabled) or blank (disabled).

12.2.25 Controlled Load L1 – Under Frequency Load Control



Relay Operation on trigger

Description	The action which the relay performs once the condition appears.
Values	open, close

Under Frequency Trigger Value

Description	Threshold value which must be breached to trigger an under frequency event.
Values	40.0 until 65.5 in increments of 0.1Hz

Under Frequency measurement period

Description	This is the duration in which the meter samples and averages the system frequency. At the end of this duration a decision is made whether to open the trigger an event or not.
Values	minimum value 0.01 seconds, maximum value 60seconds, increments of 0.02seconds.

Nominal Frequency Breach Time

Description	The frequency must stay over the under frequency trigger value for at least the duration of the breach time plus a random duration if programmed to terminate the under frequency event.
Values	hh:mm:ss in increments of 1 second (max value 18:00:00)

Minimum Random Delay

Description	This is the minimum delayed duration which is added to the Nominal Frequency Breach Time in which the relay maintains its programmed position.
Values	hh:mm:ss in increments of 1 second (max value 18:00:00)

Maximum Random Delay

Description	The meter calculates a random duration between the min and max value at each power restore. This random delay is added to the programmed position. Once that random duration expires, the module will release control of load control relay immediately. However, this is the maximum duration which could be added to the Nominal Frequency Breach Time in which the relay could maintain its programmed position.
Values	hh:mm:ss in increments of 1 second (max value 18:00:00)

A fixed delay is added to the Nominal Frequency Breach Time by programming the same delay for minimum and maximum.

Setting both values to zero (0) results in an immediate action after the override period expired.

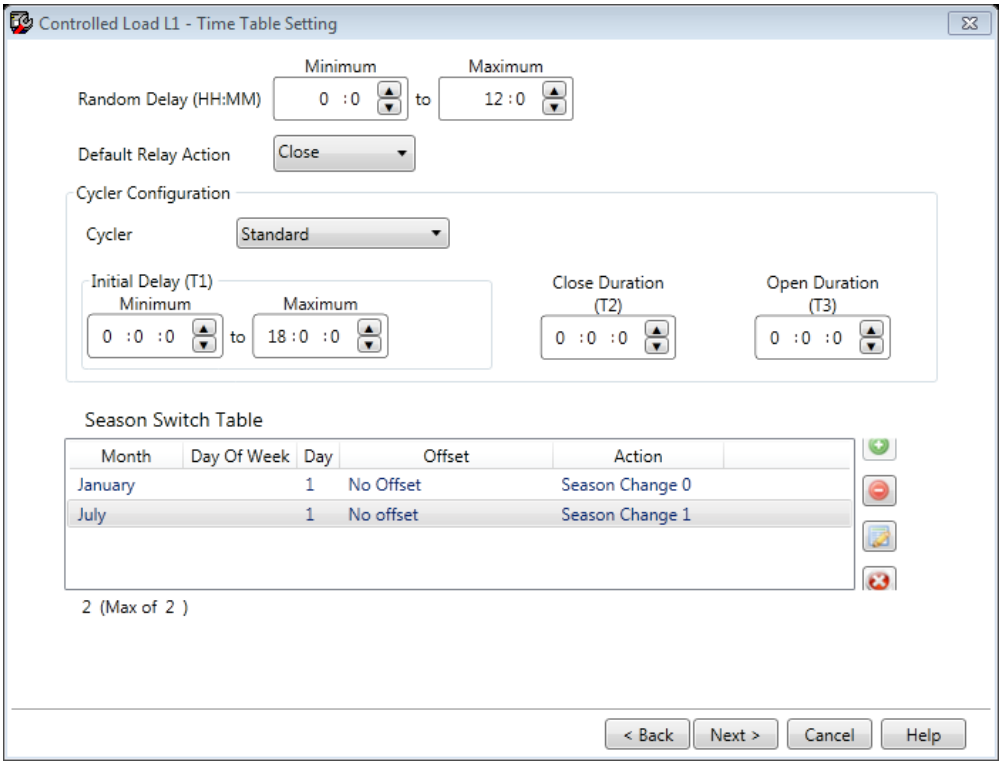
Under Frequency Initial Settings

Description	It allows the user to temporary enable or disable the Under Frequency functionality in operation either locally or remotely.
Values	✓ (enabled) or blank (disabled)

Consideration must be taken when setting these values in systems with high instability on the supply to ensure reliable relay performance and unnecessary toggling of the relay.

12.2.26 Controlled Load L1 – Time Table

This screen provides the main settings for the scheduled load control configuration with random delay, cycler settings and season settings for the load control relay in Time Table mode. In EMPWin+ version 2.12 it is not possible to disable Time Table module. This is done to support Priority Override “Start Load Cycle” command supported by R4.9.



Random Delay

Description

Provides a random delay of the value specified within the time interval of 0 to 12 hours. This random delay is settable in 1 minute increments.

The random delay settings only apply to the delayed open and delayed close relay actions.

A fixed delay is programmed by setting the minimum value equal to the maximum value.

Values

0 up to 12 hours in 1 minute increments.

Default Relay Action

Description

Determines the state of the load control relay when there is no switch times defined and if the time table is in control of the load control RSM.

Values

Open or closed

Cycler Configuration

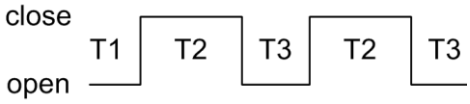
Description

The U3400 supports two kinds of cyclers. These are the standard cycler and the air conditioning cycler. The only difference between these cyclers is how they act on a “stop load cycling” switch action. While the standard cycler stops immediately and opens the relay, the air conditioning cycler checks first in which position the relay is. In case it is in close position it will not open the relay until T2 has expired. This is to protect appliances from short pulses. For detailed information refer to 9.3.2.2 Cycler.

Values Standard / Air-conditioning

Cycler Timing

The cycler causes continuous repetition of the close time (T2) and open time (T3) after the initial delay (T1) has elapsed.



0 value for “Close Duration (T2)” or “Open Duration (T1)” will disable operation of cycler. Any settings other than 0 are valid and may result in a fast toggling of the relay.

Initial Delay (T1)

Description The programmable settings for the initial delay (T1) include a minimum time delay and a maximum time delay.

Values 0h to 18h in steps of seconds

In case a fixed delay is requested then the programmed minimum value must be equal to the maximum value. Otherwise the meter calculates and records a new random delay time which is within the range of the programmable settings at each power up.

Close Duration (T2)

Description The programmable settings for the close duration (T2)

Values Default is 10 minutes. Values can be set between 1 minutes to 18h in steps of seconds

Open Duration (T3)

Description The programmable settings for the open duration (T3)

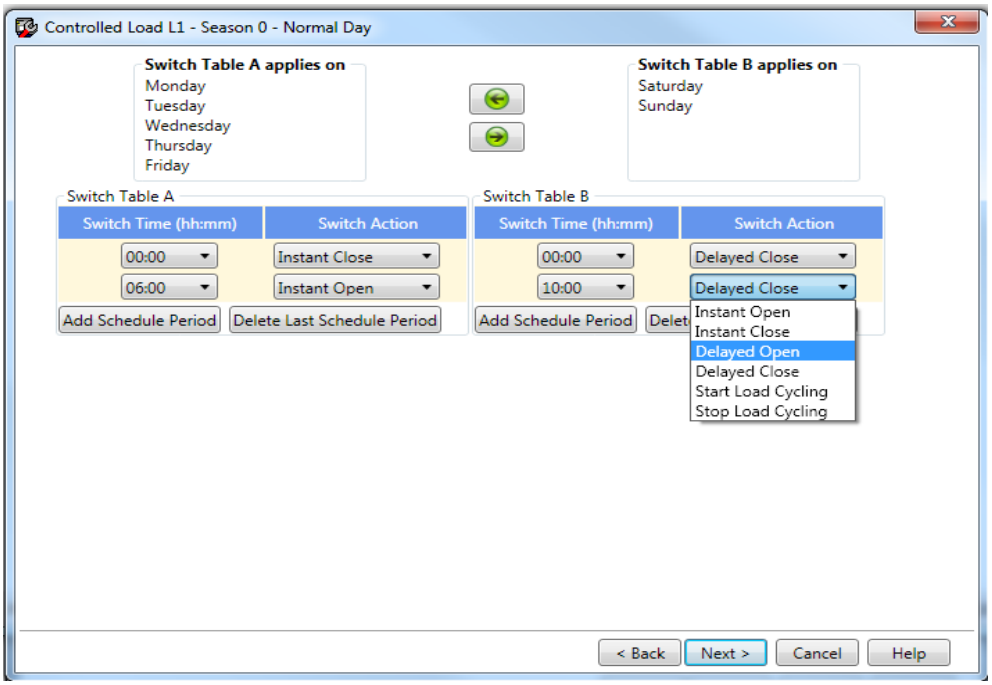
Values Default is 10 minutes. Values can be set between 1 minutes to 18h in steps of seconds

Season Switch Table

Description Two load control seasons allow the User to automatically adjust the LC relay switching times based on a seasonal change over date.

12.2.27 Controlled Load L1 – Season 0 – Normal Day

The program definition for the Season 0 and Season 1 are identical. For simplicity this manual addresses only the Season 0 control relay settings as there is no difference in the way the Season 1 is programmed.



Day allocation in Switch Table A and B

Description: Allows allocating the days to one of the two switch tables. A day can only be programmed into one of the switch tables.

Values: Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday

Switch Table A and B

Description: each of the switch tables allows a maximum of 10 load control switch actions.

The initial Add-Schedule-Period button click will add a new switch time entry box and switch action with respective drop downs. The user can select both, the switch time and the switch action.

Switch Time

Switching times, are user selectable from the drop down list and are settable in 15 minutes increments.

Values: 00:00 to 23:45 (in 15 minutes increments)

Switch Action

Description: The load control schedule action which is performed at this particular switch time.

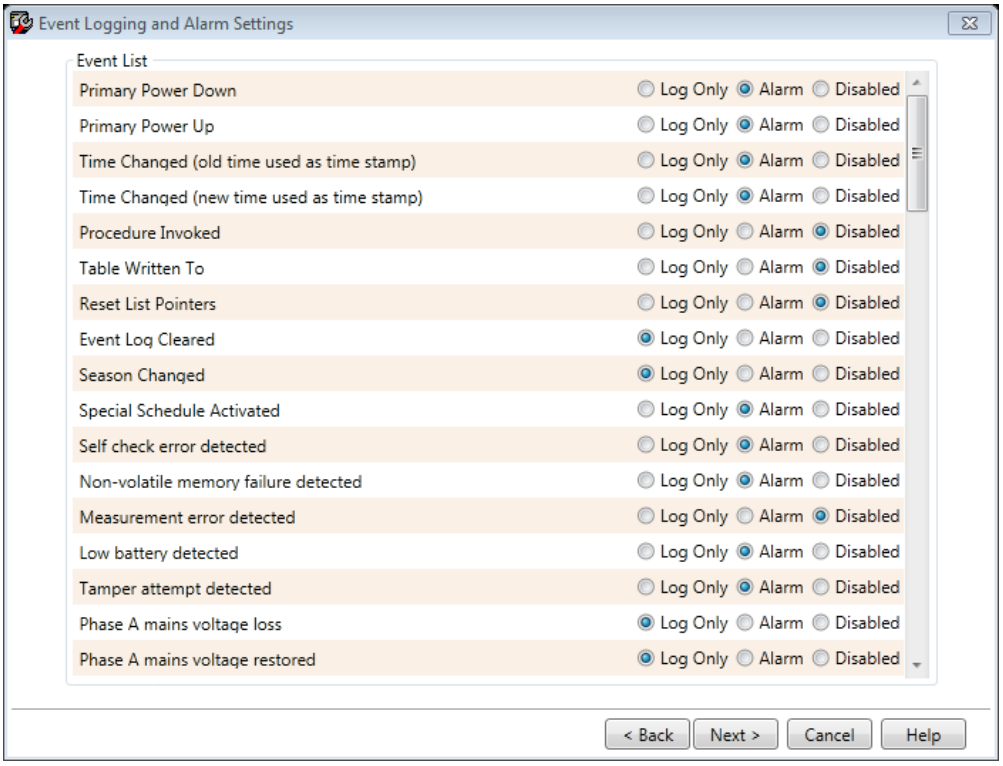
Values: Instant Open, Instant Close, Delayed Close, Delayed Open, Start Load Cycling, Stop Load Cycling.

12.2.28 Prepayment

For details please refer to document:

E350 U-series Prepayment User Manual Appendix

12.2.29 Event Logging and Alarms



Event storage The meter will store the last 500 events captured

Log Only

Description Log only- these are able to be viewed in Load Profile or Program read

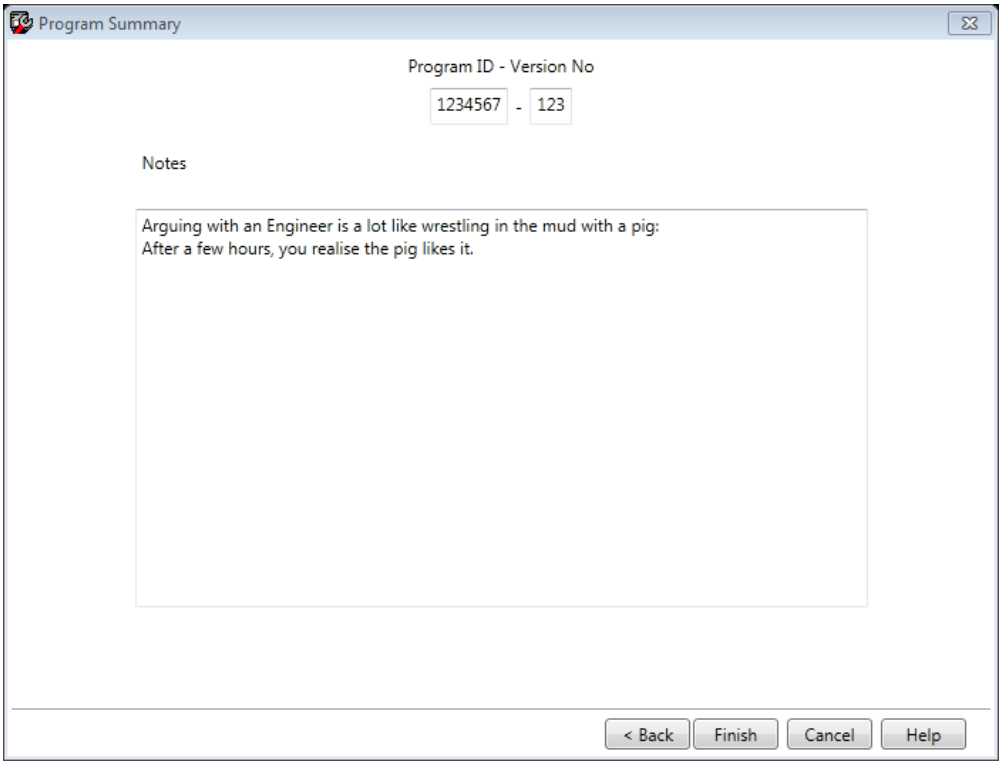
Alarms

Description Alarms- this will log as per the log only setting, but it will also alert the communications module that there has been an alarm triggered.

Disabled

Description Disabled events will not be logged or alarmed.

12.2.30 Program Summary



Program ID - Version No

Description: These two numbers are used to identify each meter program that is developed or modified. It is normal to set the Program ID as the actual number and use the version No. to reflect any changes or variations to the basic program. Major changes result in a new Program ID. Alternatively all 7 digits may be used as a single program number.

Values: The 7 digit Program ID has a value from 0000000 to 9999999. The version number is an optional value in the range from 0 to 999.

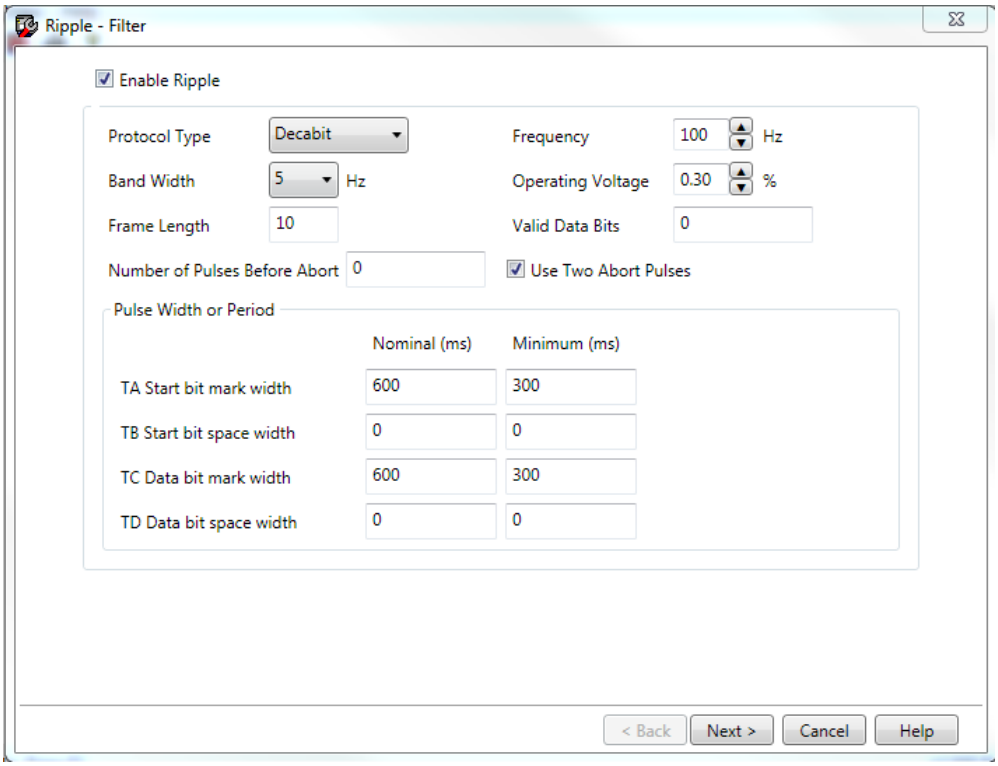
Program Saving

The program should be checked and, if satisfied, click the *Finish* button.

Once the program is saved, another screen pops up to select the template to view the meter developer program. The user can either select the default template available in the EMPWin+ folder.

12.3 Programming Ripple Control

12.3.1 Ripple – Filter



Enable Ripple

Description: "Enable Ripple" determines whether ripple is enabled or not on a ripple program side.

> The main ripple control flag is set in the meter program RSM (see 10.5.19 Controlled Load L1 – Main Configuration. If "Enable Ripple" is set but ripple control is disabled in the RSM then ripple will not work at all.

Values: ✓ (ripple enabled) or blank (ripple disabled)

Protocol Type/ Band Width/ Frequency/ Operating Voltage / Frame Length

Description: These are the filter settings which are described in section 9.2.1 General and following.

Values: refer to section 9.2.2 Filter

Valid Data Bits

Description: This number is used for track specific fail safe (see section 9.4.1.2.1 Valid Data Bits).

Values: The minimum number of "1"s that must be present in a Telegram before it is considered as valid. In Decabit this is always set to 5.

Number of Pulses before Abort

Description This is the number of pulses the receiver must see before an abort command will be recognised.

Values A numeric value which should not exceed the Frame Length of the ripple system.

Use Two Abort Pulses

Description It has the same characteristic as a single abort pulse with the exception that the pulse length needs to be double.

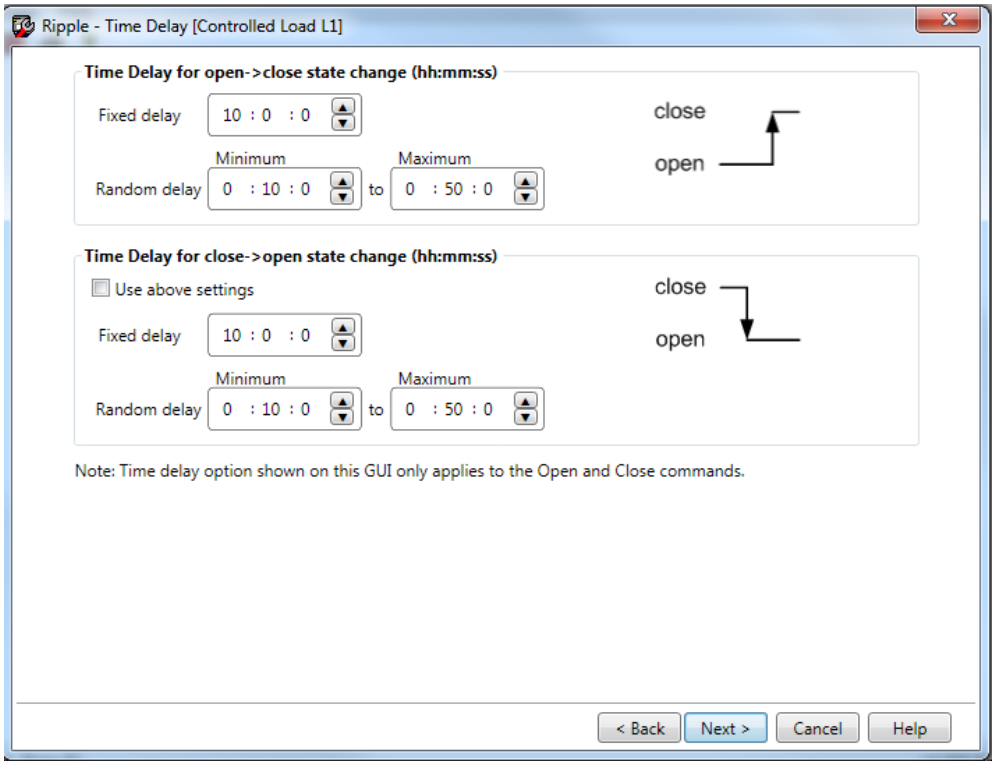
Values ✓ (enabled) or blank (disabled)

Pulse Width or Period

Description When selecting the Protocol Type these fields are filled with pre settings. However, the customer has the option to adjust any of these settings to ensure reception of the signal at sites with conducted noise on the line (refer to section 9.1.1.1 Telegram frame).

Values Depending on the protocol type (value in milliseconds)

12.3.2 Ripple – Time Delay [Controlled Load L1]



Time Delay for open -> close state change

Time Delay for close -> open state change

Description The meter supports two independent delay settings for the two relay state change actions.

These settings apply only for the Instant open/close, Fixed Delay open/close, Random Delay open/close relay actions only (refer to section 9.3.1 Relay actions and section 12.3.7 Ripple – Ripple Telegrams [Controlled Load L1]).

Values Fixed delay: hh:mm:ss in increments of 1 second (max value 12:00:00)

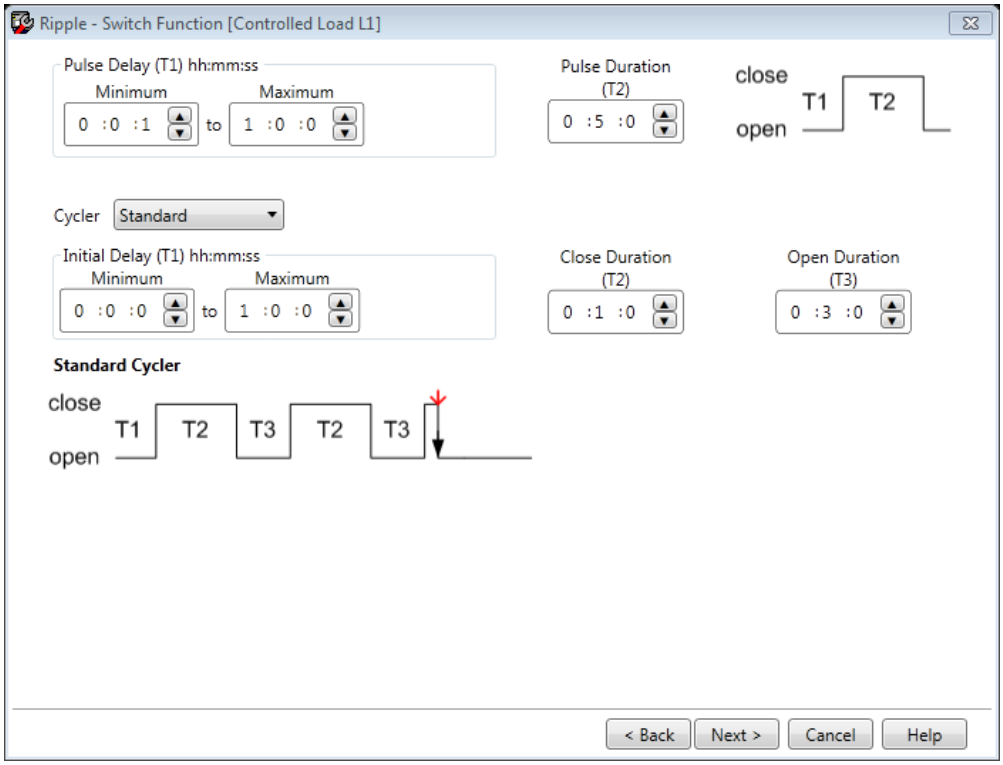
Random delay: hh:mm:ss in increments of 1 second (max value 12:00:00)

Use above settings

Description If this box is ticked then the “open->close state change” settings apply also for the “close->open state change settings”.

Values ✓ (use above settings enabled) or blank (individual settings supported)

12.3.3 Ripple – Switch Function [Controlled Load L1]



Pulse Delay (T1)

Description Sets the initial pulse delay (refer to section 9.3.2.1Pulse).

Values hh:mm:ss in increments of 1 unit (max value 18:00:00)

ⓘ A fixed delay is set by programming the same delay for Minimum and Maximum.

ⓘ Setting both values to zero (0) results in an immediate action, no delay.

Pulse Duration (T2)

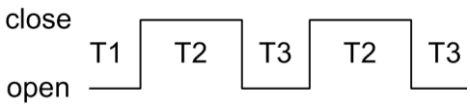
Description	Sets the duration which the pulse will last for (refer to section 9.3.2.1 Pulse) .
Values	hh:mm:ss in increments of 1 unit (max value 18:00:00)

Cycler

Description	The U3400 supports two kinds of cyclers. These are the standard cycler and the air conditioning cycler. The only difference between these cyclers is how they act on a “stop load cycling” switch action. For detailed information refer to 9.3.2.2 Cycler.
Values	Standard / Airconditioning

Cycler Timing

The cycler causes continuous repetition of the close time (T2) and open time (T3) after the initial delay (T1) has elapsed.



0 value for “Close Duration (T2)” or “Open Duration (T1)” will disable operation of cycler. Any settings other than 0 are valid and may result in a fast toggling of the relay.

Initial Delay (T1)

Description	The programmable settings for the initial delay (T1) include a minimum time delay and a maximum time delay.
Values	0h to 18h in steps of seconds

In case a fixed delay is requested then the programmed minimum value must be equal to the maximum value. Otherwise the meter calculates and records a new random delay time which is within the range of the programmable settings at each power up.

Close Duration (T2)

Description	The programmable settings for the close duration (T2)
Values	0h to 18h in steps of seconds

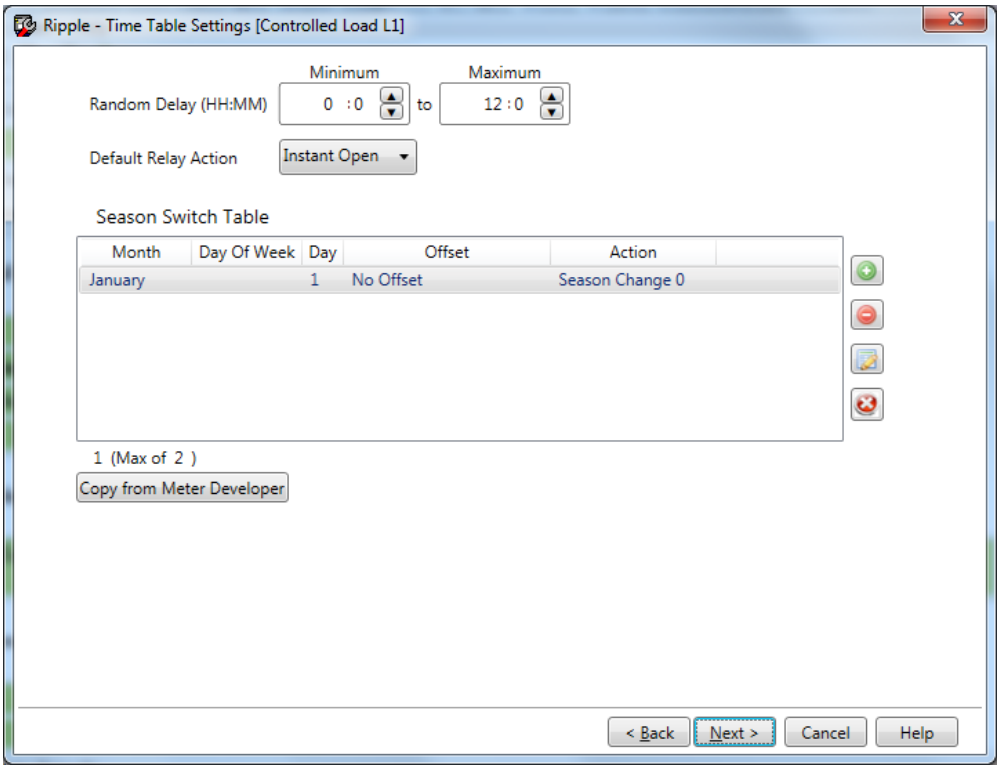
Open Duration (T3)

Description	The programmable settings for the open duration (T3)
Values	0h to 18h in steps of seconds

12.3.4 Ripple – Time Table Settings [Controlled Load L1]

This screen provides the main settings for the Time Table based load control configuration with random delay and season settings.

These settings are used for fail safe functionality or to deliberately enter Time Table mode via ripple command.



Random Delay

Description Provides a random delay of the value specified within the time interval of 0 to 12 hours. This random delay is settable in 1 minute increments.

The random delay settings only apply to the delayed open and delayed close relay actions.

A fixed delay is programmed by setting the minimum value equal to the maximum value.

Values 0 up to 12 hours in 1 minute increments.

Default Relay Action

Description Determines the state of the load control relay when there is no switch times defined and if the time table is in control of the load control RSM.

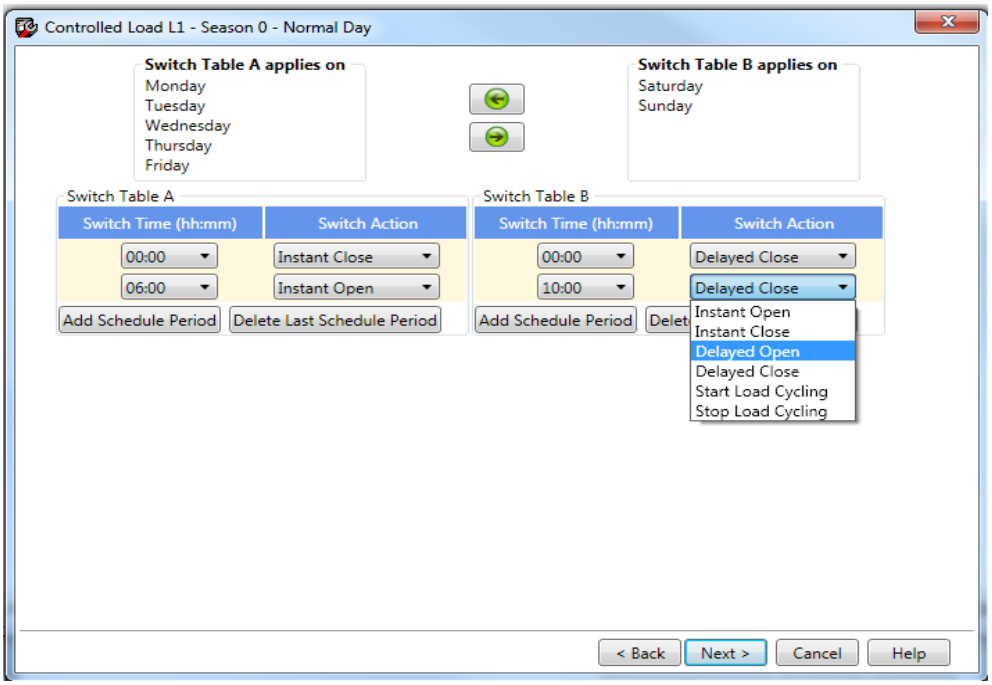
Values Open or closed

Season Switch Table

Description Two load control seasons are avalialbe.

12.3.5 Ripple – Season 0 – Normal Day [Controlled Load L1]

The program definition for the Season 0 and Season 1 are identical. For simplicity this manual addresses only the Season 0 control relay settings as there is no difference in the way the Season 1 is programmed.



Day allocation in Switch Table A and B

Description Allows allocating the days to one of the two switch tables. A day can only be programmed into one of the tables.

Values Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday

Switch Table A and B

Description each of the switch tables allows a maximum of 10 load control switch actions.

The initial Add-Schedule-Period button click will add a new switch time entry box and switch action with respective drop downs. The user can select both, the switch period and the switch action.

Switch Time

Switching times, are user selectable from the drop down list and are settable in 15 minutes increments.

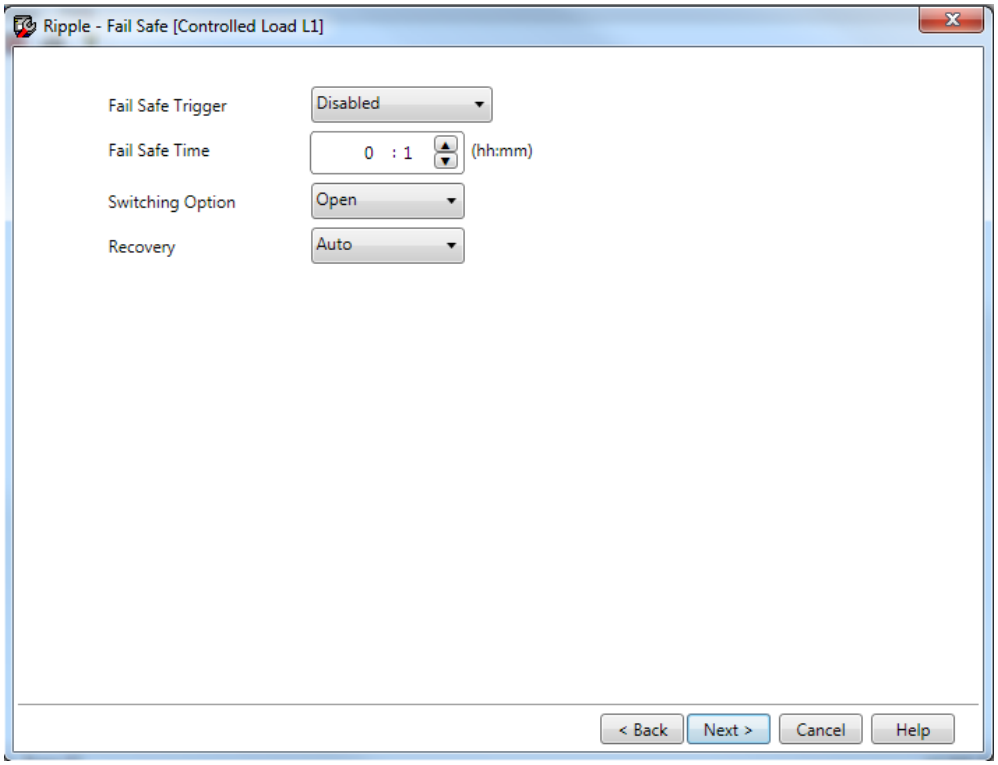
Values 00:00 to 23:45 (in 15 minutes increments)

Switch Action

Description The load control schedule action which is performed at this particular switch time.

Values Instant Open, Instant Close, Delayed Close, Delayed Open, Start Load Cycling, Stop Load Cycling.

12.3.6 Ripple – Fail Safe [Controlled Load L1]



Fail Safe Trigger

Description Refer to 9.4.1 Types of fail safe

Value Disabled, Track Specific, Device Specific, Command Specific

Fail Safe Time

Description The duration which must expire without receiving a valid telegram before entering fail safe mode.

Values hh:mm in increments of 1 minute (max value 72h)

Switching Option

Description The action which shall be performed when entering fail safe mode (refer to section 9.4.2 Fail safe actions)

Value Open, Close, Time Table

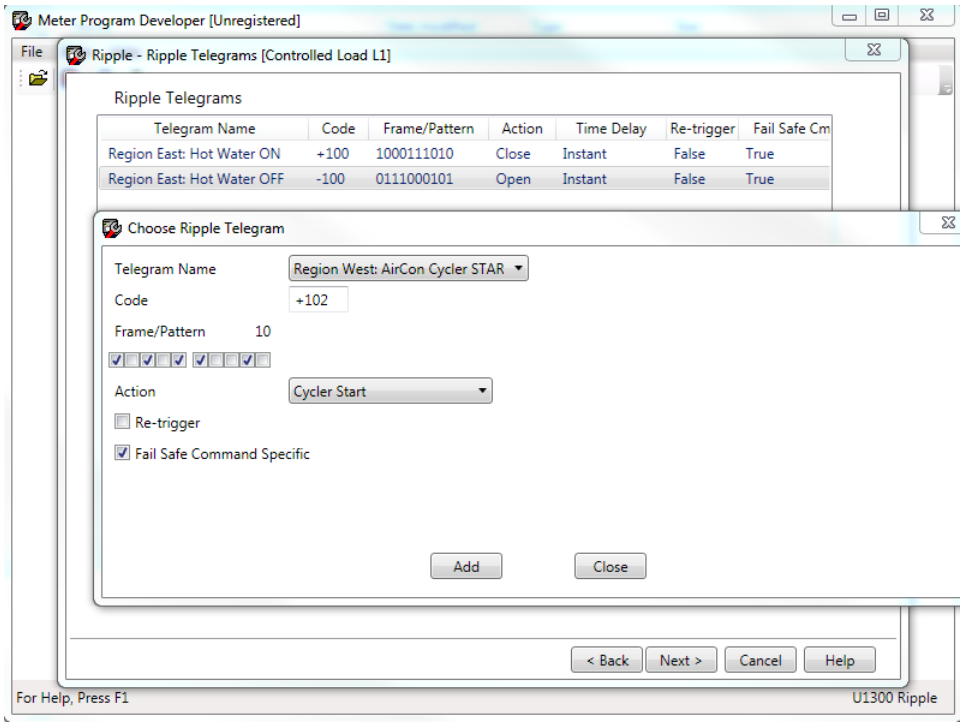
Recovery

Description The condition which must occur to exit fail safe mode (refer to section 9.4.3 Fail safe recovery)

Value Auto, Manual

12.3.7 Ripple – Ripple Telegrams [Controlled Load L1]

This screen is used to define the ripple telegrams, allocate an action to each of them. Also it's used to define telegrams which are used in fail safe.



Telegram Name

Description

Customer can define up to 255 Telegram Names in a separate configuration file which can be allocated to any ripple telegram as additional information. This is helpful when reading a ripple program on site which will provide also the name of the telegram rather than the telegram frame/pattern only.

However, this is optional and if not used it doesn't change the behaviour of the ripple receiver.

See 12.3.7.1 Telegram Name definition

ⓘ The telegram name is not hard linked to the telegram code or frame/pattern.

Code or Frame/Pattern

Description

The ripple telegram code (in this case Decabit) is the numeric form of a Frame/Pattern. One can enter the code into this field and the program translates it into the depending Frame/Pattern. Optional one can set the Frame/Pattern and the program will set the code.

ⓘ This is especially helpful when using long telegram frames such as in Semagyr. Just copy a whole string into the code field will reduce the work load and reduces the risk of failures.

Values

✓ (tick = 1 = must see);

□ (blank box = 0 = must not)

■ (full box = x = don't care = either any character other than 0 or 1)

Action

Description	The action which shall be performed once this telegram has been decoded successfully.
Value	refer to 9.3 Load Control functions

Time Delay

Description	The time delay only applies to the open and close relay actions which are set in 12.3.2 Ripple – Time Delay [Controlled Load L1]
Value	Instant, Fixed Delay, Random Delay

Re-trigger

Description	It applies only for the Switch Function refer to 9.3.2 Switch Functions
Values	✓ (enabled) or blank (disabled)

Fail Safe Command Specific

Description	this box is used to mark the ripple telegrams which must be received to reset the fail safe timer see 9.4.1.4 Command specific
Values	✓ (enabled) or blank (disabled)

12.3.7.1 Telegram Name definition

Customer has the option to define up to 255 Telegram Names which can be allocated to any ripple telegram as shown above.

However there is some important information which needs to be considered before defining the Telegram Names.

1. The Telegram Names are kept in a Customer.config.xml (user readable format) under the EMPWin+ Templates folder. The Telegram Names are not written to the meter more information below.

   Landis+Gyr propose to define all Telegram Names which could potentially be used across the entire network or a region as the first action and secondly keep this Customer.config.xml as the master file.

2. Only these tools of the EMPWin+ suite which have access to the master file will report the Telegram Name correctly either when programming or reading a meter.

   Landis+Gyr propose in order to ensure the meter program reading provides the same ripple telegram name across the entire meter network or region the master file should be copied to any computer which uses the same customer telegram configuration.

3. The meter does not store the Telegram Names as a character string but it stores an enumerator value with each ripple telegram.

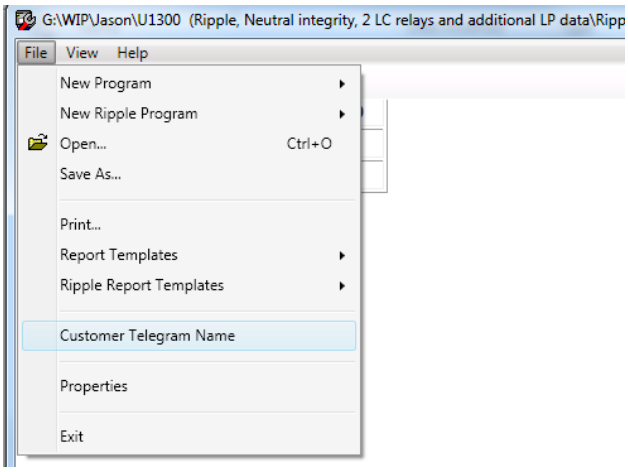
Hence, once a meter is programmed using Telegram Names any change with regards to shifting the order in the Customer.config.xml will result in wrong Telegram Name allocation when reading a previously programmed meters because the enumerator and Telegram name will not line up anymore. Amending the Telegram Name itself without changing the order will not cause a problem at all.

Landis+Gyr recommend not changing the enumerator order.

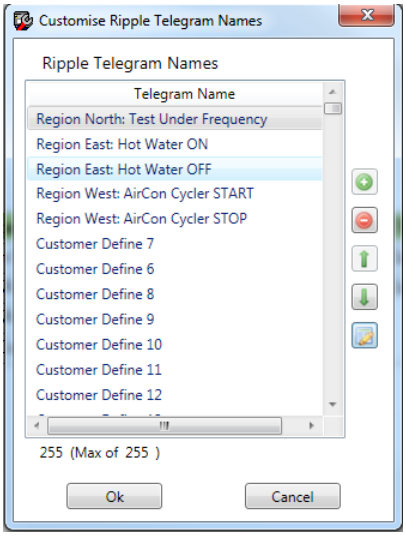
12.3.7.2 Telegram Name programing

The Customer.Config.xml file can be programmed in two ways.

1.) Using EMPWin+ MeterDev as shown below and following the steps which are self - explaining.



Using Customer Telegram Name always opens the Customer.Config.xml on that particular computer under EMPWin+ Template folder.



Enter or amend Telegram Name

Using these two buttons as shown in above window will change the order of the Telegram Name in the background.

2.) Landis+Gyr have developed an excel spread sheet which can be used by the EMPWin+ Super User. It helps to define the Telegram Names in more holistic way by directly editing the Customer.config.xml. Please ask your local sales he will provide this file.

File Name:

E350 U3400 Ripple Telegram Name Definition Tool for Super User.xlsx

12.3.7.3 Telegram Name Example

The below example shall help to understand why it is important not to change the order of Telegram Names:

Below the Customer.config.xml format which was used to program a meter. For this example we focus on Item ID 3 AirCon Cycler START

```
<?xml version="1.0" encoding="utf-8"?>
<Data>
  <Info FileType="Meter Customize file" Version="1.00" />
  <RippleCommand>
    <Item ID="1" Name="Region East: Hot Water ON" />
    <Item ID="2" Name="Region East: Hot Water OFF" />
    <Item ID="3" Name="Region West: AirCon Cycler START" />
    <Item ID="4" Name="Region West: AirCon Cycler STOP" />
  </RippleCommand>
</Data>
```

Explanation of the customer.config.xml:
“Item ID” is the enumerator which is written to meter and which is linked to the telegram code and action.
“Name” is the Telegram Name which is fix linked to the “Item ID” but only in the config file.

Meter Program read using above Customer.Config.xml

Telegram ID	Telegram Enum	Telegram Name	Code	Frame	Action
1	1	Region East: Hot Water ON	+100	1000111010	Close
2	2	Region East: Hot Water OFF	-100	0111000101	Open
3	3	Region West: AirCon Cycler START	+102	1010110010	Cycler Start
4	4	Region West: AirCon Cycler STOP	-102	0101001101	Cycler End

Note: Telegram Enum 3 is linked to Code +102, Action Cycler Start, Telegram Name AirCon Cycler START.

Now for some reason a new Telegram Name “Test Under Frequency“ has been introduced at “Item ID 1” into the Customer.Config.xml. After the change it looks now like this:

```
<Item ID="1" Name="Region North: Test Under Frequency" />
<Item ID="2" Name="Region East: Hot Water ON" />
<Item ID="3" Name="Region East: Hot Water OFF" />
<Item ID="4" Name="Region West: AirCon Cycler START" />
<Item ID="5" Name="Region West: AirCon Cycler STOP" />
```

As you can see this caused a shift of the Telegram Name relative to the Item ID and AirCon Cycler START is now linked to Item ID 4.

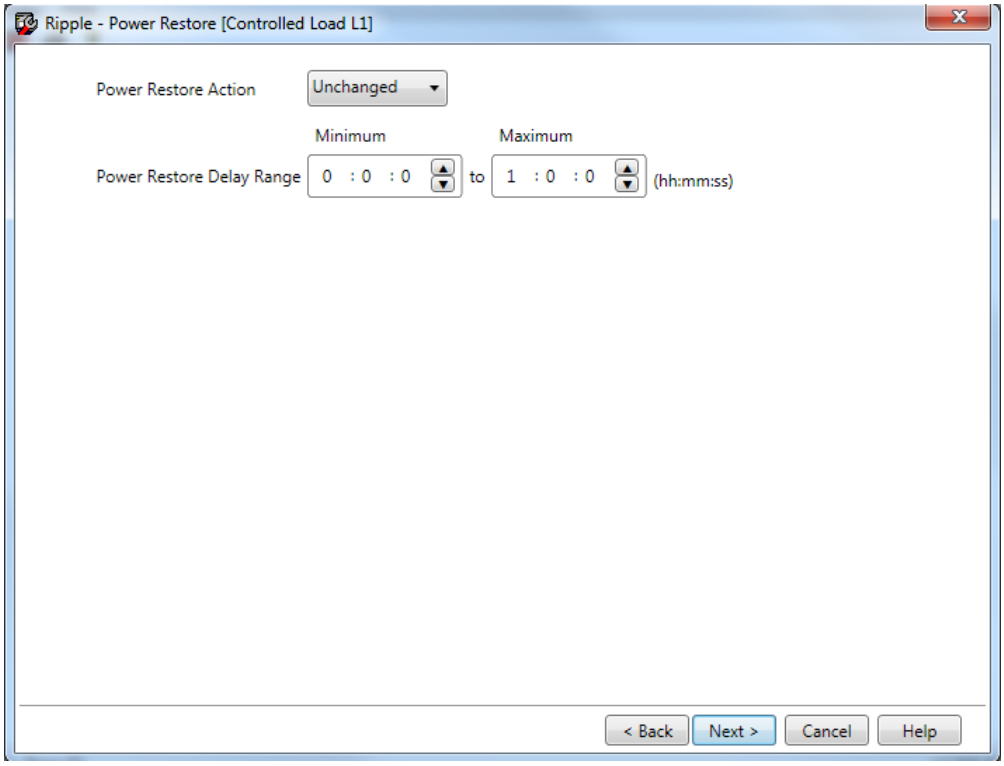
Using this changed Customer.Config.xml file to read a meter which was programmed with previous Customer.Config.xml reports the following read:

Telegram ID	Telegram Enum	Telegram Name	Code	Frame	Action
1	1	Region North: Test Under Frequency	+100	1000111010	Close
2	2	Region East: Hot Water ON	-100	0111000101	Open
3	3	Region East: Hot Water OFF	+102	1010110010	Cycler Start
4	4	Region West: AirCon Cycler START	-102	0101001101	Cycler End

As you can see the Telegram Enum 3 is still linked to Code +102, with the Action of Cycler Start <u>but</u> the Telegram Name has shifted by one according the new Customer.Config.xml. and therefore it does no longer line up with was initially intended.

As shown above **this does not affect the correct operation of the ripple receiver**. However, the Telegram Name may no longer line up with the original file.

12.3.8 Ripple – Power Restore [Controlled Load L1]



Power Restore Action

Description This is the relay action which is performed after the power restore delay has expired (refer to section 9.5 Power Restore).

Ripple has its own power restore settings which apply to the relay state machine.

Values Unchanged, Open, Close, Time Table, Memory (refer to section 9.5.1 Power restore actions).

Power Restore Delay Range

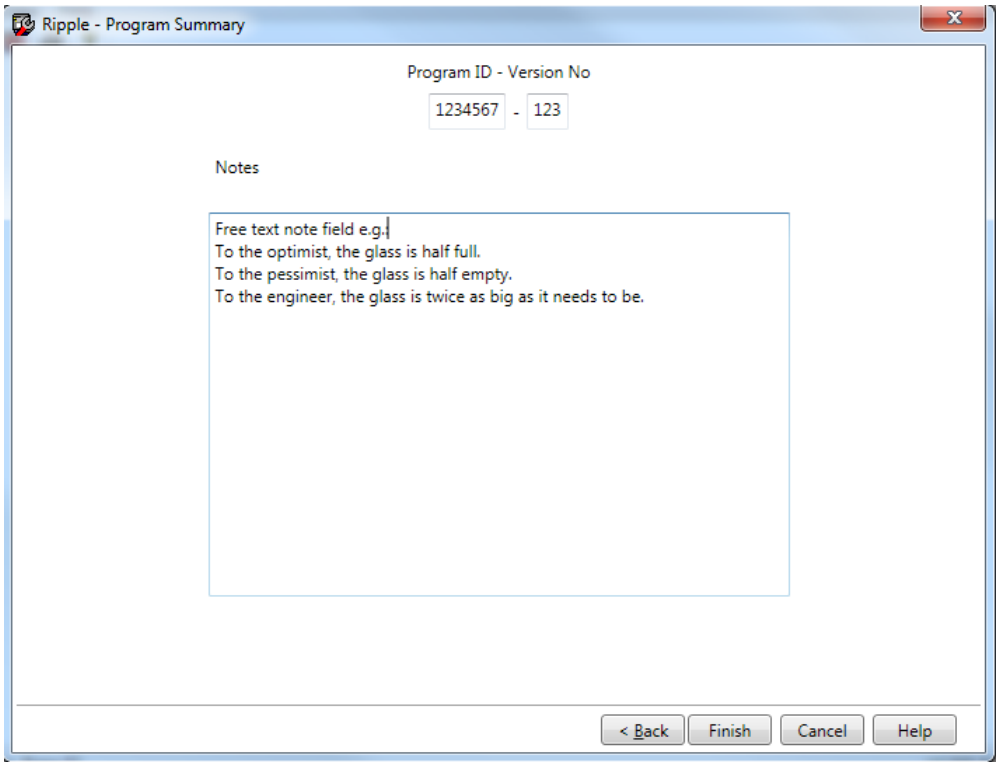
Description The meter calculates a random duration between the min and max value at each power restore. Once that random duration expires, the module will release control of load control relay immediately. However, this is the maximum duration in which the relay could maintains its programmed position. If during this waiting period a valid telegram arrives then the LC relay will be respond to that telegram immediately and this will have a higher priority then the Power Restore Action request. If no telegrams have arrived during the waiting period then on delay timeout the Power Restore Action will be requested and performed.

Values hh:mm:ss in increments of 1 unit (max value 18:00:00)

A fixed delay is set by programming the same delay for Minimum and Maximum.

Setting both values to zero (0) results in an immediate action, no delay.

12.3.9 Ripple – Program Summary



Program ID - Version No

Description These two numbers are used to identify each meter program that is developed or modified. It is normal to set the Program ID as the actual number and use the version No. to reflect any changes or variations to the basic program. Major changes result in a new Program ID. Alternatively all 7 digits may be used as a single program number.

Values The 7 digit Program ID has a value from 0000000 to 9999999. The version number is an optional value in the range from 0 to 999.

Program Saving

The program should be checked and, if satisfied, click the *Finish* button.

Once the program is saved, another screen pops up to select the template to view the meter developer program. The user can either select the default template available in the EMPWin+ folder.

12.4 Security

The U3400 meter can store three (3) different passwords. These are called, Full Access Password, Restricted Password 1, and Restricted Password 2.

The Full Access Password allows full access rights which cannot be altered. The access settings for the restricted passwords are completely programmable with options selected for Restricted 2 creating a subset of options for Restricted 1. There is a hierarchical relationship in the restricted passwords.

Password Hierarchy	
Access Level	**Description**
Full Access	Full utility access to the meter.
Restricted 2	Restricted password two is fully configurable to allow access to specific data and functions.
Restricted 1	Restricted password one has the same functionality as Restricted password two and can be configured to give the least amount of access. Parameters selected for access in Restricted one password also need to be selected in password two.

Programming the meter requires the use of the meter’s Full Access Password which allows unrestricted access to all of the meter’s programmable features. The restricted passwords will allow less access to the meter, e.g. for reading billing or Load Profile information and adjusting the meter’s time or running certain toolbox commands.

Associated with the passwords is a lockout facility. This is activated after three (3) consecutive incorrect attempts to access the meter (at the same access level) and prevents any access at that user access level for a 1 hour period.

The meter can also be programmed to reset this lockout by power cycling the meter. Eliminating the need to wait the 1 hour period for the meter to reset.

12.4.1 EMPWIN+ security access

The Meter System Supervisor should have complete control over EMSecure and its associated functions. An *Administration Password* is required for access to EMSecure and this should be setup and held by the Meter System Supervisor at the time EMSecure is installed. The Administration Password is an alphanumeric code up to 30 characters.

All password allocations are under the Meter System Supervisor's control using the Security Program with the Administration Password. This access allows the Meter System Supervisor to set up all Meter Password Groups and the Users (with their individual User Name and User Passwords) who are to have access to those meters.

The Meter System Supervisor produces a database when he runs the EMPWin+ Security program, EMSecure. The default name of this database is *emsecure.mdb.* For each individual User, the Meter System Supervisor should include a copy of this system database in the same directory as he installs the individual User's copy of the EMPWin+ programs allocated to him (refer to the EMPWin+ user manual for installation details). The default directory for the location of this database is C:\Program Files\Email Metering\EMPWin

If the password system is not required, installing EMPWin+ without the database will cause the software to adopt the default password and no access passwords will be requested. This will require all meters to have the default password (00000000) or access will not be allowed. A more detailed explanation of the Meter Security System is given in the EMPWin+ user manual

The same Administration Password, Meter Group Passwords and User Passwords should be used each time EMPWin+ is upgraded to the next version. This is done by retaining the old version of the Security database (emsecure.mdb). A prompt for this is given each time the EMSecure program is to be updated.

The Administration Password can be changed from within the EMSecure program (File/Change Password). When the program is first installed the default password should be changed immediately.

Entering a blank password eliminates the requirement for password entry and anyone can get access to the security system if this is done.

Detailed instructions for installing and upgrading the EMPWin+ Security Program (EMSecure) are given in the EMPWin+ user manual.

Meter Passwords

Access to a particular meter can only be obtained if the User has an allocated password which is one of the 3 passwords used by that meter. The User need not know the actual password he is using, only the name of the Meter Group he has been authorised to access.

The following parameters are required for the EMSecure program only:

Group Name

Description The name of the Meter Group allocated by the Meter System Supervisor. As explained above each Group Name has 3 meter passwords which can be programmed into meters. If the meter system uses only one set of meter passwords then only one group name need be created.

Values Up to 30 alphanumeric characters (case sensitive)

User Name

Description The name allocated by the Meter System Supervisor to an individual who has access to the Meter System. The User Name and User Password are a password combination which must be entered by the User to gain access to the software for communicating with the meters.

Values Up to 30 alphanumeric characters (case sensitive)

User Password

Description The password allocated by the Meter System Supervisor to an individual who has access to the Meter System. This is coupled with the above User Name and is required for access to the EMPWin+ Communications program.

Values Up to 10 alphanumeric characters (case sensitive)

Full Access Password

The Full Access password must be used when performing full programming of a meter. The restricted passwords should be made available to other Users who do not need full read/write access to the meter. The meter passwords consist of 8 numeric digits in the range from 0 to 9. Meters with the default Landis+Gyr program will have the Full Access password of 00000000.

Description This is the highest level access password for the specified meter group. Authorised main password users have full read/write access rights to carry out all types of meter reading, control, configuration and program modifications for the specified meter group.

Values Integer number must be 8 digits.

Restricted Password 1 and 2

These two passwords and their associated list of Toolbox / Data Read access and Program Configuration access can be independently configured. If the meter is accessed with a restricted password then the meter will determine what functions are permitted. For example, if the meter is accessed with Restricted1 (which might have been configured to disable time/date updates) then it doesn't matter what communication software is used, the meter will not allow its time or date to be modified.

Description A separate list of Meter Programming Access and Toolbox Function Access rights for each of the restricted password levels can be programmed by the

	Meter System Supervisor (Administration Password holder) only, see Security program below.
Values	Integer number must be 8 digits.

13 Disposal

The components used to manufacture the device can, in the main, be broken down into constituent parts and sent for suitable recycling or disposal. When the product is removed from use, the whole product must be sent to a professional electronic waste treatment process. The waste treatment company must be accepted by the officials.

End processing of the product and recycling of its components must always be carried out in accordance with the local laws and instructions given by the officials of the country where the end processing and recycling are done.

By request, Landis+Gyr will give more information about the environmental influence of the product.

This product must not be disposed of in regular waste. Use a professional electronic waste treatment process.

The following are general guidelines and should NOT take priority over local disposal and environmental policies which should be adhered to without compromise.

Component parts	Disposal
Printed circuit boards, LCD, LEDs	Electronic waste: disposal according to local regulations.
Battery	Hazardous waste: disposal according to local regulations.
Metal components	Sorted and delivered to collective recycling point
Plastic components	Sorted and delivered for re-granulation if at all possible

14 Terms and abbreviations

The following terms and abbreviations are used in this document.

Term	Description
AMI	Advanced Metering Infrastructure
AMM	Advanced Metering Infrastructure
ANSI	American National Standards Institute
AS	Australian Standard
DC	Data Concentrator
HAN	Home Area Network
HES	Head End System
IEC	International Electrotechnical Commission
IHC	In Home Controller
IHD	In Home Display
Gridstream AIM	Landis+Gyr's Advanced Metering Management (AMM) system that includes applications for automated meter reading (AMR) and metering data management (MDM).
LAN	Local Area Network
LC	Load Control
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LP	Load Profile
MP	Metering Point
NMS	Network Management System
PV	Photo Voltaic
QoS	Quality of Supply
RF	Radio Frequency
RTC	Real Time Clock
SSN	Silver Springs Network
WAN	Wide Area Network
ZigBee	Radio Communication protocol

Appendix A – Specifications

Accuracy to AS62053.21	
Active Energy	Class 1
Reactive Energy	Class 2
Operating Range	
Voltage (phase to neutral)	240VAC ± 20%
Current	(Ib) 10A – (Imax) 100A
Starting current at 240 V	0.004 Ib or 40mA
Frequency	49-51Hz
Voltage element burden at 240 V	<10VA, <2W
Typical current element burden	0.07 VA
Creep at no load	No more than 1 pulse
Absolute Maximum Ratings	
Current, continuous	100A
Current, short Duration	3000A for 0.5cycle
Voltage continuous	$288V_{AC}$
Voltage, short duration	415V for 1 minute
Insulation withstand	4kV for 1 minute
Impulse voltage withstand	10kV, in accordance with AS62052.11
Min. load control relay switches	30,000
Min. disconnect relay switches	10,000
Battery	
Battery life	Typically 10 years (mains connected) depending on operating conditions such as temperature. Typically 5 years (no power supply) depending on operating conditions such as temperature.
Impermeability	
According AS60529	IP53
Temperature	
Specified operating range	-10°C to +60°C
Limit range of operation	-20°C to +70°C
Storage temperature	-25°C to +70°C
Humidity	
Humidity	Up to 95% non-condensing
Hardware	
Terminal cable size	4 - 35mm^2 Brass Terminal max cable size 16mm^2 Steel Terminal max cable size 35mm^2
Screw Torque	Steel terminals nominal 2.5 Nm, max 3.0 Nm Brass terminals nominal 2.5 Nm, max 3.0 Nm.

	LC terminals 2.0 Nm +/- 10%
Weight	approx. 1.6 kg
Mounting holes	SDM-Mounting
Terminal screws	slotted
Dimensions	
Standard terminal cover	242 X 175 X 109mm
Extended terminal cover	284 X 175 X 109mm
Energy Output LED	
Pulse rate	once every 1Wh/ 1 varh
Program Test LED	Can be programmed for Import or Export wh or varh
General Characteristics	
Time keeping accuracy	Less than 2ppm/day at 23°C using RTC chip.
Communication, optical port	Complies with AS1284.10.2 (ANSI standard interface), 1200 to 38400 baud
Disconnect Relay	100A
Load Control Relay	Max 31.5A
Enclosure construction	10% glass filled polycarbonate with ultraviolet and flame retardant stabilisers are used for the U3400 meter base. The viewing window for the LCD is made of clear ultraviolet stabilised polycarbonate.
Energy Register Resolution	1 Wh for active energy 1 varh for reactive energy
Communication Ports	
Optical port Baud Rate	9600 to 38400kbps
Meter and Communication module	ANSI C12.18/21 Protocol

Appendix B – Event Table

Event	Description
Primary Power Down	Meter power down
Primary Power Up	Meter power up
Time Changed (old time used as time stamp)	Meter time has been changed (time stamped using existing meter time)
Time Changed (new time used as time stamp)	Meter time has been changed (time stamped using new time)
Procedure Invoked	Procedure Invoked
Table Written To	Meter Tables Written To
Event Log Cleared	History of events cleared
Season Changed	Changing from one season to another
Special Schedule Activated	TOU special recurring /non recurring dates
Self check error detected	Failed self diagnostics
Non-volatile memory failure detected	Hardware error
Low battery detected	Battery level is low
Tamper attempt detected	Terminal cover area has been accessed
Phase A mains voltage loss	Loss of mains voltage has occurred
Phase A mains voltage restored	Mains voltage has been restored
Phase B mains voltage loss	Loss of mains voltage has occurred
Phase B mains voltage restored	Mains voltage has been restored
Phase C mains voltage loss	Loss of mains voltage has occurred
Phase C mains voltage restored	Mains voltage has been restored
QOS threshold I breach start	QOS voltage breach has occurred. Shows voltage and date and time stamp -eg: Triggered value is 260.1 - Maximum breach
QOS threshold I breach end	QOS voltage breach has finished. Shows voltage and date and time stamp - eg: Voltage - Worst value is 261.5 - Maximum breach
QOS threshold II breach start	QOS voltage breach has occurred. Shows voltage and date and time stamp -eg: Triggered value is 260.1 - Maximum breach
QOS threshold II breach end	QOS voltage breach has finished. Shows voltage and date and time stamp - eg: Voltage - Worst value is 261.5 - Maximum breach
QOS threshold III breach start	QOS voltage breach has occurred. Shows voltage and date and time stamp -eg: Triggered value is 260.1 - Maximum breach
QOS threshold III breach end	QOS voltage breach has finished. Shows voltage and date and time stamp - eg: Voltage - Worst value is 261.5 - Maximum breach
Service Disconnect open	Disconnect breaker has been opened
Service Disconnect close	Disconnect breaker has been closed
Service Disconnect armed for open	Disconnect breaker is in the arm state, ready for local user to use button on meter

Service Disconnect armed for close	Disconnect breaker is in the arm state, ready for local user to use button on meter
Controlled Load open	Load control relay has been opened
Controlled Load close	Load control relay has been closed
Primary group command accepted	Primary load control group action accepted from initiated group command
Primary group command rejected	Primary load control group action rejected result from initiated group command
Secondary group command accepted	Secondary load control group action accepted from initiated group command
Secondary group command rejected	Secondary load control group action rejected result from initiated group command
Tertiary group command accepted	Tertiary load control group action accepted from initiated group command
Tertiary group command rejected	Tertiary load control group action rejected result from initiated group command
Service Disconnect open due to auto-disconnect	Auto-disconnect breaker disconnected due to criteria exceeded
Service Disconnect open due to supply capacity	Disconnect breaker disconnected due to either the Normal or Emergency supply capacity criteria being exceeded
Service Disconnect close after supply capacity timeout.	Disconnect breaker re-connected due to the supply capacity time out period elapsing
Controlled Load override.	When a communications command has been sent to open the load control relay
Multiple events logged today	50 events logged (which are turned on for logging/alarm)
Controlled Load override timer expired.	The override time specified in the meter program has expired
Controlled Load closed due to boost.	Boost button pressed and relay has closed
Reprogramming	The meter has been partially or fully programmed
Scroll button pressed	Button 1 pressed to scroll displayable items
Alt button pressed	Button 1 pressed for a minimum of 1sec to enable the alternate displays
Boost button pressed	Button 2 pressed for a countdown period 3 seconds to initiate the boost function
Service Disconnect close acknowledged	Following an arm state- button has been pressed
Service Disconnect open acknowledged	Following an arm state- button has been pressed
Primary group command table write	When data has been written to the load control group tables
Secondary group command table write	When data has been written to the load control group tables
Tertiary group command table write	When data has been written to the load control group tables
Failed password on login	During a login attempt the wrong password or encryption key was used
Multiple failed logins	After three sequential login attempts with the wrong password this event will be created
Communication lockout	When an incorrect password or key is used

	three or more times consecutively for a particular user level the meter enters password lockout mode. In this mode the meter refuses to allow the particular user to login for a period of one hour.
Firmware changed	The firmware has been a firmware change in the meter
Master group command accepted	An accepted command to action a relay without grouping eg EMPWin+ toolbox function
Master group command rejected	A rejected command to action a relay without grouping e.g. EMPWin+ toolbox function
Local Parameter Change	Can be any change to the meter via the optical port- tables written to, reprogramming etc
Remote Power Fail	This event occurs if a disturbance is detected on the incoming power supply. It also sets a pin for the remote port hardware to enable early detection of a possible power failure.
Communications Update	This event is triggered each time that the "Comms Display" item (Alt2 Display option only) is shown on the display in response to a procedure write or when modifying the Internal or External antenna selection for 3G or 3G Modem enabling via a manufacturing procedure command.
Load Profile Status Change	The Load Profile status has changed from enabled to disabled or vice versa.
Multi Source Tamper	This event can be triggered by either removing the meter's terminal cover or if the load side voltage is detected as being inconsistent with the open state of the Breaker.
Extended Load Control	This event is triggered when the LC relay is operated by the Power Down LC logic or Under Frequency LC logic.
PredicTemp™	A message is sent to communications module when the sensed internal temperature has traversed the set limits.
Ripple	The meter records all LC switches that originate from the ripple module. It provides information like the action, any delay (if applied) and the LC targeted.
Critical Table Changed	Critical Table change event, logged when the group ID is changed and maybe when other IDs are changed too or when relay present in MT68 is changed
Security Events: • ANSI Protocol • KF2 Protocol	For each of these headings the first ever and last five login failures due to using the wrong password are logged. The Security Events are factory reset and

	are not cleared by any EMPWin+ operation.
Prepayment	Refer to: E350 U-series Prepayment User Manual Appendix
Breaker State Mismatch	
NSMP LoS	
Pulse Output Threshold Control	

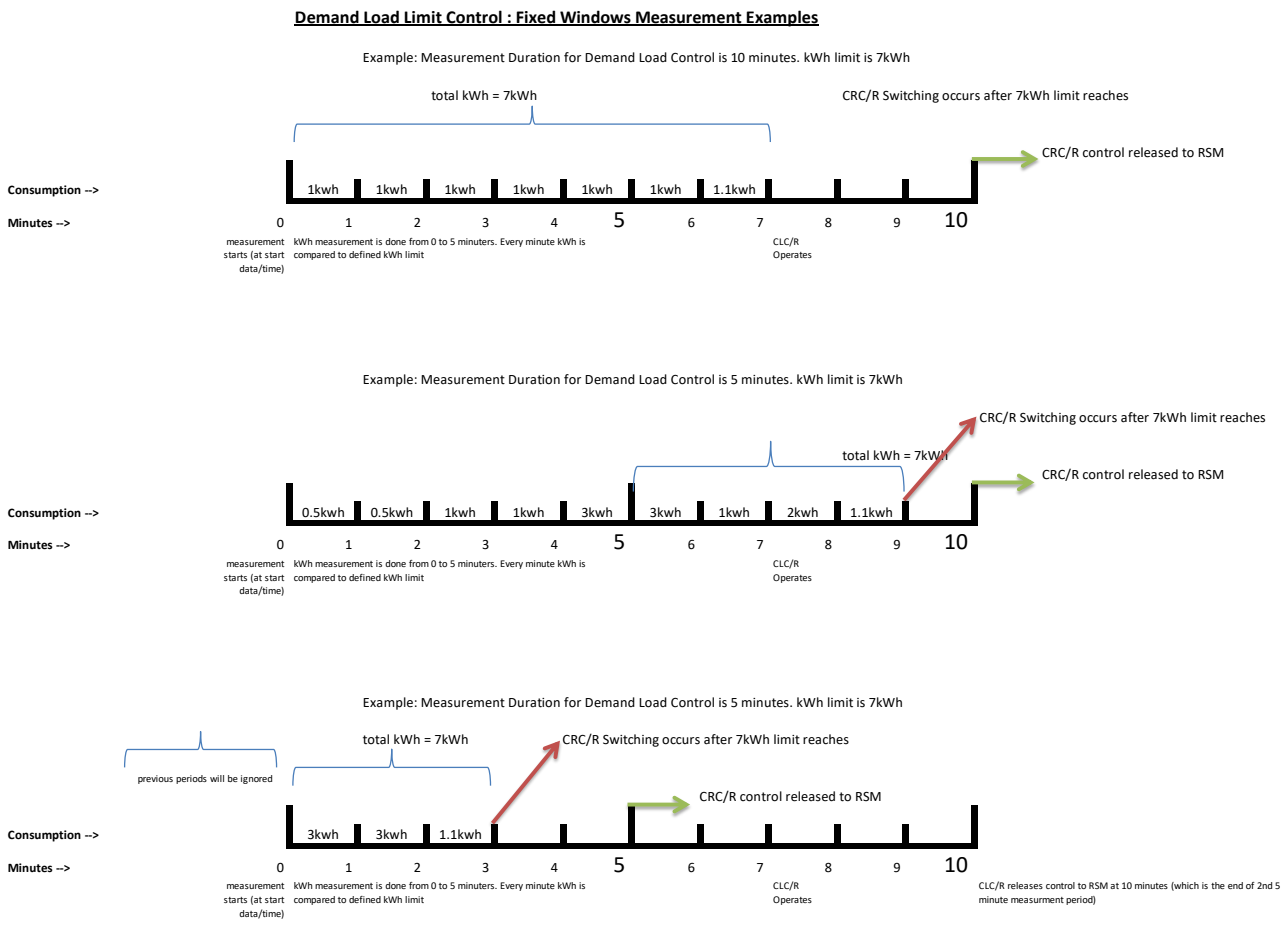
Appendix C – Standards

The U3400 meter complies in full or in part with the following standards relating the Meter hardware, mechanicals, metrology, communication, radio, Radiation/ Protection and other regulatory standards.

Standards	
NMI M6	Pattern Approval and Initial Verification of Electricity Meters and Associated Transformers: Definitions, Metrological and Technical Requirements
AS 60529	Degrees of protection provided by enclosures for electrical equipment
AS 62052-11-2005	Electricity metering equipment (AC) – General requirements, tests and test conditions. Part 11: Metering equipment
AS 62052-21-2005	Electricity metering equipment (a.c.) - General requirements, tests and test conditions - Part 21: Tariff and load control equipment
AS 62053.21-2005	Electricity metering equipment (AC) - Particular requirements - Static meters for active energy (classes 1 and 2) (IEC 62053 21 Ed.1.0 (2003) MOD)
AS 62053.23-2006	Electricity metering equipment (ac) - Particular requirements - Static meters for reactive energy (classes 2 and 3)
IEC 62053-61 Ed. 1.0	Electricity metering equipment (a.c.) - Particular requirements - Part 61: Power consumption and voltage requirements
IEC 62054.11-2006	Particular requirements for electronic ripple control receivers
AS 62054.21-2006	Electricity metering (ac) - Tariff and load control - Particular requirements for time switches
AS1284.10.2-2006	Data exchange for meter reading, tariff and load control - Direct local data exchange via hand-held unit (HHU) - ANSI Standard interface
AS 1284.11-1995	Single-phase multifunction watthour meters
AS 4140-1995	Metering and Utility information exchange
EGR part D(COP4)	Electricity Governance Rules of New Zealand - Part D Metering Arrangements.
DPI Specification Release 1.1	Department of Primary Industries ADVANCED METERING INFRASTRUCTURE Minimum AMI Functionality Specification (Victoria) Release 1.1

Appendix D – Accessories

Device part number	Description
CD-EMPWin+	EMPWin+ Software for configuring and reading the U3400 meter.
AIP200	AIP200 Optical Probe (ANSI and IEC) Fitted with 9 Pin D Connector and 1.5 metre Cable
AIP200-USB	AIP200 Optical Probe (ANSI and IEC) with in-built Smart Connect and USB connector on 1.5 metre cable
FTMS-AIP300	AIP300 Optical probe with Bluetooth and USB connector
849710	U3400 Short terminal Cover Assembly
849711	U3400 Long terminal Cover Assembly
849263	U3400 Communications Cover
	Landis+Gyr communication modules. Please contact your sales represantive

end of the E350 U3400 – User Manual

Appendix E – DLLC Use Cases